

The Winner's Curse in Test-Blind Admissions^{*}

Phi Adajar

Massachusetts Institute of Technology

Job Market Paper

Nagisa Tadjfar

University of British Columbia

Kartik Vira

Institute for Fiscal Studies

October 6, 2026

[Click here for the latest version](#)

High-stakes final exams play two roles in university admissions: they give students an incentive to exert effort, and they give universities information with which to screen applicants, shaping the equilibrium match. We study these forces using the staggered rollout of unconditional offers in the UK, which let some students matriculate without requirements on their age-18 exams (A-levels). Students who accept an unconditional offer score 0.14SD below those who decline one, conditional on observables. However, using a two-way fixed effects specification, we find that exposure to unconditional offers has precisely zero impact on students' age-18 exam performance, indicating that the observed gap is driven by private information. Exposure to unconditional offers increases university matriculation by 1.1 percentage points, mostly at middle-tercile universities. A structural model that incorporates the value students place on certainty of admission shows that, in equilibrium, recipients' gains come at the expense of non-recipients. Relative to a ban, unconditional offers lower total student welfare by 0.7%, hurting the three highest-achieving quartiles of students. Post-qualification admissions, under which students apply after receiving their scores, raise welfare by 12.5% through better sorting. Our results highlight how the distributional effects of admissions policies depend on student information, informing current policy debates in the UK.

^{*}Contact: phiadajar@mit.edu. We are grateful to Nikhil Agarwal, Esther Duflo, and Parag Pathak for their guidance through this project. We thank Joshua Angrist, David Autor, Peter Blair, Glenn Ellison, Amy Finkelstein, Nathan Hendren, numerous participants at the MIT Labor Lunch, the MIT Applied Microeconomics Seminar, and the CSQIEP mentoring conference for helpful feedback on this work. This work was undertaken in the UK Office for National Statistics (ONS) Secure Research Service using data from the ONS and other owners. All analysis and conclusions are those of the authors, and do not necessarily reflect the view of any other party. We acknowledge support from the George and Obie Shultz Fund at MIT, the National Science Foundation, the NBER PhD Fellowship for Global Math Talent, and the Jerry A. Hausman Graduate Dissertation Fellowship.

1 Introduction

In 2024, nearly three million students across the US, France, and Japan — 79% of these countries' college entrants — were admitted to college before or without a final exam score.¹ When universities choose which students to admit, it can change the course of those students' lives — and thousands of colleges do so without considering a key component of a student's academic record. These test-optional and test-blind systems often allow students to know if and where they can attend university much earlier, but may also lead to students studying less. Further, universities that use test-optional or test-blind admissions have fewer ways to evaluate students, making their selection problem more difficult. Media narratives and research on this topic have largely focused on the use of standardized tests at elite colleges, but these policies are relevant for students across the ability distribution and across the world.

Whether universities should or should not use final exams to admit students depends on the answers to three questions. When university entrance no longer depends on exams, do students learn less? How do test-blind admissions change if and where students attend university, considering both direct impacts on students subject to these policies and spillovers to other applicants? And finally, given what students value — including both match quality and certainty — are there admissions systems which outperform a test-blind mechanism?

We answer these questions with evidence from universities in the United Kingdom, which used a form of test-blind admissions during the 2010s. After applying to programs, students receive tentative offers which are conditional on age-18 standardized test scores (A-levels); they convert to admittance so long as students meet the relevant conditions. In 2014, some programs began making *unconditional offers*; rather than having admittance rely on future age-18 test scores, programs guaranteed admittance for students regardless of their scores on the exam. Students who ranked these offers gained certainty about their placement, but possibly at the expense of attending a more-selective program if they earned high grades. Policymakers criticized their rapid proliferation, concerned that these offers shifted students toward worse universities and reduced students' academic effort in studying for their exams. These offers were eventually banned by universities themselves in 2021.

Unconditional offers were used by academically less-selective programs to enroll higher-ability students, but the expected gains in student achievement did not materialize. These programs targeted unconditional offers toward high-achieving students with 0.4SD higher teacher-predicted grades on average. Extrapolating from students' high academic characteristics observed at the time of offer, programs would have expected the realized age-18 test scores of incoming students to increase by 0.06SD in the year they began using these offers. However, enrolling students, conditional on observables, scored 0.14SD *lower* on their age-18 exams than students who received the offer but declined. As a result, programs experienced a *decrease* in achievement of 0.02SD when they began making unconditional offers.

¹In the US, we count students attending community colleges or students enrolling at other colleges without submitting the ACT or SAT. In France, we count *lycée* students who receive *Parcoursup* offers before taking their *baccalauréat* exams; offers are contingent only on passing this exam. In Japan, we count students who enroll in private universities through the *sōgō-gata* or *gakkō suisen-gata* selection paths, which rarely require the Common Test for University Admissions.

This taker–decliner achievement gap is consistent with two factors. There is a *moral hazard* explanation: students may reduce test-taking effort after taking an unconditional offer, knowing that their end-of-year grades no longer matter. On the other hand, students may have *private information*: students who expect to do poorly on their end-of-year exams, and so to miss the grade requirements of their other offers, may self-select into taking these offers. Critics, including the Education Secretary, argued that these offers reduced students’ effort and distorted students’ decisions, and Universities UK cited lower motivation among takers when it restricted them in 2021. Effectively, these narratives claim that the taker–decliner gap was entirely due to moral hazard, even though it may have also been caused by private information.

We distinguish between these two explanations with reduced-form evidence. To test for moral hazard, we estimate the effect on age-18 test scores of unconditional offer *exposure*, defined as applying to a program that uses these offers. We focus on the first year that an application set is exposed to unconditional offers; first-year exposure is plausibly exogenous if students are unaware of when programs begin making these offers, and we provide evidence consistent with this assumption. We estimate the effect of first-year exposure on test scores using a two-way fixed effects design which controls for students’ application sets, interacted with an indicator for post-first-year usage to isolate first-year treatment effects.

Using this specification, we find that exposure to unconditional offers is inconsistent with the moral hazard hypothesis. We estimate a statistically precise null effect of unconditional offer exposure on later test scores. This finding is robust to a battery of alternative specifications. We conclude that contrary to popular belief, the gap in achievement between unconditional offer takers and decliners is driven by private information, not moral hazard.

Though they did not affect test scores, unconditional offers did increase university attendance and the quality of universities attended. Students exposed to unconditional offers are 1.1 percentage points more likely to matriculate into university. This increase is driven by attendance at middle-tercile universities. Among the set of students that attend university, exposure to unconditional offers marginally raises the quality of university attended.

We then turn to the general equilibrium effects of unconditional offers, and consider how they affect students who do *not* receive unconditional offers, and if there are policies which are better for welfare. To answer these questions, we estimate a structural model of university applications. Our model includes both private student information and an unconditional-offer premium: the value a student places on an offer being guaranteed, whether from risk aversion, lower stress, or settling her plans early. Given the absence of moral hazard in our reduced-form results, we can simplify the model by assuming that students’ effort is not reduced in response to unconditional offers. The model features a four-stage process mimicking the UK university application system in which students hold private information about their own eventual exam performance. We recover the unconditional-offer premium and the extent of private information: the model estimates that the unconditional-offer premium is small for the average student, and that students know two-thirds of the variance of their age-18 test scores before taking them.

We then turn to estimating welfare under various admissions designs: a full ban on unconditional offers, unconditional offers used at their 2018 levels, and post-qualification admissions, under which students apply to university only *after* receiving their age-18 grades. Post-qualification

admissions were debated in the UK in 2021 and are again under discussion as AI makes predicted grades harder to defend ([Harvey-Brewin, 2026](#)). In aggregate, unconditional offers at 2018 levels decrease student utility by 0.7%, benefiting the lowest-achieving quartile of students at the expense of the other three. However, post-qualification admissions lead to an increase in welfare of 12.5%, largely through improved allocation, with benefits accruing to higher-achieving students. Programs largely benefit from post-qualification admissions, as well; however, the most-selective quartile of programs is marginally hurt, which may lower the chance of policy reforms occurring.

Our counterfactual simulations and reduced-form analysis contribute to the growing literature on the optimal design of college admissions systems. The shift to test-optional university admissions in the wake of COVID-19 — and the shift back to test-required admissions a few years later — have led to a rapid proliferation of papers analyzing the efficacy of such policies and impacts on students and cohort composition, mostly at selective institutions ([Sacerdote, Staiger and Tine, 2025](#); [Avery, Shi and Magouirk, 2025](#)). [Dessein, Frankel and Kartik \(2025\)](#) focuses on universities’ choice to be test-optional or test-required; in their model, inter-university competition leads to an adversely selected applicant pool. [Doğan and Erdil \(2025\)](#) discusses the benefits of post-qualification admissions from a theoretical lens. [Borghesan \(2025\)](#) uses a structural model to derive equilibria when universities stop requiring the SAT. Our contribution here is twofold. We provide reduced-form evidence on test-blind admissions from less-selective universities, allowing us to test the broader impacts of such policies, and do so in a setting that informs the global debate.

Our setting also allows us to study how admissions policies induce or deter student effort. A small literature shows that affirmative action can raise effort exerted by high-school students who benefit from such policies ([Grau, 2018](#); [Akhtari, Bau and Laliberté, 2024](#); [Cotton, Hickman and Price, 2022](#)). A recent paper from [Krishna et al. \(2026\)](#) shows that changing grade reporting on centralized tests can also induce effort. Existing evidence on how test-blind policies change student effort is correlational ([Office for Students, 2019](#)); we provide the first causal estimate, separating the effort response from selection.

The timing of admissions has been studied through a large theoretical literature; in matching markets where students are screened by multiple schools, enrollees are adversely selected. Most directly, [Chade \(2006\)](#) develops a model with imperfect signals in which being matched leads to an “acceptance curse”; this is further developed by [Lee \(2009\)](#) in the university admissions setting.² In these papers, inter-university competition leads to adverse selection; in our setting, selection occurs primarily through students’ private information about a bad future signal. In contrast to [Lee \(2009\)](#), who shows that the winner’s curse leads to efficiency-improving early admissions, we demonstrate empirically that selection on students’ private information acts *against* early admissions. Finally, this literature is largely theoretical; we provide reduced-form evidence resulting from quasi-experimental variation in offer timing.

We also contribute to a literature on preference estimation in the context of application systems. We build upon the work of [Agarwal and Somaini \(2018\)](#), [Larroucau and Rios \(2020\)](#), [Bleemer \(2024\)](#), and [Kapor \(2025\)](#), which model students’ application choice of portfolios or rank-order lists. While the literature abstracts away from the timing of offers, we incorporate offer types into

²See also [Chade, Lewis and Smith \(2014\)](#) and [Kloosterman and Troyan \(2020\)](#); the latter focuses on student inferences after being accepted.

the student-side choice problem directly. Further, we discuss how preferences under correlated admissions systems (such as those in [Ali and Shorrer, 2025](#)) can be estimated using the methodology of [Larroucau and Rios \(2020\)](#), which explicitly requires independent admissions.

Our decomposition of the taker–decliner achievement gap relates to a literature focused on insurance and consumer credit differentiating adverse selection and moral hazard. The positive correlation test of [Chiappori and Salanié \(2000\)](#) describes how correlation between insurance take-up and outcomes is a sign of private information, but distinguishing the two forces requires additional exogenous variation, such as the field experiment studied by [Karlan and Zinman \(2009\)](#).³ We bring this literature to the new domain of college admissions.

2 Empirical Setting, Data, and Descriptive Results

2.1 University applications in the UK

In many countries, including Germany, Australia, and China, universities use *post-qualification* admissions systems, meaning that students take local or national standardized examinations and only after receiving the scores do students apply to and enroll in universities.⁴ In this sense, both students and universities finalize the matching process after all the relevant information (i.e., test scores) is observed by both parties.

In contrast, the United Kingdom uses *pre-qualification* admissions. Students apply to university in January during their final year of secondary school (Year 13) with grades from exams taken at age 16, as well as teacher-predicted grades for age-18 exams that students have not completed when they apply. Students receive offers and rank choices before taking their exams and receiving their final grades. As a result, programs have historically made “conditional offers”, which are contingent on students achieving certain grades in their final exams; programs can withdraw offers to students who do not meet these conditions. This means that students face uncertainty about their university enrollment until they receive their final grades.

Approximately 250,000 18-year-old students from England, Wales, and Northern Ireland⁵ apply every year to university through the centralized Universities and Colleges Admissions Service (UCAS) platform. Because tuition fees for domestic students are uniform across universities⁶, the relevant tradeoffs for UK students are over academic fit and preferences for geography, amenities, and other factors. [Table 1](#) summarizes the process as of 2012.⁷

³See also the canonical models of [Akerlof \(1970\)](#) and [Rothschild and Stiglitz \(1976\)](#); further theoretical work in [Chiappori et al. \(2006\)](#); and [Adams, Einav and Levin \(2009\)](#), [Einav, Finkelstein and Cullen \(2010\)](#), and [Einav and Finkelstein \(2011\)](#) for additional empirical studies.

⁴In Germany, this is the *Abitur*; in Australia, the *ATAR*; in China, the *gāokǎo*.

⁵We exclude Scottish students, as they face different exam and tuition fee systems to students in the rest of the UK.

⁶Tuition fees for domestic students are capped in England, Wales, and Northern Ireland, and essentially all universities charge fees at the cap. In England and Wales, the cap was set at £9,000 per year in 2012, and following irregular below-inflation increases, it currently stands at £9,790. Students have access to standardized government loans with income-contingent repayment plans for tuition and living costs.

⁷There have been several changes in the university applications process during our analysis window. In 2007, the first year of our data, students could apply to up to 6 programs. This was reduced to 5 in 2008. In the late 2010s, the UK government phased out an intermediate exam taken at age 17 prior to university applications; refer to [Tadjfar and Vira \(2025\)](#) for more details.

Table 1: Timeline of UK university applications process as of 2012

January	Students apply to up to 5 programs, specific majors at specific universities. ^a Applications include age-16 and age-17 standardized test grades and teacher predictions of age-18 grades.
February–May	Programs make offers of places. Offers are contingent on the student meeting conditions on their age-18 grades.
June	Students rank up to two programs from which they receive offers; all others are forfeited. ^b Students then take age-18 exams.
August	Age-18 exam results are released, and students match to programs. Programs must accept all students who ranked their offer and met the offer conditions, and have discretion to accept students who failed to meet conditions. Students who are not accepted at either ranked program may go unmatched or join an after-market clearing process. ^c

^a There are approximately 35,000 such programs across 160 different universities.

^b These are termed the *firm* and *insurance* choice; we call them the first and second choice throughout.

^c 71% of students who ranked programs are admitted to their first choice, 10% to their second choice, 10% go unmatched, and 9% are admitted to a different program through the clearing process.

2.2 Unconditional offers

In the 2010s, universities increasingly began using *unconditional offers*, a form of test-blind admissions. In contrast to conditional offers, an unconditional offer admits students to a program regardless of their age-18 exam results. Historically, these offers were primarily reserved for students applying to university after completing secondary schooling (such as students re-applying after a gap year), as these students have already completed their exams and their final grades are included in the initial application. In 2014, programs increasingly began to make unconditional offers to applicants who had *not* completed their exams.⁸ Programs also began to make *binding unconditional offers*,⁹ where the offer had no conditions only if the student ranked it as their first choice. We pool together these two types of unconditional offers and refer to them as unconditional offers throughout. Figure 1 shows the rapid growth in unconditional offers from 2010 through 2019 for our analysis sample: in that final year, 38% of applicants received at least one unconditional offer.

Reactions to the growth of unconditional offers varied greatly. Critics, including the Education Secretary and the Association of School and College Leaders,¹⁰ argued that offers were primarily a tool for universities to increase enrollment rather than a genuine benefit to students, and that the pressure to commit to a university before sitting age-18 exams discouraged academic effort.¹¹ A particular concern was that binding unconditional offers pressured students to forfeit their higher-

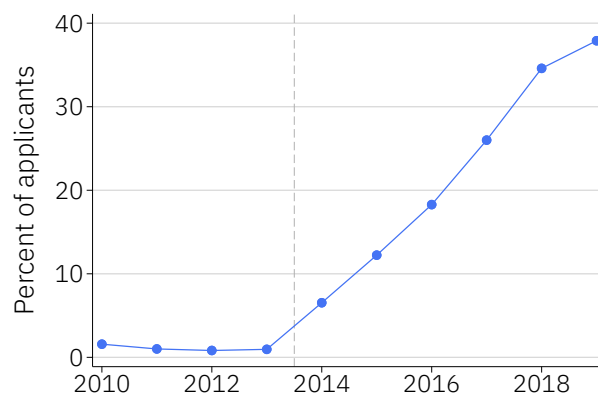
⁸One reason behind this change is the lifting of strict enrollment caps. In 2014, enrollment caps were lifted at all universities (except alternative providers: private “for-profit” entities or other independent institutions that receive no UK government funding; they enroll less than 2% of UK undergraduates and are not considered in our results). In 2015, enrollment caps were lifted at all universities including alternative providers.

⁹In the UK, these were referred to as *conditional unconditional offers*, as they were conditional on the student ranking the offer first; we use the term ‘binding unconditional’ for clarity.

¹⁰In the UK, ‘college’ refers to secondary education institutions such as sixth form colleges and vocational colleges, equivalent to grades 11 and 12 in the US; they do *not* refer to universities or other tertiary education institutions.

¹¹See, for example, coverage in *The Guardian* in 2018 and 2019.

Figure 1: Percent of applicants receiving an unconditional offer, 2010–2019.



quality options, effectively limiting their ability to benefit from strong age-18 performance. Universities defending the practice argued that unconditional offers helped them attract high-achieving students and that anxious students valued a guaranteed admission offer; a 2019 UCAS survey found that 70% of applicants supported unconditional offers, largely because of how they reduced stress.

The widespread usage of unconditional offers ended in 2021 amidst various bans. In 2020, amid concerns about pressure on students and disruption caused by the COVID-19 pandemic (which led to the cancellation of age-18 examinations that year), the UK government temporarily banned binding unconditional offers. Following this, in 2021, Universities UK, a group representing 142 universities, introduced a “Fair Admissions Code of Practice.” Under this code, universities could no longer make unconditional offers to students who had not yet completed their age-18 examinations, effectively ending the practice for mainstream applicants.¹² Universities UK cited several reasons for this self-imposed ban, most notably the “unfair pressure” created by binding unconditional offers and that unconditional offer takers were “less likely to report feeling motivated to perform well at school” (Universities UK, 2020).

2.3 Data sources

Our analysis draws on the Longitudinal Education Outcomes (LEO) database, produced by the UK Department for Education (DfE) and accessed via the Secure Research Service (SRS) of the Office for National Statistics, which allows us to link three administrative datasets from different data providers using anonymized individual identifiers (Office for National Statistics, 2025, 2026a,b).¹³

We observe the universe of university applications to UK universities between 2007 and 2021 in the Universities and Colleges Admissions Service (UCAS) dataset. The application data includes the program identifiers (including both university and major), whether each program made an offer and the type of offer (conditional or unconditional), the student’s ranking of their offers (first and second choices), and whether the student was ultimately admitted. The data also includes

¹²The [code of practice](#) also introduced requirements around transparency in admissions criteria and restrictions on the use of other incentive-based admissions practices.

¹³Due to the sensitive nature of this data, all analysis was conducted in person in the UK at the Institute for Fiscal Studies or at SRS-designated Saferooms or Safepods.

teacher-predicted and realized age-18 grades (A-levels, BTEC examinations), and the subjects students take in Years 12 and 13. Programs rely heavily on teacher-predicted age-18 grades to make initial offers, but recognize that these predictions are upward-biased and less predictive of university performance than realized scores (Murphy and Wyness, 2020). The National Pupil Database (NPD) provides student-level data on educational attainment for students in England completing secondary school between 2009 and 2021. The NPD data provides additional demographic characteristics, age-16 standardized test results (GCSEs), age-17 results (AS-levels), and realized age-18 grades (A-levels, BTEC examinations). Our research also uses data from the Higher Education Statistics Agency (HESA), which provides student-level data on university enrollment, degree program, degree type, program start and completion dates, and degree classification. The typical undergraduate degree lasts three years, after which students receive a degree classification — a coarse grade analogous to GPA, ranging from first-class honors through 2:1, 2:2, and third-class honors — which plays a significant role in graduate labor market outcomes (Britton et al., 2022). Our HESA sample spans academic years 2006–07 through 2020–21, and we observe outcomes for students matriculating from schools in England, Wales, Northern Ireland, and Scotland.

We restrict our main analysis sample to first-time applicants in the standard UCAS main scheme aged 18 as of August 31 of the application year. We also exclude students who apply only through alternative UCAS routes, or who cannot be linked to HESA data.¹⁴ We limit our analysis to students applying in the 2010–2019 application years. For the reduced-form analysis in Section 3, we further restrict our sample to students with application sets in which no constituent program had used unconditional offers in or before 2013 and all constituent programs each averaging at least 20 offers per year between 2010 and 2012. Additional details on the construction of variables are provided in Appendix B.

2.4 Targeting and take-up

The rise in unconditional offers is driven by smaller and academically less-selective programs. Table A.1 reports characteristics of programs where more than 5% of offers made were unconditional in some year between 2014 and 2019. Programs making unconditional offers were 39 percentage points *less likely* to be in the top tercile of university rankings, 11 percentage points less likely to be STEM programs, admitted students in 2010–2012 whose age-18 exams scored 0.5SD lower, and had lower historical rates of being ranked as a student's first choice.

What types of students did programs target with unconditional offers? Columns (1)–(3) of Table 2 show the mean age-16 and age-18 grades (converted to z-scores) of students who received unconditional offers, compared to those who did not. Students who received unconditional offers were stronger than non-recipients on all academic characteristics; the largest difference occurs in teacher-predicted age-18 scores, where recipients scored 0.40SD higher than non-recipients. However, the difference between takers and decliners among recipients indicates that on the basis of observables, weaker students self-selected into taking unconditional offers.¹⁵ This gap is

¹⁴UCAS programs that do not link to HESA data are generally specialist programs at arts academies or music conservatoires rather than traditional university programs.

¹⁵We define take-up as ranking a binding unconditional offer first, or ranking a standard unconditional offer first or second.

especially large in the realized age-18 exam scores. As reported in column (6), unconditional offer takers scored 0.28SD lower on their age-18 exams than recipients who declined them; we call this the taker–decliner achievement gap.¹⁶

Table 2: Mean standardized test scores of unconditional offer recipients and takers

	Receiving an unconditional offer			Take-up, among offered		
	(1) No offer	(2) Received offer	(3) Difference	(4) Decliner	(5) Taker	(6) Difference
Age-16 English (z)	−0.007 (0.003)	0.086 (0.006)	0.093*** (0.007)	0.129 (0.013)	0.076 (0.006)	−0.053*** (0.015)
Age-16 math (z)	0.022 (0.003)	0.062 (0.007)	0.040*** (0.008)	0.114 (0.015)	0.050 (0.007)	−0.064*** (0.017)
Teacher-predicted age-18 (z)	−0.299 (0.003)	0.101 (0.006)	0.400*** (0.007)	0.251 (0.014)	0.068 (0.007)	−0.183*** (0.015)
Achieved age-18 (z)	−0.282 (0.003)	−0.037 (0.007)	0.245*** (0.008)	0.191 (0.014)	−0.089 (0.007)	−0.280*** (0.016)

* $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$. Means, with standard errors in parentheses, among students in the first year in which any program in their application set makes unconditional offers. Columns (1)–(3) compare students who receive an unconditional offer with those who do not; columns (4)–(6) compare, among recipients, those who rank an unconditional offer against those who do not. A student ranks an unconditional offer if they rank an unconditional offer first or second, or a binding unconditional first. N is the full regression sample, covering all cohorts and exposure groups, and ranges from 1.33m to 1.63m across rows with the availability of each score.

2.5 Enrolling student composition

Programs targeted unconditional offers towards high-achieving students, but academically weaker students accepted them; we examine how these two forces affected programs’ actual enrolling class.

Our analysis focuses on how enrolling class composition changed between the first year of usage and the years before unconditional offer use. Define $Z_{st} \in \{0, 1\}$ as an indicator for whether program s gave more than 5% unconditional offers in year t , and let t_s^{first} be the first year in which $Z_{st} = 1$. Table 3 reports coefficients from estimating

$$Y_{st} = \beta_1 Z_{st}^{\text{first}} + \beta_2 Z_{st}^{\text{later}} + \alpha_s + \lambda_t + \varepsilon_{st}, \quad (1)$$

where Y_{st} is an outcome of program s in year t , and α_s and λ_t are program and year fixed effects. The exposure indicator is split in two: $Z_{st}^{\text{first}} = \mathbf{1}\{t = t_s^{\text{first}}\}$, the first year a program makes unconditional offers, and $Z_{st}^{\text{later}} = Z_{st} \mathbf{1}\{t > t_s^{\text{first}}\}$, every subsequent year in which it makes them. The sample is program-years between 2010 and 2019, excluding programs that made unconditional offers before 2014 or averaged fewer than 20 offers per year in 2010–2012. The coefficient β_1 effectively compares enrolling class composition in the first year of unconditional offer usage to the pre-unconditional-offer period.

Panel A of Table 3 suggests that programs started making unconditional offers in a “bad year” to offset fewer applications. In the year unconditional offers were first used, programs had 21.4 fewer

¹⁶ The same qualitative pattern holds, with a smaller magnitude, among students exposed in a later year; see Table A.2 in the appendix.

Table 3: Change in program characteristics after adopting unconditional offers

	(1) First year of unconditional offers	(2) Later years of unconditional offers
<i>Panel A. Applications, offers, and enrollment</i>		
Applications	-21.40*** (2.049)	-19.12*** (1.906)
Offers made	-19.87*** (1.736)	-15.94*** (1.614)
Enrollment	1.863*** (0.355)	3.856*** (0.330)
Rate ranked first	0.035*** (0.002)	0.039*** (0.002)
Offer rate	0.003 (0.002)	0.001 (0.002)
<i>Panel B. Test scores of the enrolling class (z)</i>		
Age-16 English	0.012** (0.005)	0.008 (0.005)
Age-16 math	0.012** (0.006)	0.010* (0.005)
Teacher-predicted age-18	0.070*** (0.006)	0.070*** (0.006)
Expected age-18	0.055*** (0.005)	0.056*** (0.004)
Achieved age-18	-0.024*** (0.006)	-0.042*** (0.006)

* $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$. Each row is a separate regression of the named program-year outcome on indicators for the first year and for later years in which the program makes unconditional offers, with program and year fixed effects; columns (1) and (2) report the two coefficients. Expected age-18 attainment is a forecast of the enrolling class's achieved score from the characteristics observed at the time of application, contrasting with actual age-18 attainment. N ranges from 45,183 to 46,826 across rows.

applications compared to years without unconditional offer usage. Despite receiving fewer applications, these programs saw a statistically significant *increase* in enrollment of 1.9 students while holding the aggregate offer rate (including both conditional and unconditional offers) constant.

Panel B shows that programs enrolled students with better academic attributes observable at the time of application. The first three rows of Panel B show that in their first year of making unconditional offers, programs enrolled students with 0.01SD higher age-16 English and math scores and 0.07SD higher teacher-predicted grades. Absent any student private information or moral hazard effects, projecting these observables onto realized age-18 test scores leads to expected age-18 scores that are 0.06SD higher, as shown in the fourth row of Panel B.¹⁷

However, the actual academic performance of these students did not match the expected improvement. The last row of Panel B shows that the realized age-18 test scores of enrolling students were in fact 0.02SD *lower* than in cohorts where programs did not use unconditional offers. This large discrepancy between *expected* age-18 grades and *realized* age-18 grades suggests that students either had substantial private information about how they will fare in the actual exams relative to their teacher-predicted grades and sorted into accepting the unconditional offers accordingly, exerted *less* effort preparing for the exams than they otherwise would have in the absence of an

¹⁷We estimate this by regressing actual age-18 grades on teacher-predicted age-18 grades, age-16 grades, and demographic variables using results from in-sample students between 2010 and 2013. The fitted values from this regression are used as the expected age-18 achievement.

unconditional offer, or a combination of both.

3 Causal Effects of Unconditional Offers

Table 3 poses a puzzle: in the year a program began making unconditional offers, its enrolling class was *stronger* on test scores observable at the time of offer-making than in other years, yet its realized age-18 scores were *lower*. The taker–decliner achievement gap in Table 2 shows this more directly: takers scored 0.28SD worse than decliners on their age-18 exams.

There are two reasons that the taker–decliner achievement gap alone is not enough to conclude that unconditional offers hurt students. First, this gap is descriptive, not causal, and may reflect selection. Second, it does not account for how unconditional offers change where students enroll. This section addresses both issues.

3.1 Effects on test scores

Three components make up the taker–decliner achievement gap: the causal change in achievement when a student takes an unconditional offer, holding observables and unobservables constant (moral hazard, equal to the average treatment effect on treated students); the difference in students’ unobservable characteristics (private information); and the difference in students’ observable characteristics (the composition effect). Formally, let T_i indicate take-up by student i , $Y_i(1)$ and $Y_i(0)$ be i ’s age-18 score with and without take-up, and X_i be the characteristics programs observe when making offers. The taker–decliner achievement gap δ decomposes as ¹⁸

$$\delta \equiv \mathbb{E}_Y[Y \mid T = 1] - \mathbb{E}_Y[Y \mid T = 0] \quad (2)$$

$$= \text{Moral Hazard} + \text{Private Information} + \text{Composition Effect}, \quad (3)$$

where

$$\text{Moral Hazard} \equiv \mathbb{E}_Y[Y(1) - Y(0) \mid T = 1],$$

$$\text{Private Information} \equiv \mathbb{E}_X[\mathbb{E}_Y[Y(0) \mid T = 1, X] - \mathbb{E}_Y[Y(0) \mid T = 0, X] \mid T = 1],$$

$$\text{Composition Effect} \equiv \mathbb{E}_X[\mathbb{E}_Y[Y(0) \mid T = 0, X] \mid T = 1] - \mathbb{E}_Y[Y(0) \mid T = 0].$$

Subscripts give the variable being averaged: $\mathbb{E}_Y$ averages age-18 scores and $\mathbb{E}_X$ averages characteristics, each among the students in its conditioning set, so the outer $\mathbb{E}_X[\cdot \mid T = 1]$ weights by takers’ characteristics.

Of key interest is the *conditional* taker–decliner achievement gap,

$$\delta_X \equiv \delta - \text{Composition Effect} = \text{Moral Hazard} + \text{Private Information}, \quad (4)$$

which represents the portion of δ that cannot be predicted solely on the basis of observable characteristics; in effect, δ_X is the taker–decliner achievement gap that came as a surprise to programs.

¹⁸ $\mathbb{E}_X[\mathbb{E}_Y[Y(0) \mid T = 1, X] \mid T = 1] = \mathbb{E}_Y[Y(0) \mid T = 1]$. The right-hand sum then simplifies to $\mathbb{E}_Y[Y(1) \mid T = 1] - \mathbb{E}_Y[Y(0) \mid T = 0]$, which equals δ since $Y = TY(1) + (1 - T)Y(0)$.

We estimate δ_x by estimating the gap in age-18 test scores between takers and decliners with increasingly strict controls for observable characteristics, and report results in Table 4. Conditioning on all characteristics observed at the time of application, the gap between takers and decliners shrinks to -0.139SD . If we “over-condition” by including controls for application sets (unobserved by programs), the gap shrinks to -0.127SD , indistinguishable from the -0.139SD gap with only observed characteristics. This correlation between take-up and outcomes conditional on observables indicates the presence of either private information or moral hazard; in Appendix C, we show that they cannot be disentangled without further information.¹⁹ As described in Section 2.2, programs and policymakers attributed the taker–decliner achievement gap to moral hazard.

Table 4: The taker–decliner achievement gap

	$\hat{\delta}$	$\hat{\delta}_x$		
	(1)	(2)	(3)	(4)
	No controls	+ Academic attainment	+ Demographics, school	+ Application-set FE
Uncond. take-up	-0.280^{***} (0.016)	-0.133^{***} (0.011)	-0.139^{***} (0.011)	-0.127^{***} (0.038)
Composition effect		-0.147	-0.141	-0.153
N	1,630,305	1,522,513	1,519,370	308,671

* $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$. Each column regresses the achieved age-18 score on an indicator for take-up — an unconditional offer ranked first or second, or a binding unconditional ranked first — adding, cumulatively, blocks of the characteristics a program observes when it makes the offer. Column (1) estimates the raw taker–decliner achievement gap $\hat{\delta}$; columns (2)–(4) estimate $\hat{\delta}_x$, the gap net of covariates. The composition effect row reports $\hat{\delta} - \hat{\delta}_x$: the part of the gap a program could have predicted at the point of offer. Column (4) additionally absorbs the application set, which programs do not observe. Standard errors are clustered at the application set.

To estimate the moral hazard effect of unconditional offers, we use quasi-random variation in *exposure* to unconditional offers resulting from the idiosyncratic timing of unconditional offer adoption at the program level. Quasi-random variation in unconditional offer exposure allows us to separate the treatment effect of unconditional offers on test scores, which corresponds to moral hazard, from selection into take-up of unconditional offers, which corresponds to private information.

With this identifying variation, we can recover the causal effect of exposure on an outcome using a two-way fixed effects specification. Let $a(i)$ and $t(i)$ denote the application set and year of student i . Each student appears in a single year, so both are attributes of i , and we write a and t for them below. Define $Z_{at} \in \{0, 1\}$ as an indicator for whether any program in application set a gave more than 5% unconditional offers in year t , and let t_a^{first} be the first year in which set a is exposed. We then estimate

$$Y_i = \alpha_{a,P_{at}} + \gamma_1 Z_{at}^{\text{first}} + \gamma_2 Z_{at}^{\text{later}} + \lambda_t + X_i' \beta + \varepsilon_i, \quad (5)$$

where Y_i is the age-18 test score of student i , X_i a vector of academic and demographic controls (age-16 grades, teacher-predicted age-18 grades, socioeconomic quintile, and gender), and λ_t year fixed effects. The term $\alpha_{a,P_{at}}$ contains application set $\times$ post-first-exposure fixed effects, where $P_{at} = \mathbf{1}\{t > t_a^{\text{first}}\}$ indicates years after a set’s first exposure. The exposure indicator is split in two: $Z_{at}^{\text{first}} = \mathbf{1}\{t = t_a^{\text{first}}\}$, the first year an application set is exposed to unconditional offers, and

¹⁹See also Chiappori and Salanié (2000), which makes this point in the context of insurance markets.

$Z_{at}^{\text{later}} = Z_{at} \mathbf{1}\{t > t_a^{\text{first}}\}$, every subsequent year in which the set is exposed. Standard errors are clustered at the application set level.²⁰

Our primary coefficient of interest is γ_1 , the causal effect of first-year exposure to unconditional offers in one's application set. The coefficient γ_1 is causal under the assumption of strict exogeneity of treatment for first-year exposure. The core intuition for why this is likely to hold in this setting is that the first year of exposure comes as a "surprise" to students. Programs did not announce that they were using unconditional offers in advance, so students had little ability to predict which programs would introduce unconditional offers in that year. This assumption is consistent with the balance tests in Table A.3; in the year that unconditional offers began, application sets had no changes in observable characteristics in five of the six tests. In contrast, if students preferentially applied to programs that had given unconditional offers in the past, then γ_2 would not capture only the treatment effect of exposure, but also changes in applicant composition.

Figure A.1 shows the growth of both first-year and later exposure. Exposure spreads quickly once it begins: by 2018, 66.8% of students are exposed to unconditional offers through a program in their application set. From 2016 onward, most of those students are in sets exposed in an earlier year rather than newly exposed, and first-year exposure is the weaker treatment: 17.1% of first-year-exposed students receive an unconditional offer, against 31.0% of those exposed in later years.

Estimates from Equation (5) are reported in Table 5, and find strong evidence against the moral hazard hypothesis. Exposure to unconditional offers has a precisely estimated null effect on students' age-18 test scores. The precision of our estimates allows us to rule out declines of more than 0.010SD in final exam scores.

Table 5: Effect of exposure to unconditional offers

	(1) Receives an offer	(2) Uncond. take-up	(3) Achieved age-18 (z)
First exposure	0.166*** (0.004)	0.143*** (0.004)	0.005 (0.007)
Constant	0.014*** (0.002)	0.013*** (0.002)	0.259*** (0.002)
<i>N</i>	368,874	366,288	366,288

* $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$. Each column is a separate regression on an indicator for the first year in which any program in the student's application set makes unconditional offers. Column (1) is *receipt*: whether the student is made an unconditional offer at all. Column (2) is *take-up*: an unconditional offer ranked first or second, or a binding unconditional ranked first — a choice the student makes over the offers they hold. All columns include academic and demographic controls, year fixed effects, and application set $\times$ post-first-exposure fixed effects. The sample is years 2010–2019, main scheme applicants, programs averaging at least 20 offers per year over 2010–2012, with no program in the application set using unconditional offers in 2013 or earlier. Standard errors clustered at the application set, of which there are 97,730. Columns (2) and (3) are $\hat{\pi}$ and $\hat{\rho}$ in Table 6.

Estimates in Table 5 represent an intent-to-treat effect of exposure on test scores; however, as described in Equation (2), moral hazard is the average treatment effect on the treated. We can de-

²⁰We define application sets as unique combinations of {major $\times$ university} pairs applied to, aggregating across different programs within the same major at the same institution.

rive this quantity by interpreting [Table 5](#) as components of an instrumental variable specification, instrumenting take-up with exposure. Doing so requires the standard instrumental variable hypotheses. Relevance holds from column (2) of [Table 5](#), with $F = 1,555$. The exclusion restriction requires that exposure move achieved scores only through take-up. This is plausible: decliners would have no reason to change their behavior in response to having had such an offer available given that they declined. Independence follows from the strict exogeneity argument described above. As noted by [Bloom \(1984\)](#), given that unexposed students have close to zero take-up,²¹ the IV estimate corresponds to the average treatment effect on the treated — exactly the moral hazard term of interest. Under these assumptions, we find that

$$\text{Moral Hazard} = \mathbb{E}_Y[Y(1) - Y(0) \mid T = 1] = \frac{\rho}{\pi} = 0.033\text{SD}. \quad (6)$$

We conclude that take-up of unconditional offers does not cause students to achieve lower grades.

We now test for the presence of private information in our setting. Starting from [Equation \(4\)](#), express private information as the wedge

$$\text{Private Information} = \delta_X - \frac{\rho}{\pi}, \quad w \equiv \rho - \pi\delta_X = -\pi \text{Private Information}. \quad (7)$$

Testing for private information is then equivalent to testing $w > 0$. We report the results of this exercise in [Table 6](#). The wedge is positive and precisely estimated, $\hat{w} = 0.024$ with a bootstrapped standard error of 0.007, rejecting $w \leq 0$: the causal effect of take-up — moral hazard — cannot explain why takers underperform decliners.

From these results, we can bound the share of the conditional gap which can possibly be explained by moral hazard. Let $\theta = (\rho/\pi)/\delta_X$ be the share of the conditional taker–decliner achievement gap composed of moral hazard.²² We can bound θ by finding the smallest θ_0 at which $w(\theta_0) = \rho - \theta_0\pi\delta_X$ is significantly positive using a one-sided t -test. We find that $\theta \leq 0.346$: at most a third of the difference between takers and decliners is due to moral hazard, and the remainder is due to students’ private information.

These results are robust to alternate sample definitions, alternate thresholds for what counts as using unconditional offers, dropping the post-first-exposure block, and clustering at the least-selective university in the application set rather than at the application set, as reported in [Table A.4](#). As shown in [Table A.6](#), the decomposition is also robust to relaxing the assumption of no always-takers. Additionally, we estimate treatment effects of exposure on age-18 test scores using a quantile regression, and report results in [Table A.5](#); because results are insignificant across the ability distribution, we conclude that the null effect in [Table 5](#) is not driven by an increase in age-18 scores for high-achievers and a decrease for low-achievers, or vice versa.

²¹In [Table A.6](#), we relax this assumption as a robustness check.

²²The estimate of δ_X differs slightly between [Table 4](#) and [Table 6](#); the former is estimated on all unconditional offer recipients, while the latter only during the year of first exposure.

Table 6: Decomposing the taker–decliner achievement gap

	Estimate	SE	95% CI
<i>Panel A. Estimated inputs</i>			
$\hat{\pi}$, effect of exposure on take-up	0.143	(0.003)	[0.138, 0.151]
$\hat{\rho}$, effect of exposure on the achieved score (z)	0.005	(0.007)	[−0.010, 0.019]
$\hat{\delta}_X$, taker–decliner achievement gap net of covariates (z)	−0.134	(0.011)	[−0.156, −0.114]
<i>Panel B. Test for private information</i>			
$\hat{w} = \hat{\rho} - \hat{\pi}\hat{\delta}_X$	0.024	(0.007)	[0.009, 0.039]
<i>Panel C. Implied split of the gap</i>			
Moral hazard (ATT) = $\hat{\rho}/\hat{\pi}$	0.033	(0.048)	[−0.068, 0.128]
Private information = $\hat{\delta}_X - \hat{\rho}/\hat{\pi}$	−0.167	(0.049)	[−0.260, −0.064]
Moral-hazard share $\theta = (\hat{\rho}/\hat{\pi})/\hat{\delta}_X$	−0.244	(0.365)	≤ 0.346

Table reports decomposition of the taker–decliner achievement gap using the test described in [Section 3.1](#), yielding point estimates and bounds on θ , the share of the gap composed of moral hazard. The gap is estimated on 1,700,457 students, while exposure effects are estimated on 366,288 students in non-singleton application sets. Standard errors are from a cluster bootstrap over application sets with 200 replications. The last column reports two-sided 95% percentile intervals, except in the θ row, where it reports the one-sided 95% upper bound obtained by inverting $w(\theta_0)$: the largest share not rejected by the data. Robustness checks reported in [Table A.4](#).

3.2 Effects on matriculation and degrees

Unconditional offers do not impact students’ academic achievement. Do they affect which programs students attend? Using [Equation \(5\)](#), we estimate the causal effect of unconditional offer exposure on students’ trajectories through higher education.

We first examine results from university enrollment and degree attainment. Results are shown in panel (a) of [Figure 2](#) and column (1) of [Table 7](#). On net, unconditional offer exposure leads to an increase in university matriculation by 1.1pp, with the increase concentrated at middle-tercile universities. Recall that the effect of exposure on unconditional offer receipt is 16.6pp; again treating these results as an IV, we infer that students receiving an unconditional offer are 6.4pp more likely to matriculate at university.

Exposure also increases the quality of the program students attend. We define a program’s average achievement as the average age-18 test scores of students in that program during 2010–2013, before unconditional offers began. We then estimate [Equation \(5\)](#) with the dependent variable of program quality a student enrolls at, restricted to students that enroll in university. Among university enrollees, exposure to unconditional offers leads students to matriculate at programs whose average achievement is 0.008SD higher.

Finally, we turn to the long-term impacts of unconditional offer exposure on students’ educational trajectories. To do so, we estimate [Equation \(5\)](#) on years of university attended and degree completion by age 24, restricted to students for whom we observe data at age 24. Results are reported in panel (b) of [Figure 2](#) and columns (3) and (4) of [Table 7](#). Students exposed to unconditional offers spend an additional 0.052 years enrolled in university; however, this does not translate to a significant increase in university degree completion. Panel (b) of [Figure 2](#) shows that there is a significant decrease of degree completion at bottom-tercile universities; there is suggestive evi-

dence that this is offset by an increase in degree completion at middle- and top-tercile universities, though estimates are not significant at the 5% level.

Figure 2: Effect of unconditional offer exposure on matriculation and degree attainment, by university tercile. (a) Matriculation at ages 18–21, and ever; (b) holding an undergraduate degree by ages 21–24.

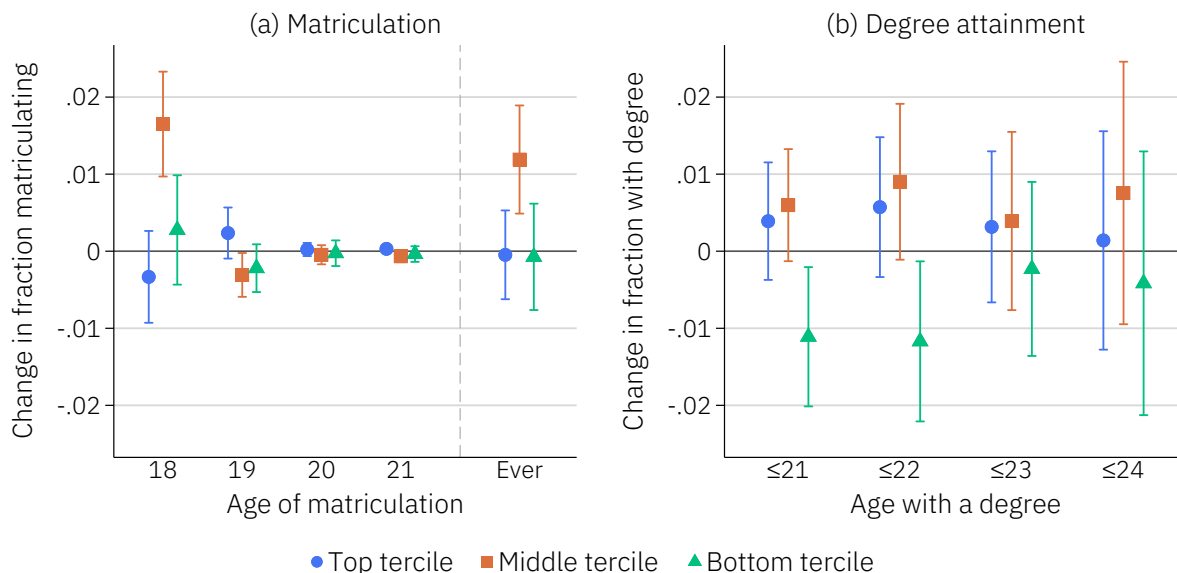


Table 7: Effect of unconditional offer exposure on matriculation, program quality, years enrolled, and degree attainment

	(1) Ever matriculates	(2) Program quality (z)	(3) Years enrolled by 24	(4) Degree by 24
First exposure	0.011*** (0.003)	0.008** (0.004)	0.052* (0.028)	0.005 (0.009)
Mean of dep. var.	0.926	0.301	3.044	0.798
<i>N</i>	368,870	296,990	126,810	126,810

* $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$. Each column is a separate estimation of Equation (5) on the outcome in its header, with the controls and fixed effects of Table 5, reporting the coefficient on first-year exposure; standard errors are clustered on the application set. Columns (1) and (4) are probabilities; column (2) is the average achieved z-score of the program a student matriculates at, as measured by average student achievement in 2010–2013, on a sample restricted to university matriculants; column (3) is years enrolled by age 24. *N* is rounded to the nearest 10.

4 A Structural Model of Application and Enrollment

Section 3 shows that unconditional offers have small positive effects on students' university matriculation and no effect on their academic achievement. Three questions remain that the reduced form cannot answer: how much students value an offer being unconditional (the unconditional-offer premium), how one student's unconditional offer affects other students, and how the match would change under other admissions systems. We answer them with a structural model of the UK university admissions market, a four-stage game that mirrors the institutional timeline in Table 1. We estimate students' preferences using techniques from Larroucau and Rios (2020) and

programs' offer and admission rules with probits, and we model students' private signals as in [Kapoor \(2025\)](#).

The reduced-form results shape the model. Because unconditional offers do not lower students' grades, the model omits moral hazard, and private information is identified from how students' rank-order choices covary with their realized grades. For example, if students who rank more-selective programs also outperform their expected age-18 grades, we infer that they possess private information about their academic ability.

4.1 Structural setup

The market has four stages. In Stage 1, each student applies to up to five programs, which form her portfolio A_i . In Stage 2, programs make conditional and unconditional offers, and student i holds the offer set $B_i \subseteq A_i$. In Stage 3, she observes a private signal of her age-18 exam performance and ranks up to two options, $C_i \subseteq B_i \cup \{s_0\}$, where s_0 is the outside option of not enrolling. In Stage 4, she takes her age-18 examinations and enrolls at her highest-ranked program that admits her, or takes the outside option, whose utility is normalized to zero.

We describe the stages in reverse order, because each choice depends on what the student expects later. Let $\mathcal{I}$ index students and $\mathcal{S}$ index programs (university $\times$ major).

4.1.1 Stage 4: admission and enrollment

In the final stage, students take their age-18 examinations, which reveal their performance $r_i \sim N(0, \sigma_r^2)$. A conditional offer that a student ranked converts to a place if and only if her augmented score clears the program's final bar,

$$z_i^{\text{offer}} \gamma_s^{\text{admit}} + \alpha_s r_i + \mu_i \geq \bar{\pi}_s^R, \quad (8)$$

where z_i^{offer} collects the characteristics programs see at application (age-16 grades, teacher-predicted age-18 grades, age-18 test composition, Low-SES, and indicators for missing data), γ_s^{admit} are program s 's final-stage weights on them, and $\bar{\pi}_s^R$ is its conversion cutoff. The term $\mu_i \sim N(0, 1)$ is the part of an application that every program sees and we do not, such as the teacher recommendation letter and the personal statement. An unconditional offer converts with certainty. A binding unconditional offer is unconditional only if ranked first; ranked second, it carries the program's usual condition.

4.1.2 Stage 3: ranking

In Stage 3, students observe a private signal of their age-18 exam performance and rank up to two options, (c_1, c_2) , from their offer set B_i and the outside option s_0 ; offers they do not rank are forfeited. Student i 's utility for enrolling at s is

$$U'_{is} = U_{is} + z'_i \lambda_i \cdot \mathbf{1}[s \text{ is an unconditional offer}] + \varepsilon_{is}, \quad (9)$$

where U_{is} is her Stage-1 utility ([Section 4.1.4](#)), $\varepsilon_{is} \sim N(0, \sigma_\varepsilon^2)$ is an i.i.d. shock, and $z'_i \lambda_i$ is the unconditional-offer premium. The vector z_i contains a constant, the student's standardized teacher-

predicted age-18 grades, and an indicator for being female. The shock ε_{is} is realized after the offer set is observed and captures what the student learns between applying and ranking, such as impressions from campus visits.

The signal is $\psi_i \sim N(0, \sigma_\psi^2)$, where $r_i = \psi_i + e_i$ and $e_i \sim N(0, \sigma_r^2 - \sigma_\psi^2)$ is independent of ψ_i . The student observes ψ_i ; programs do not. The share $\sigma_\psi^2 / \sigma_r^2$ measures how well a student can forecast her own age-18 exam performance.

Because she knows neither r_i nor μ_i , admission is uncertain. Students know the offer process, including every program's weights and cutoffs, and so hold true beliefs about their chances of admission.²³ A student also learns about μ_i from her offer set B_i . Write $a_{is} \equiv \Pr(\text{admit } s \mid \psi_i, B_i)$ for the probability that Equation (8) holds for program s , and $a_{i,c_2 \setminus c_1} \equiv \Pr(\text{admit } c_2, \text{ reject } c_1 \mid \psi_i, B_i)$. The student chooses (c_1, c_2) to maximize the expected value of eventual enrollment,

$$\max_{c_1, c_2 \in B_i \cup \{s_0\}} W_i(c_1, c_2), \quad W_i(c_1, c_2) = a_{ic_1} U'_{ic_1} + a_{i,c_2 \setminus c_1} U'_{ic_2}. \quad (10)$$

4.1.3 Stage 2: offers

In Stage 2, programs decide which applicants receive offers. Program s scores applicant i by

$$\pi_{is} = z_i^{\text{offer}} \gamma_s + \mu_i, \quad (11)$$

where z_i^{offer} and μ_i are as in Section 4.1.1, and the offer-stage weights γ_s can differ from γ_s^{admit} because r_i is not yet known. Program s makes an offer if and only if $\pi_{is} \geq \bar{\pi}_s$, a threshold set to fill expected capacity, and the offer is unconditional if and only if $\pi_{is} \geq \bar{\pi}_s^U$, where $\bar{\pi}_s^U > \bar{\pi}_s$, so the highest-scoring offer holders receive unconditional offers. Because μ_i enters every program's score, offers are nested: a student with an offer from a program that sets a high bar on μ_i also has offers from every program she applied to that sets a lower one.

4.1.4 Stage 1: initial applications, portfolio choice

In Stage 1, each student applies to up to five programs. Student i 's indirect utility from enrolling at program s is

$$U_{is} = \delta_s - 0.01 d_{is} + \beta_i^c \text{close}_{is} + \beta^m m_{is} + v_{is}, \quad v_{is} \sim N(0, \sigma_v^2), \quad (12)$$

with v_{is} independent across student-program pairs and $U_{i0} = 0$. The program fixed effects δ_s absorb everything about a program that all students value alike; Table 8 reports their mean as the constant. Distance d_{is} is measured in kilometers from the student's home postcode to the institution, and its coefficient is normalized to -0.01 , so a unit of utility is worth 100km of avoided travel at the margin. The indicator $\text{close}_{is} = \mathbb{1}[d_{is} < 40]$ marks programs the student can attend while living at home, and m_{is} marks programs in her home UK region.²⁴

²³As discussed in Agarwal and Somaini (2018), utility and beliefs about admission probability cannot be jointly estimated, so we pin the latter at its true value.

²⁴The 12 International Territorial Level 1 regions, as defined by the Office for National Statistics.

The value of living at home varies across students:

$$\beta_i^c = \bar{\beta}^c + \Gamma^c \text{Low-SES}_i + \eta_i^c, \quad \eta_i^c \sim N(0, \sigma_\eta^2), \quad (13)$$

where Low-SES_i marks students from the two most deprived socioeconomic quintiles, who are likely to value living at home more.

When applying, a student knows z_i^{offer} but not μ_i , so her chance of an offer from s is

$$p_{is} = \Pr[\text{offer from } s \mid z_i^{\text{offer}}] = \Phi(z_i^{\text{offer}} \gamma_s - \bar{\pi}_s). \quad (14)$$

She chooses the portfolio

$$A_i = \arg \max_{A \subseteq \mathcal{S}, |A| \leq 5} V(A), \quad V(A) = \mathbb{E}_B[W_i(C_i(B))], \quad (15)$$

where $C_i(B)$ is her optimal ranking given offer set B (Section 4.1.2) and the expectation is over offer sets.

4.2 Model assumptions

Solving Equation (15) exactly requires valuing millions of portfolios, each of which involves finding the optimal ranking for every possible offer set. Two assumptions make the portfolio value a simple function of admission probabilities and utilities.

Assumption 1 (Portfolio evaluation). When choosing a portfolio, students value it as though all held offers can be ranked. Further, they do not anticipate the arrival of their private signal ψ_i nor the Stage-3 idiosyncratic shock.

A portfolio is then worth the expected utility of the best program that admits the student, so she need not solve the ranking problem for every offer set, while she still faces the risk that a conditional offer does not convert. A fully rational student would instead reason through every possible offer set, grade outcome, and preference shock.

Assumption 2 (Nested admission). When choosing a portfolio, students treat admission events as nested across programs: a student admitted to a program is also admitted to every program with a higher probability of admitting her.

Offers are nested in the model, because the common holistic component μ_i enters every program's offer score (Equation (11)). Nesting is natural in the UK context, where admission depends heavily on academic grades; in the data, 92.5% of students who receive an offer from a given program receive offers from all less-selective programs they apply to.

Implication 1 (Portfolio valuation). Under Assumption 1 and Assumption 2, index the $K = |A_i| \leq 5$ programs in A_i so that $q_{is_{(1)}} \leq q_{is_{(2)}} \leq \dots \leq q_{is_{(K)}}$, where q_{is} is the probability that s makes student i an offer and admits her, and set $q_{is_{(0)}} \equiv 0$. Then

the expected utility of A_i is

$$V(A_i) = \sum_{B \subseteq A_i} P_i(B) \max_{b \in B} U_{ib} = \sum_{k=1}^K (q_{is_{(k)}} - q_{is_{(k-1)}}) \max_{j \geq k} U_{is_{(j)}},$$

where $P_i(B)$ is the probability that the programs admitting student i are exactly those in B .

The first equality follows from [Assumption 1](#). The second follows from [Assumption 2](#): a student admitted to $s_{(k)}$ is also admitted to every $s_{(j)}$ with $j > k$, so with probability $q_{is_{(k)}} - q_{is_{(k-1)}}$ her admitted set is $\{s_{(k)}, \dots, s_{(K)}\}$. Together they make $V(A_i)$ linear in admission probabilities, and under [Implication 2](#) below the inner maximum is $U_{is_{(k)}}$.

The relevant probability is the *total* chance of admission q_{is} , not the offer probability alone: a program that makes many offers but sets a demanding grade condition is not a safe choice. Because μ_i enters both [Equation \(11\)](#) and [Equation \(8\)](#), receiving an offer and meeting its condition are positively dependent, so students compute q_{is} as a joint probability rather than as the product of the two marginals:

$$q_{is} = p_{is} [g_{is} + (1 - g_{is}) \Pr(\text{admit } s \mid \text{offer from } s)],$$

where g_{is} is the probability that an offer from s is unconditional, and the conditional admission probability integrates over μ_i truncated at the offer threshold and over r_i given z_i^{offer} .

Implication 2 (No dominated programs). Under [Assumption 1](#) and [Assumption 2](#), let A_i be an optimal portfolio in which every program strictly raises $V(A_i)$, ordered as in [Implication 1](#) with distinct admission probabilities $q_{is_{(1)}} < \dots < q_{is_{(K)}}$. Then

$$U_{is_{(1)}} > U_{is_{(2)}} > \dots > U_{is_{(K)}}.$$

Moreover, if $q_{is} < q_{is'}$ and $U_{is} < U_{is'}$ for some program s' , then $s \notin A_i$.

This implication imposes a utility ordering on the programs a student applied to even though no ranking among them is observed. It follows from [Implication 1](#) and portfolio optimality. The condition that every program strictly raises $V(A_i)$ is needed: with no application cost, a program that adds nothing to $V(A_i)$ can still fill a spare slot.

Assumption 3 (No moral hazard). Realized performance r_i is independent of B_i , conditional on A_i and z_i^{offer} .

In line with our results from [Section 3](#), we assume that students do not change their effort in response to the offers they hold.²⁵

4.3 Estimation procedure

We estimate the model in three blocks, in order: the supply side in Stages 2 and 4 (offer and conversion probabilities), Stage 1 (student utilities, from applications), and Stage 3 (the coefficient λ_i on

²⁵Moral hazard, if present, would make a stronger case for post-qualification admissions in [Section 4.6](#).

the unconditional-offer premium and the signal variance σ_ψ^2 , from rankings). Each block conditions on the blocks before it; [Appendix F.6](#) discusses the validity of this sequential estimator, together with convergence diagnostics, and [Appendix F](#) gives further details.

4.3.1 Supply side

We estimate two probabilities separately: receiving an offer, which we observe for every applicant, and converting an offer into a place, which we observe only for students who rank the program, the “at-risk” sample. Both are Firth-penalized probits, which handle the perfect separation at the most and least selective programs. The offer probit is fit separately for each program and cohort; in cells where every applicant or no applicant received an offer, we set the offer probability to 0.999 or 0.001. Although μ_i is a common shock, each probit is correctly specified, because μ_i is integrated out of every marginal probability. Let these probabilities be $\hat{p}_{is}$ and $\hat{a}_{is}$, respectively:

$$\hat{a}_{is} = \Phi(z_i^{\text{offer}'} \hat{\gamma}_s^{\text{admit}} + \hat{\alpha}_s \hat{r}_i + \hat{c}_s^R), \quad \hat{p}_{is} = \Phi(z_i^{\text{offer}'} \hat{\gamma}_s + \hat{c}_s),$$

where the fitted intercepts $\hat{c}_s = -\hat{\pi}_s$ and $\hat{c}_s^R = -\hat{\pi}_s^R$ recover the offer and admission cutoffs of [Equation \(14\)](#) and [Equation \(8\)](#).

Students’ Stage-1 admission probability combines the two as in [Implication 1](#):

$$\hat{q}_{is} = \hat{p}_{is} [\hat{g}_{is} + (1 - \hat{g}_{is}) \hat{a}_{is}^o],$$

where $\hat{g}_{is}$ is the probability that an offer from s is unconditional, implied by s ’s unconditional-offer cutoff, and $\hat{a}_{is}^o$ is the conversion probability conditional on holding the offer. Because the offer and conversion indices share μ_i , we compute $\hat{a}_{is}^o$ from their bivariate normal distribution, with correlation $1/\sqrt{1 + \hat{\alpha}_s^2 \sigma_r^2}$, rather than setting it to $\hat{a}_{is}$.

4.3.2 Demand side, Stage 1

Following [Larroucau and Rios \(2020\)](#), an optimal portfolio survives every “one-shot swap” (OSS): replacing one chosen program with an unchosen one does not raise its value,

$$V(A_i; q_i, U_i) \geq V((A_i \setminus \{s\}) \cup \{\ell\}; q_i, U_i) \quad \text{for all } s \in A_i, \ell \notin A_i. \quad (16)$$

For a student who applies to fewer than five programs, adding a program must not raise the value of her portfolio either: $V(A_i; q_i, U_i) \geq V(A_i \cup \{\ell\}; q_i, U_i)$ for all $\ell \notin A_i$. With independent admission chances, these inequalities characterize the optimal portfolio.²⁶ With correlated admissions, they are necessary but not sufficient, because rejection by one program is informative about the others; the exact conditions are the dynamic-programming inequalities of [Ali and Shorrer \(2025\)](#). Written as linear inequalities in utilities, those conditions compare the observed portfolio with a number of alternative portfolios that grows combinatorially with the number of programs, which is too many to impose in our sampler, so we use the OSS inequalities instead; our counterfactual

²⁶[Larroucau and Rios \(2020\)](#) show that, with independent admission chances, a portfolio that no one-shot swap improves is optimal, so it matches the outcome of the Marginal Improvement Algorithm of [Chade and Smith \(2006\)](#).

simulations solve the exact portfolio problem. [Appendix E](#) characterizes the gap between the two sets of inequalities.²⁷

The observed portfolio has no closed-form choice probability, so we sample Stage-1 utilities for every student–program combination by Markov chain Monte Carlo. Student utilities must satisfy inequalities $\mathcal{L}_i U_i > 0$, where $\mathcal{L}_i$ consists of three blocks of inequalities: a non-negativity constraint, $U_{is} > 0$ for all $s \in A_i$; the ranking constraint implied by [Implication 2](#); and the one-shot swap inequalities described in [Equation \(16\)](#) (addition inequalities for students who apply to fewer than five programs). We draw from the joint posterior of the Stage-1 parameters and the utilities $\{U_i\}$ by Gibbs sampling; [Appendix F.2](#) gives the blocking, and [Appendix F.7](#) gives the priors and run lengths.

We treat q_{is} as latent and resample it with the utilities, proposing the offer-probit coefficient vector from its asymptotic distribution and accepting the proposal if the observed portfolio still satisfies the OSS (or addition) inequalities. The conversion-probit component of q_{is} is held fixed at its point estimate. To account for possible mis-specification of believed admissions odds, we use only the OSS (or addition) constraints and non-negativity constraints to define $\mathcal{L}_i$ when redrawing q .²⁸

4.3.3 Demand side, Stage 3

The ranking stage conditions on the Stage-1 estimates of [Section 4.3.2](#), drawing the live-at-home coefficients β_i^c and the program fixed effects δ from their Stage-1 posterior so that Stage-1 uncertainty carries into the Stage-3 estimates. We estimate the Stage-3 parameters $(\sigma_\varepsilon^2, \lambda_i, \sigma_\psi^2)$ by Gibbs sampling subject to the student’s revealed ranking: that student i ranked (c_1, c_2) rather than any other pair from B_i implies

$$W_i(c_1, c_2) \geq W_i(c'_1, c'_2) \quad \text{for every other pair } (c'_1, c'_2), \quad (17)$$

with U'_{is} defined as in [Equation \(9\)](#). There are at most 26 such inequalities. As in Stage 1, we draw U'_{is} for every student–program combination subject to these constraints.

The signal ψ_i is latent. Each sweep draws ψ_i by a Metropolis step given the observed first and second choices and the take-up decision, recomputing the conversion probabilities in [Equation \(17\)](#) under the student’s posterior, and then draws σ_ψ by slice sampling under a half- t prior, given $\{\psi_i\}$ and the realized $\{r_i\}$, keeping every observed ranking feasible. The remaining parameter draws are conjugate.

4.3.4 Sample

We use the same sample restrictions as in [Section 3](#), with the additional restriction that students must have applied only to mathematics programs. The supply-side probits are fit on the full appli-

²⁷A separate interpretation of our methodology is that by using the OSS inequalities, rather than the full dynamic-programming inequalities, our parameters are set-identified, rather than point-identified. In practice, given the standard errors of our estimates, the identified set is a good approximation for point-identification.

²⁸If the test for accepting a redrawn q also required the ranking constraints, every accepted draw would preserve the estimated order of programs by selectivity, and by [Implication 2](#) each student’s utilities over her chosen programs would have to follow that order. We therefore leave the ranking constraints out of the test, so a student’s believed order of programs can differ from the estimated one; for accepted draws, all constraints, including the ranking constraints, are rebuilt under the new q .

cant population, for additional precision. The Stage-1 sampler is run for 50,000 iterations (5,000 discarded as burn-in) and the Stage-3 sampler for 250,000 iterations (25,000 discarded); [Appendix F.7](#) gives priors and further implementation detail.

4.4 Identification

Supply-side identification is standard: each per-program probit is identified by the excluded instrument of distance, following [Kapor \(2025\)](#). We focus on the demand side, where the argument follows [Larroucau and Rios \(2020\)](#), [Agarwal and Somaini \(2020\)](#), and [Kapor \(2025\)](#).

We pin the scale of students' preferences by fixing the distance coefficient to -0.01 , so that every parameter is measured relative to 100km of avoided travel. Utility's location is pinned by the outside option, normalized to zero.

Two features identify student preferences. First, distance is excluded from programs' utility over students, so it acts as a special regressor ([Agarwal and Somaini, 2020](#)), anchoring its units free of supply-side contamination. Second, the remaining utility parameters are identified from the OSS inequalities ([Larroucau and Rios, 2020](#)); here, identification comes from year-to-year variation in cutoffs and grade requirements, which "trace out" regions of utility.

Private information is identified from differences in Stage-3 behavior among students holding the same menu of offers with the same observable characteristics. Three margins in the data identify σ_ψ^2 : the take-up rate of unconditional offers, the taker–decliner achievement gap, and the gap in selectivity between students' first and second choices.

The unconditional-offer premium is identified by variation in the availability of unconditional offers that is unrelated to student preferences. As in [Section 3](#), the supply-side rollout of unconditional offers is plausibly exogenous to student characteristics, so comparing the rankings of exposed and unexposed cohorts identifies the coefficient λ_i .

4.5 Results

[Table 8](#) reports the estimates. Because the distance coefficient is normalized to -0.01 , every parameter is a willingness to travel, which we report in kilometers.

High-SES students prefer to live away from home: their mean value of living at home is -41km , precisely estimated. This mean masks heterogeneity, both idiosyncratic and related to SES: the standard deviation of η_i^c is 259km , and Low-SES students value living at home at 132km on average. Students also prefer a university in their home region of the UK, valuing region match at 71km .

The unconditional-offer premium falls steeply with predicted ability and is larger for women. At mean predicted ability, a male student values an unconditional offer at 12km (posterior SD 20.4km), and a female student values it at 118km . The premium falls by 118km per standard deviation of predicted ability, so it is negative for men more than 0.10SD above mean predicted ability and for women more than 1.0SD above it.

Although the unconditional-offer premium is small for the average student, private information still shapes who takes one: students with low signals preferentially take unconditional offers. We estimate that age-18 achievement has a variance of $\sigma_r^2 = 0.331$ and that students privately observe

Table 8: Structural parameter estimates

<i>Preferences over programs</i>			<i>Unconditional-offer premium coefficients λ_i</i>		
	Estimate	SD		Estimate	SD
Distance (per km)	−0.01	(anchor)	At mean predicted ability	0.119	0.204
Can live at home	−0.409	0.030	Gradient, per SD of score	−1.181	0.180
× Low-SES Γ^c	1.732	0.054	Female	1.062	0.309
SD across students σ_η	2.591	0.023	Ranking shock σ_ε	4.820	0.041
Same region	0.712	0.019			
Constant	−1.303	0.022			
Idiosyncratic shock σ_v	2.917	0.011			
			<i>Information</i>		
			Private-signal variance σ_ψ^2	0.221	0.006
			Variance of age-18 scores σ_r^2	0.331	—
			Share privately observed	0.67	—

Posterior means and posterior standard deviations from the Stage-1 and Stage-3 samplers of [Section 4.3.2](#) and [Section 4.3.3](#), for students applying only to mathematics programs. Distance is normalized to −0.01, so utility is in hundreds of kilometers. Only live-at-home carries a random coefficient and a Low-SES interaction. σ_η , σ_v , σ_ε are square roots of sampled variances, with delta-method SDs; the posterior SD is recovered as $\text{MCSE} \times \sqrt{\text{ESS}}$. Rows without an SD are fixed or derived quantities for which the sampler reports no posterior standard deviation. Convergence statistics are reported in [Table A.7](#); notably, the convergence diagnostics for σ_ψ^2 indicate that its SD is likely understated.

two-thirds of it ($\sigma_\psi^2/\sigma_r^2 = 0.67$). This aligns with [Section 3](#): because students hold much private information about their own ability, they can self-select into unconditional offers on the basis of their private signals.

4.6 Counterfactuals

We re-solve the market under three designs: a full ban on unconditional offers, as in 2010–2013; unconditional-offer shares held at each program’s 2018 level, the year with the most unconditional offers in this sample; and post-qualification admissions, in which students apply after age-18 results are released, so r_i is common knowledge. Post-qualification admissions are of particular interest because they are the only design in which students’ private signals play no role in admissions. Students receive the unconditional-offer premium only under 2018 shares.

We report welfare relative to the ban, in kilometer-equivalents, counting unplaced students at the outside option’s zero utility so that designs placing different numbers of students remain comparable. [Appendix F.8](#) details each design.

[Table 9](#) reports each design’s welfare change relative to the ban: for students who do and do not receive an unconditional offer under 2018 shares, by quartile of achieved attainment, and in aggregate. Panel A reports mean student utility in hundreds of kilometers, Panel B the part of that utility that comes from the unconditional-offer premium, and Panel C the change in the number of students placed.

Unconditional offers benefit recipients but, on average, hurt everyone else, shifting utility and placements from high-achieving to low-achieving students. Average utility falls by 9.3km (−0.7%): recipients gain 34.5km and non-recipients lose 21.7km. By achieved attainment, only the lowest quartile gains, and the second quartile loses most. Excluding the premium (Panel A minus Panel B), every quartile loses utility from its match, and the three upper quartiles also lose places (Panel C); the lowest quartile gains only through the unconditional-offer premium (Panel B). We read this as selection: low-achieving students, expecting to do poorly, take unconditional offers, while high-

Table 9: Welfare under counterfactual application system designs

	Uncond. offer in 2018		By achieved attainment				
Design	Yes	No	Q1	Q2	Q3	Q4	All students
<i>Panel A. Total average student utility</i>							
Unconditional shares of 2018	+0.345	−0.217	+0.086	−0.220	−0.190	−0.047	−0.093
Post-qualification admissions	+1.791	+1.629	−0.050	+1.817	+2.359	+2.535	+1.665
<i>Panel B. Utility from the unconditional-offer premium</i>							
Unconditional shares of 2018	+0.277	—	+0.235	+0.028	−0.003	−0.014	+0.061
<i>Panel C. Students placed</i>							
Unconditional shares of 2018	+278	−388	+39	−66	−63	−20	−110
Post-qualification admissions	+113	+401	−197	+357	+292	+63	+514
Students	6,926	24,480	7,852	7,851	7,851	7,852	31,406

Each cell is the mean difference from the ban on unconditional offers, per simulated market: in student utility, measured in hundreds of kilometers (Panels A and B), or in the number of students placed (Panel C). Unplaced students have utility 0. Panel A includes the utility from the match and the unconditional-offer premium $z'_i \lambda_i$ for students placed on an unconditional offer; Panel B reports the premium alone. Q1–Q4 are quartiles of achieved age-18 attainment. The first two columns split students by whether they receive an unconditional offer under 2018 shares; each student keeps that group in every design. Under the ban, mean utility is 13.34 and 25,202 students are placed. Differences are averaged over 30 simulations, paired within simulation. Details on counterfactuals are discussed in [Appendix F.8](#).

achieving students forgo them to compete on their exam results; after the exams, the low achievers already hold places, so the high achievers find fewer places left.

Post-qualification admissions, in contrast, raise welfare overall, mostly by improving the match: students gain 166.5km in utility (12.5%). The top quartile's placement rate barely changes, but its students attend programs they prefer.²⁹ The second quartile's placement rate rises by 4.5pp. One explanation is that, at application, these students look like the lowest quartile, and neither receives offers; when admission waits for exam results, their performance earns them places. The lowest quartile loses: 197 fewer of its students are placed, and its average utility falls by 5.0km.

Even if post-qualification admissions improve the match for students, programs would have to agree to adopt them. [Table 10](#) reports how each design changes program enrollment and the mean age-18 score of entering students, by adoption status in 2018, by quartile of programs' average achievement in 2010–2013, and in aggregate. Under 2018 shares, programs that use unconditional offers enroll 145 more students and programs that do not enroll 255 fewer; enrollment declines in every selectivity quartile, with minor changes in achievement except at adopters, whose entering classes score 0.068SD lower. This evidence is suggestive of why universities may have collectively banned unconditional offers in 2021: adopters gained students at non-adopters' expense, enrollment decreased overall, and lower-achieving students were more likely to attend university.

Post-qualification admissions, in contrast, raise enrollment at most programs but not in the most selective quartile. After their Stage-3 shocks, some high-achieving students re-optimize toward programs outside the top quartile. The top quartile's enrollment falls by only 0.3%, but the

²⁹Students may not have even applied to these programs at baseline, as they have not received their Stage-3 idiosyncratic shock. Under post-qualification admissions, students' behavior can reflect these preferences.

Table 10: Where students enroll, by program

	Uncond. offers in 2018		By program selectivity				
Design	Adopters	Non-adopters	Q1	Q2	Q3	Q4	All programs
<i>Panel A. Students enrolled</i>							
Unconditional shares of 2018	+145	−255	−11	−17	−8	−74	−110
Post-qualification admissions	+74	+440	+122	+268	+157	−33	+514
<i>Panel B. Mean age-18 scores of incoming students (SD)</i>							
Unconditional shares of 2018	−0.068	+0.010	+0.002	−0.028	−0.013	+0.008	−0.004
Post-qualification admissions	+0.084	+0.002	+0.108	+0.196	+0.060	−0.073	+0.012
Programs	16	44	15	15	15	15	60
Mean enrolled under the ban	3,127	22,075	2,812	3,387	7,099	11,904	25,202
Mean attainment under the ban (SD)	0.577	1.005	−0.068	0.393	0.984	1.332	0.952

Each cell is the mean difference from the ban on unconditional offers, per simulated market, in the number of students a program enrolls (Panel A) and in the average achieved age-18 scores of its incoming class (Panel B). Adopters are programs with a positive unconditional-offer share in 2018. Q1–Q4 are quartiles of programs' average incoming scores in 2010–2013 (Q4 is the most selective). Differences are averaged over 30 simulations, paired within simulation. Details on counterfactuals are discussed in [Appendix F.8](#).

achievement of its entering students falls by 0.073SD. Less selective programs gain both enrollment and student quality, particularly the second-lowest quartile. Post-qualification admissions, while largely beneficial for students and programs, hurt the most selective programs.

5 Conclusion

The design of admissions systems matters. In the context of UK university admissions, unconditional offers increase students' matriculation rates with no negative impact on their achievement, a finding that goes against the prevailing narratives. Using this finding, we develop a model to evaluate the total welfare impacts of such offers, accounting for student heterogeneity, private information, and a preference for certainty in admissions. In aggregate, unconditional offers lead to a net loss in student welfare and transfer utility from high-achieving students to low-achieving students. We then show that post-qualification admissions can improve student welfare by 12.5%, with benefits accruing to higher-achieving students.

The UK Department for Education considered moving to post-qualification admissions in 2022, but ultimately decided against it over many concerns, particularly the compressed timeline that universities would need to admit students. Our findings suggest that this debate should be reopened given the large benefits to students; other recent media has also suggested this given the rise of artificial intelligence ([Harvey-Brewin, 2026](#)).

Our results suggest several avenues for future research. How much do our results extrapolate to other non-education settings, such as labor market unraveling?³⁰ How strong is private information in these settings? Does private information act as a force against the unraveling of these markets? Development and estimation of dynamic best-response models incorporating private information

³⁰Empirical work has documented unraveling in several different markets: [Niederle and Roth \(2003\)](#) and [Herrmann et al. \(2022\)](#) in medical fellowship hiring, [Fréchette, Roth and Ünver \(2007\)](#) in college football scheduling, and [Avery et al. \(2007\)](#) in law clerk hiring.

could shed light on these issues.

On a separate note, private information about ability is necessary for the strong selection into take-up that is present in our setting. A recent literature suggests that parents often have inaccurate beliefs about children's ability ([Dizon-Ross, 2019](#); [Wang, Cheng and Smyth, 2022](#)) — so where does students' private information come from? Perhaps there is a wedge between parent and child perceptions of ability, or perhaps the structure of secondary education in the UK has allowed students to gain stronger signals of their own performance. Identifying reasons behind and ways to close this gap could help improve the efficiency of familial education investments.

References

- Adams, William, Liran Einav, and Jonathan Levin. 2009. "Liquidity Constraints and Imperfect Information in Subprime Lending." *American Economic Review*, 99(1): 49–84.
- Agarwal, Nikhil, and Paulo Somaini. 2018. "Demand Analysis Using Strategic Reports: An Application to a School Choice Mechanism." *Econometrica*, 86(2): 391–444.
- Agarwal, Nikhil, and Paulo Somaini. 2020. "Revealed Preference Analysis of School Choice Models." *Annual Review of Economics*, 12(1): 471–501.
- Akerlof, George A. 1970. "The Market for 'Lemons': Quality Uncertainty and the Market Mechanism." *The Quarterly Journal of Economics*, 84(3): 488–500.
- Akhtari, Mitra, Natalie Bau, and Jean-William Laliberté. 2024. "Affirmative Action and Precollege Human Capital." *American Economic Journal: Applied Economics*, 16(1): 1–32.
- Ali, S. Nageeb, and Ran I. Shorrer. 2025. "Hedging When Applying: Simultaneous Search with Correlation." *American Economic Review*, 115(2): 571–598.
- Avery, Christopher, Christine Jolls, Richard Posner, and Alvin Roth. 2007. "The New Market for Federal Judicial Law Clerks." *NBER Working Papers*, w13213.
- Avery, Christopher, Lena Shi, and Preston Magouirk. 2025. "Test-Optional College Admissions: ACT and SAT Scores, Applications, and Enrollment Changes." National Bureau of Economic Research w34260, Cambridge, MA.
- Bleemer, Zachary. 2024. "Top Percent Policies and the Return to Postsecondary Selectivity." *Unpublished*.
- Bloom, Howard S. 1984. "Accounting for No-Shows in Experimental Evaluation Designs." *Evaluation Review*, 8(2): 225–246.
- Borghesan, Emilio. 2025. "The Heterogeneous Effects of Changing SAT Requirements in Admissions: An Equilibrium Evaluation." *Unpublished*.
- Britton, Jack, Ian Walker, Ben Waltmann, and Yu Zhu. 2022. "How Much Does It Pay to Get Good Grades at University?" Institute for Fiscal Studies.
- Callaway, Brantly, and Pedro H. C. Sant'Anna. 2021. "Difference-in-Differences with Multiple Time Periods." *Journal of Econometrics*, 225(2): 200–230.
- Chade, Hector. 2006. "Matching with Noise and the Acceptance Curse." *Journal of Economic Theory*, 129(1): 81–113.
- Chade, Hector, and Lones Smith. 2006. "Simultaneous Search." *Econometrica*, 74(5): 1293–1307.
- Chade, Hector, Gregory Lewis, and Lones Smith. 2014. "Student Portfolios and the College Admissions Problem." *The Review of Economic Studies*, 81(3): 971–1002.
- Chiappori, Pierre-André, and Bernard Salanié. 2000. "Testing for Asymmetric Information in Insurance Markets." *Journal of Political Economy*, 108(1): 56–78.
- Chiappori, Pierre-André, Bruno Jullien, Bernard Salanié, and François Salanié. 2006. "Asymmetric Information in Insurance: General Testable Implications." *The RAND Journal of Economics*, 37(4): 783–798.
- Cotton, Christopher S., Brent R. Hickman, and Joseph P. Price. 2022. "Affirmative Action and Human Capital Investment: Evidence from a Randomized Field Experiment." *Journal of Labor Economics*, 40(1): 157–185.
- Dessein, Wouter, Alex Frankel, and Navin Kartik. 2025. "Test-Optional Admissions." *American Economic Review*, 115(9): 3130–3170.
- Dizon-Ross, Rebecca. 2019. "Parents' Beliefs about Their Children's Academic Ability: Implications for Educational Investments." *American Economic Review*, 109(8): 2728–2765.
- Doğan, Battal, and Aytek Erdil. 2025. "Widening Access in University Admissions." *Journal of Economic Theory*, 224: 105973.
- Einav, Liran, Amy Finkelstein, and Mark R. Cullen. 2010. "Estimating Welfare in Insurance Markets Using Variation in Prices." *Quarterly Journal of Economics*, 125(3): 877–921.

- Einav, Liran, and Amy Finkelstein.** 2011. "Selection in Insurance Markets: Theory and Empirics in Pictures." *Journal of Economic Perspectives*, 25(1): 115–138.
- Fréchette, Guillaume R., Alvin E. Roth, and M. Utku Ünver.** 2007. "Unraveling Yields Inefficient Matchings: Evidence from Post-Season College Football Bowls." *The RAND Journal of Economics*, 38(4): 967–982.
- Grau, Nicolás.** 2018. "The Impact of College Admissions Policies on the Academic Effort of High School Students." *Economics of Education Review*, 65(1): 58–92.
- Harvey-Brewin, Gianina.** 2026. "Admissions on Predicted A-Level Grades Look Foolish in AI Era." *Times Higher Education*.
- Herrmann, Amanda C., Cheryl Hanau, Donald Karcher, Douglas C. Miller, Alexandra Murtha, Ashley E. Sanders, Charles Timmons, and Karen L. Kaul.** 2022. "The Pathology Fellowship Application Crisis: The Current State and Suggestions for Remediation." *Academic Pathology*, 9(1): 100029.
- Kapor, Adam.** 2025. "Transparency and Percent Plans." *Econometrica*, 93(6): 2123–2157.
- Karlan, Dean, and Jonathan Zinman.** 2009. "Observing Unobservables: Identifying Information Asymmetries With a Consumer Credit Field Experiment." *Econometrica*, 77(6): 1993–2008.
- Kloosterman, Andrew, and Peter Troyan.** 2020. "School Choice with Asymmetric Information: Priority Design and the Curse of Acceptance." *Theoretical Economics*, 15(3): 1095–1133.
- Krishna, Kala, Sergey Lychagin, Wojciech Olszewski, Ron Siegel, and Chloe Tergiman.** 2026. "Pareto Improvements in the Contest for College Admissions." *The Review of Economic Studies*, 93(1): 629–663.
- Kulpa, Wladyslaw.** 1997. "The Poincaré-Miranda Theorem." *The American Mathematical Monthly*, 104(6): 545–550.
- Larroucau, Tomas, and Ignacio Rios.** 2020. "Do "Short-List" Students Report Truthfully? Strategic Behavior in the Chilean College Admissions Problem." *Unpublished*.
- Lee, Sam-Ho.** 2009. "Jumping the Curse: Early Contracting with Private Information in University Admissions." *International Economic Review*, 50(1): 1–38.
- McCulloch, Robert, and Peter E Rossi.** 1994. "An Exact Likelihood Analysis of the Multinomial Probit Model." *Journal of Econometrics*, 64(1-2): 207–240.
- Murphy, Richard, and Gill Wyness.** 2020. "Minority Report: The Impact of Predicted Grades on University Admissions of Disadvantaged Groups." *Education Economics*, 28(4): 333–350.
- Niederle, Muriel, and Alvin E. Roth.** 2003. "Unraveling Reduces Mobility in a Labor Market: Gastroenterology with and without a Centralized Match." *Journal of Political Economy*, 111(6): 1342–1352.
- Office for National Statistics.** 2025. "GRading and Admissions Data England-Ofqual-DfE-UCAS."
- Office for National Statistics.** 2026a. "Longitudinal Education Outcomes SRS Iteration 2.1 Standard Extract - England."
- Office for National Statistics.** 2026b. "Universities and Colleges Admissions Service - UK."
- Office for Students.** 2019. "Data Analysis of Unconditional Offer-Making." Office for Students.
- Rothschild, Michael, and Joseph Stiglitz.** 1976. "Equilibrium in Competitive Insurance Markets: An Essay on the Economics of Imperfect Information." *The Quarterly Journal of Economics*, 90(4): 629.
- Sacerdote, Bruce, Douglas Staiger, and Michele Tine.** 2025. "How Test Optional Policies in College Admissions Disproportionately Harm High Achieving Applicants from Disadvantaged Backgrounds." National Bureau of Economic Research w33389, Cambridge, MA.
- Tadjfar, Nagisa, and Kartik Vira.** 2025. "Prediction or Prejudice? Standardized Testing and University Access." *Unpublished*.
- Tierney, Luke.** 1994. "Markov Chains for Exploring Posterior Distributions." *The Annals of Statistics*, 22(4).
- Universities UK.** 2020. "Fair Admissions Review." Universities UK.
- Wang, Haining, Zhiming Cheng, and Russell Smyth.** 2022. "Parental Misbeliefs and Household Investment in Children's Education." *Economics of Education Review*, 89: 102284.

A Supplementary Information

Table A.1: Characteristics of programs that adopt unconditional offers

Panel A. Program types

	(1) Top tercile	(2) Middle tercile	(3) Bottom tercile	(4) STEM	(5) Arts	(6) Enrollment
Uses unconditional offers	−0.389*** (0.012)	0.082*** (0.013)	0.307*** (0.013)	−0.112*** (0.013)	0.035*** (0.011)	−1.927** (0.975)
Constant	0.477*** (0.010)	0.271*** (0.009)	0.252*** (0.009)	0.398*** (0.010)	0.174*** (0.008)	32.19*** (0.745)

Panel B. Baseline program selectivity, enrolling class characteristics

	(7) Offer rate	(8) Rate ranked first	(9) Age-16 English (z)	(10) Age-16 math (z)	(11) Teacher- predicted age-18 (z)	(12) Achieved age-18 (z)
Uses unconditional offers	0.092*** (0.006)	−0.065*** (0.004)	−0.355*** (0.013)	−0.457*** (0.015)	−0.538*** (0.016)	−0.530*** (0.016)
Constant	0.758*** (0.005)	0.297*** (0.003)	0.298*** (0.011)	0.413*** (0.012)	0.038*** (0.014)	0.173*** (0.013)

* $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$. Each column is a separate regression of the named program characteristic, measured in the baseline year, on an indicator for whether more than 5% of a program's offers are unconditional. Observations are programs; the sample is programs averaging at least 20 offers per year over 2010–2012 that did not use unconditional offers in 2013 or earlier. $N = 4,990$, except in columns (9)–(12), where it is 4,976. Standard errors in parentheses.

Table A.2: Characteristics of unconditional offer recipients and takers, by timing of exposure

	Receiving an unconditional offer			Take-up, among offered		
	(1) No offer	(2) Received offer	(3) Difference	(4) Decliner	(5) Taker	(6) Difference
<i>Panel A. First-year exposure</i>						
Age-16 English (z)	−0.007 (0.003)	0.086 (0.006)	0.093*** (0.007)	0.129 (0.013)	0.076 (0.006)	−0.053*** (0.015)
Age-16 math (z)	0.022 (0.003)	0.062 (0.007)	0.040*** (0.008)	0.114 (0.015)	0.050 (0.007)	−0.064*** (0.017)
Teacher-predicted age-18 (z)	−0.299 (0.003)	0.101 (0.006)	0.400*** (0.007)	0.251 (0.014)	0.068 (0.007)	−0.183*** (0.015)
Achieved age-18 (z)	−0.282 (0.003)	−0.037 (0.007)	0.245*** (0.008)	0.191 (0.014)	−0.089 (0.007)	−0.280*** (0.016)
<i>Panel B. Later-year exposure</i>						
Age-16 English (z)	−0.083 (0.002)	−0.113 (0.003)	−0.030*** (0.004)	−0.091 (0.007)	−0.119 (0.004)	−0.028*** (0.006)
Age-16 math (z)	−0.106 (0.003)	−0.206 (0.003)	−0.100*** (0.005)	−0.175 (0.008)	−0.213 (0.004)	−0.039*** (0.007)
Teacher-predicted age-18 (z)	−0.442 (0.003)	−0.227 (0.003)	0.215*** (0.005)	−0.151 (0.008)	−0.243 (0.004)	−0.092*** (0.007)
Achieved age-18 (z)	−0.395 (0.003)	−0.331 (0.003)	0.064*** (0.005)	−0.154 (0.008)	−0.368 (0.005)	−0.214*** (0.007)

* $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$. Columns are defined as in Table 2. Panel A repeats the first-exposure figures; Panel B reports the same quantities for students in a year after the first in which any program in their application set makes unconditional offers.

Table A.3: Balance of student characteristics at first exposure to unconditional offers

	(1) Age-16 English (z)	(2) Age-16 math (z)	(3) Independent school	(4) Female	(5) Low-SES	(6) Predicted age-18 score (z)
First year exposed	0.001 (0.007)	0.007 (0.007)	0.041 (0.026)	−0.001 (0.005)	−0.002 (0.004)	0.022*** (0.008)
Later years exposed	−0.007 (0.014)	0.014 (0.016)	−0.044 (0.055)	−0.004 (0.008)	0.011 (0.010)	0.017 (0.017)

* $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$. Each column re-estimates Equation (5) with a student characteristic measured before application as the outcome, and without the academic and demographic controls. Indicators for missing age-16 and teacher-predicted scores are estimated alongside and not reported. Observations are students; the sample is years 2010–2019, main-scheme applicants whose application sets contain no program that used unconditional offers in 2013 or earlier and whose programs each average at least 20 offers per year over 2010–2012. $N = 370,973$, except in column (5), where it is 368,907. Standard errors are clustered at the application set level.

Figure A.1: Time series of exposure to unconditional offers. (a) Share of students in each exposure group over time; (b) share of students with an unconditional offer, split by exposure group.

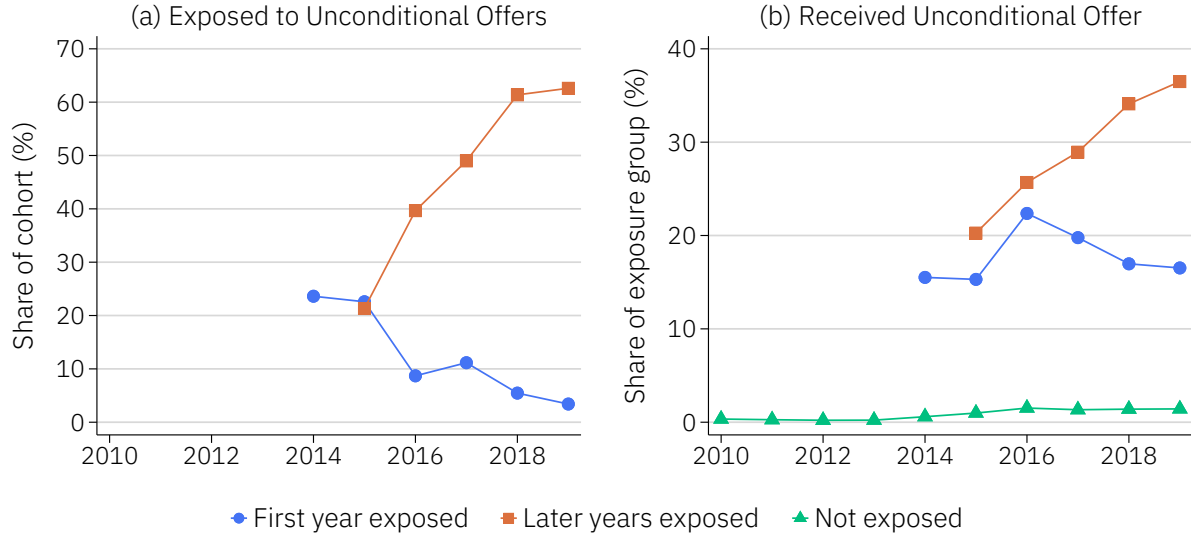


Table A.4: Robustness of the decomposition

	Main	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\hat{\pi}$	0.143	0.143	0.177	0.141	0.146	0.144	0.144	0.149	0.145
$\hat{\rho}$	0.005	0.005	0.001	0.004	0.005	0.004	0.003	0.002	0.004
$\hat{\delta}_x$	-0.134	-0.134	-0.122	-0.128	-0.133	-0.138	-0.139	-0.134	-0.136
$\hat{w}$	0.024	0.024	0.022	0.022	0.025	0.024	0.023	0.022	0.024
	(0.007)	(0.007)	(0.008)	(0.009)	(0.007)	(0.007)	(0.007)	(0.008)	(0.007)
t	3.43	3.50	2.83	2.29	3.70	3.52	3.27	2.70	3.43
ATT	0.033	0.033	0.003	0.026	0.036	0.026	0.019	0.012	0.026
PI	-0.167	-0.167	-0.125	-0.154	-0.169	-0.165	-0.159	-0.146	-0.162
$\hat{\theta}$	-0.244	-0.244	-0.023	-0.205	-0.271	-0.191	-0.140	-0.088	-0.194
θ cap	0.346	0.327	0.561	0.650	0.281	0.352	0.420	0.563	0.365
Clusters	97,730	97,730	97,004	59,709	103,379	100,998	93,898	5,007	81,892

$\hat{\pi}$ is the first stage, $\hat{\rho}$ the reduced form and $\hat{\delta}_x$ the taker-decliner achievement gap; $\hat{w} = \hat{\rho} - \hat{\pi}\hat{\delta}_x$, with its standard error in parentheses and t its t -statistic; $ATT = \hat{\rho}/\hat{\pi}$ is the moral-hazard effect and $PI = \hat{\delta}_x - \hat{\rho}/\hat{\pi}$ private information; $\hat{\theta}$ is the moral-hazard share of the gap and θ cap its one-sided 95% upper bound, obtained by inverting $w(\theta_0)$ in every column. *Main* is the specification of Table 6: a 5% threshold for a program counting as using unconditional offers, programs averaging at least 20 offers per year over 2010–2012, clustering on the application set, and standard errors from a 200-replication cluster bootstrap. The following columns report a variety of robustness checks, changing elements of our primary specification. (1) replaces the bootstrap with standard errors propagated linearly from the three clustered inputs under an assumption of independence. (2) raises the threshold for using unconditional offers from 5% to 10%. (3) restricts to 2010–2015. (4), (5) and (6) set the baseline offer requirement to a mean of at least 0, 10 and 30 offers per year rather than 20. (7) clusters on the least-selective university in the application set rather than on the set itself, which is the only column that speaks to correlation across sets that share a university. (8) drops observations after a set's first year of exposure, and is equivalent to $ATT(g, g)$ from Callaway and Sant'Anna (2021). Columns (2)–(8) all assume independence, as in (1); the gap in standard errors between (1) and Main shows the implication of this assumption.

Table A.5: Treatment effects across the distribution of achieved attainment

Quantile	(1) Smoothing (−.01, .01)	(2) Smoothing (−.05, .05)	(3) Smoothing (−.1, .1)
0.1	−0.005 (0.019)	−0.005 (0.018)	−0.023 (0.020)
0.2	−0.008 (0.015)	−0.026* (0.015)	−0.029** (0.014)
0.3	−0.008 (0.014)	−0.003 (0.014)	−0.029** (0.013)
0.4	0.003 (0.015)	−0.006 (0.014)	−0.016 (0.014)
0.5	0.014 (0.014)	0.018 (0.014)	0.003 (0.014)
0.6	0.007 (0.016)	0.017 (0.016)	0.020 (0.016)
0.7	0.005 (0.016)	0.015 (0.017)	0.026 (0.016)
0.8	0.041** (0.018)	0.029* (0.017)	0.055*** (0.018)
0.9	0.039** (0.018)	0.027 (0.018)	0.018 (0.019)

* $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$. Each column reports the coefficient on first exposure from a quantile regression of standardized achieved age-18 attainment on the same controls and fixed effects as Table 6's $\hat{\rho}$ (year fixed effects, application-set $\times$ post-first-exposure fixed effects, academic and demographic controls), estimated separately at each decile 0.1–0.9, with standard errors in parentheses. Columns differ in the smoothing bandwidth of the smoothed quantile regression, given in each column header. $N = 368,874$.

Table A.6: Always-taker robustness of the decomposition

Assumed always-taker effect	0 SD	-0.05 SD	-0.10 SD	-0.15 SD
ATT	0.030 (0.045)	0.026 (0.045)	0.022 (0.045)	0.018 (0.045)
PI	-0.164 (0.046)	-0.160 (0.046)	-0.156 (0.046)	-0.152 (0.046)
$\hat{\theta}$	-0.224 (0.332)	-0.193 (0.332)	-0.162 (0.332)	-0.131 (0.332)
θ cap	0.324	0.355	0.386	0.417

Relaxes the assumption in [Section 3.1](#) that there are no always-takers, and recomputes bounds on the share of the conditional gap which is composed of moral hazard. [Table 5](#), column (2) puts this share at $\hat{p}_A = 0.0129$ (SE 0.00201), against a complier share of $\hat{\pi} = 0.143$; write $\xi_C = \hat{\pi}/(\hat{\pi} + \hat{p}_A) = 0.917$ and $\xi_A = \hat{p}_A/(\hat{\pi} + \hat{p}_A) = 0.083$ for the resulting weights on compliers and always-takers among the treated. Each column assumes always-takers have the stated moral-hazard effect (in place of the complier LATE of [Table 6](#)) and recomputes $ATT = \xi_A(\text{assumed effect}) + \xi_C(\hat{\rho}/\hat{\pi})$, $PI = \hat{\delta}_X - ATT$, and $\theta = ATT/\hat{\delta}_X$. Rather than treat the assumed effect as known with certainty, we give it a standard error scaled for the always-takers' smaller population. Standard errors on ATT, PI and θ propagate this alongside the analytic clustered standard errors of $\hat{\pi}$, $\hat{\rho}$ and $\hat{\delta}_X$, under an assumption of independence between all four, extending the approach of column (1) of [Table A.4](#). θ cap is the one-sided 95% upper bound on θ , obtained the same way as in [Table A.4](#) by inverting $w(\theta_0)$ on the resulting blended estimate. $\hat{\pi}$, $\hat{\rho}$, $\hat{\delta}_X$, $\hat{w}$ and its t -statistic are unaffected by the assumed always-taker effect and are omitted; see [Table 6](#) or [Table A.4](#)'s Main column for their values. Sample includes 1,700,457 students for the gap, 366,288 for the exposure effects, and 97,730 application-set clusters.

Table A.7: Convergence diagnostics for the structural estimates

	ESS	$\hat{R}$	Geweke
<i>Preference stage — 50,000 draws</i>			
Constant	107	1.046	-2.13
Can live at home	2,331	1.002	1.64
× Low-SES	2,042	1.002	0.97
σ_η^2	384	1.006	-0.57
Same region	368	1.002	0.18
σ_v^2	118	1.007	0.53
Belief acceptance		0.630	
Converged		5 of 6	
<i>Signal stage — 250,000 draws</i>			
λ_i at mean ability	52,510	1.000	-0.17
Gradient, per SD	49,711	1.000	0.26
Female	54,743	1.000	0.66
σ_ε^2	5,781	1.000	-0.19
σ_ψ^2	73	1.026	4.11
Belief acceptance		0.938	
Converged		4 of 5	

Diagnostics for [Table 8](#). A parameter counts as converged when split- $\hat{R} \leq 1.01$ and the effective sample size is at least 20; the sampler separately flags $|Geweke| > 2$. In the preference stage only the constant fails, and it is weakly identified by construction. In the signal stage only σ_ψ^2 fails, and its posterior standard deviation in [Table 8](#) is likely understated.

B Data and Variable Construction

In this section, we describe in detail the definition and construction of variables.

Table B.1: Variable definitions

Variable	Definition
Unconditional offer	Indicator equal to 1 if an offer is marked as unconditional using UCAS flags, defined at the offer level.
Major	A specific JACS3.0 principal subject code; the Joint Academic Coding System (v3.0) was the standard subject-classification scheme in UK higher education prior to 2019.
Program	A particular combination of university and major. Programs may consist of multiple courses (the true unit of application); we aggregate offers and admission across all courses in a given program.
Application set	The set of programs applied to by a given student, defined at the student level.
Uses unconditional	Indicator equal to 1 if more than 5% of a program's offers in a given year are unconditional offers, defined at the year-program level.
Exposed to unconditional offers	Indicator equal to 1 if any program in a given application set uses unconditional offers, defined at the application set level.
First exposure	Indicator for whether an application set is exposed to unconditional offers for the first time, defined at the year-application set level.
Later exposure	Indicator for whether an application set is exposed to unconditional offers in a period after first exposure, defined at the year-application set level.
Take-up	Indicator for take-up of an unconditional offer, which occurs either through ranking a binding unconditional offer first, or ranking a standard unconditional offer first or second. Referred to as "Take-up" or "Uncond. take-up" in the exhibits. Defined at the student level.
Age-18 scores (z)	A student's aggregate achievement on their age-18 examinations (which may include both A-level and BTECs). If a student took more than three such exams, we consider the three exams that were most taken by other students in their primary major of application. Grades are converted to a numerical value using "tariff points": A* = 56, A = 48, B = 40, C = 32, D = 24, E = 16, U = 0. BTEC grades are converted using the same UCAS Tariff scale, applied to whichever qualification size a student holds (e.g., for a BTEC National Extended Diploma, sized to three A-levels: D*D*D* = 168, D*D*D = 160, D*DD = 152, DDD = 144, DDM = 128, DMM = 112, MMM = 96, MMP = 80, MPP = 64, PPP = 48; for a BTEC National Diploma, sized to two A-levels: D*D* = 112, DD = 96, DM = 80, MM = 64, MP = 48, PP = 32). We normalize scores to mean 0, standard deviation 1, within groups of students taking the same number of A-levels and the same number of BTEC exams.
Teacher-predicted age-18 scores (z)	A student's aggregate achievement, as predicted by their teachers, on their age-18 examinations. A small minority of students have a completed A-level exam at the time of their applications; for these students, we replace the relevant predicted grade with their achieved grade. Variable construction proceeds identically to Age-18 scores (z).

Continued on next page

Table B.1 – continued from previous page

Variable	Definition
Teacher-predicted age-18 scores (categorical)	A student's aggregate achievement, as predicted by their teachers, on their age-18 examinations. Identical to Teacher-predicted age-18 scores (z), but rather than standardizing the scores, we create an indicator variable for each unique combination of (score 1, test type 1, score 2, test type 2, score 3, test type 3).
Age-16 test scores (z)	Student's achievement on their age-16 examinations (GCSEs). We convert percentiles on the exam to a z score using the inverse normal CDF. Additionally includes a separate indicator for missing values.
Socioeconomic quintile	Quintile (1–5, 1 most deprived) of the Index of Multiple Deprivation (IMD) associated with a student's postcode.
Low-SES	Indicator for whether a student's postcode belongs to the bottom two quintiles of IMD.
Female	Indicator in UCAS data of whether a given student is female.
Independent school	Indicator for whether a student attends an independent school.
Expected achievement (z)	Fitted values from a regression of Age-18 scores (z) on Teacher-predicted age-18 scores (categorical), Age-16 test scores (including indicator for missing values), Low-SES, and Female using data from 2010–2013. Absent moral hazard, this is the best prediction available of student achievement given their prior observable characteristics.
Distance (d_{is})	Straight-line distance in kilometers between a student's postcode and the program's institution.
Commuting range	Indicator equal to 1 if $d_{is} < 40\text{km}$, i.e. close_{is} in Section 4.1.4 .
Region match (m_{is})	Indicator for whether a student's home region and a program's region are the same, using the 12 International Territorial Level 1 (ITL1) regions defined by the Office for National Statistics.
Matriculation	Indicator for whether a student enrolls at a university in a given year (e.g., at age 18, 19, ...).
Years enrolled	Cumulative number of years a student is observed enrolled in HESA by a given age (e.g., by age 24), used in Section 3.2 .
Degree completion / classification	Indicator for completing a degree by a given age, and the classification received (first-class honors, 2:1, 2:2, third-class honors), from HESA.

C A Model of Unconditional Offers

Takers score 0.139SD below decliners with the same academic attainment, demographics, and school ([Table 4](#), column 3). Two forces could produce this gap: students who know they would place higher may decline their offers (private information), or takers may prepare less for the examination (moral hazard).

We build a stylized model of the unconditional-offer market in which both forces are present. It shows that unconditional offers can be made, and taken, alongside either force; that either force alone produces a gap between takers and decliners of the kind we observe, so the gap does not tell

them apart; and that although adopters anticipate the gap, the programs above the highest adopter end up with weaker intakes than with conditional offers alone.

A unit mass of students is matched to four programs $k \in \{a, b, c, d\}$, each with capacity $\frac{1}{4}$. A student's ability is $A = v_1 + v_2$, with v_1 and v_2 independent and uniform on $[0, 1]$: v_1 is public (age-16 scores, teacher-predicted grades), while v_2 is unobserved by all parties until the age-18 examination. With probability p , independent of ability, a student privately observes v_2 after receiving her offers and before ranking them; otherwise she observes nothing. A student who takes an unconditional offer prepares less for the examination: her realized ability is $A - s$, with $s \geq 0$. Because her place is secured, the loss falls on the program that admitted her and on no one's placement.³¹ Finally, a share q of unconditional-offer holders decline for reasons unrelated to ability, such as a preferred offer elsewhere, and rank their conditional offers instead.³² Programs are aware of p and s , but not which students received a signal. Programs maximize the total realized ability of their intake.³³ A student who enrolls at k receives utility u_k , with $u_a > u_b > u_c > u_d$, and evaluates lotteries with a strictly increasing, strictly concave U .

Timing follows the UK admissions cycle. Students apply to all programs before v_2 is realized. Programs can make two types of offers: offers which are conditional on the revelation of v_2 , and unconditional, in which admission is guaranteed without regard to v_2 . Students can either rank their conditional offers, or commit to one unconditional offer and forfeit all other offers.

An equilibrium is a profile in which each program's set of unconditional offers maximizes its payoff taking the admission thresholds as given,³⁴ each student's take-up decision maximizes her expected utility, and conditional admission clears the market. Strategies are the set of students given unconditional offers by each program. In equilibrium this set is an interval of observable ability, the program's *unconditional-offer range*. The remaining conditional admission thresholds are set by market clearing. To simplify the analysis, we assume that students break any ties in favor of unconditional offers, and that programs make unconditional offers only when they are strictly profitable. We characterize the equilibrium of this market in [Proposition C.1](#). Four equilibria, one for each combination of the two forces, are shown in [Figure C.1](#). In each, one or both of the middle-tier programs make unconditional offers to a range of observably strong students, and the remainder of the class is admitted conditionally.

Proposition C.1 (Equilibrium unconditional-offer ranges). With $u = (3, 2.5, 2, 1)$, $U = \ln$, and $q = 0.2$, as in [Figure C.1](#), in every equilibrium:

- (i) Programs a and d never make unconditional offers. If b or c makes unconditional offers, it offers to an interval of observable ability, $[\underline{x}_k(p, s), \bar{x}_k]$; it breaks even at the floor, and uninformed students stop accepting at the ceiling. When

³¹In this model we take no stand on the cost of shirking to the student herself, which matters to the extent that attainment affects later outcomes ([Section 3.2](#)); in the model, students value only their placement.

³²The share q multiplies the value of every unconditional offer and every taker mass by $1 - q$; it changes no sign, floor, or ceiling at the margin, but at positive scale it narrows b 's unconditional-offer range and can widen c 's ([Appendix D](#)). We set $q = 0.2$ in [Figure C.1](#), so that every panel has decliners to compare takers with.

³³This is equivalent to maximizing average quality and enforcing capacity $1/4$. Because total enrolled ability is fixed when $s = 0$, the game among programs is then constant-sum, and every equilibrium is purely redistributive; when $s > 0$, total attainment falls by s times the mass of takers.

³⁴A program values each offer by its effect on its intake at the prevailing admission thresholds and does not account for how its own offers move those admission thresholds ([Appendix D.1](#)).

both offer, the two intervals are disjoint, with c 's below b 's.

- (ii) Unconditional-offer ranges are widest at $p = s = 0$. Program k makes unconditional offers if and only if $s < \hat{s}_k(p)$, where the exit threshold $\hat{s}_k$ is strictly decreasing in p , positive at $p = 0$ whenever k offers at all, and reaches zero at some $p < 1$.

Proof. See [Appendix D.6](#). □

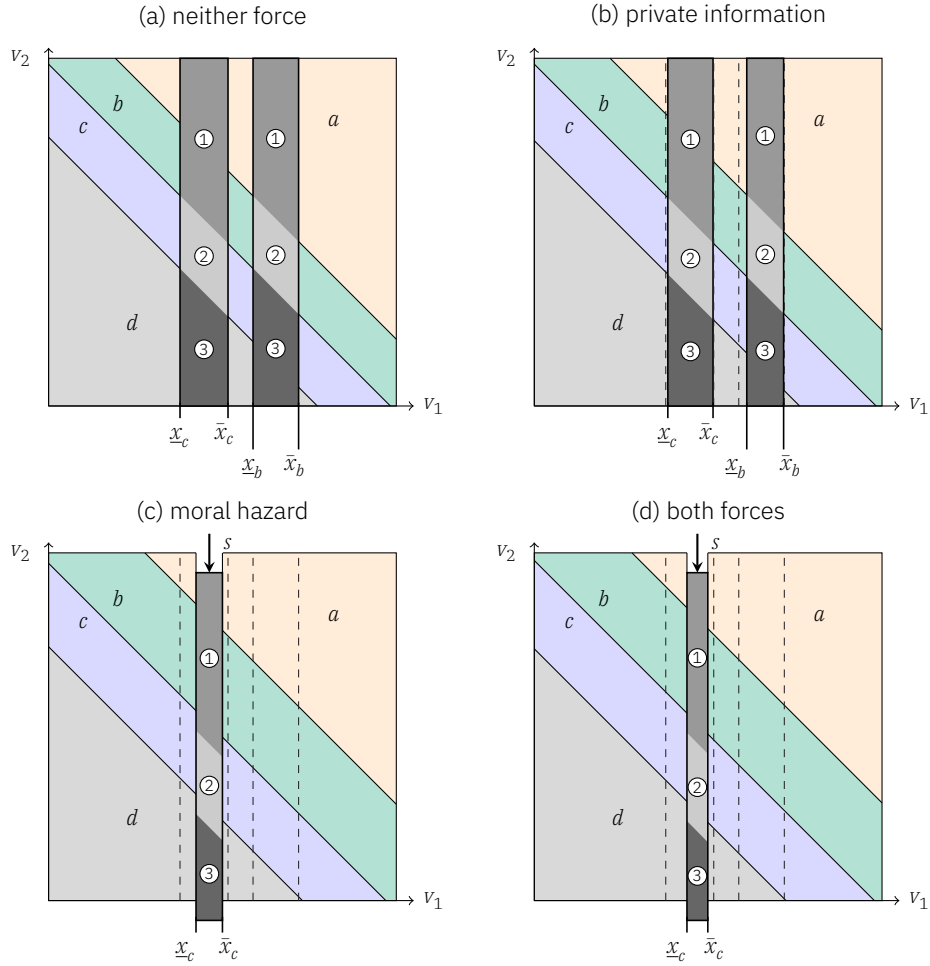
An equilibrium exists ([Proposition D.5](#)), and it is unique at every parameter point we compute.³⁵ Which program's exit threshold lies inside the other's depends on exact utility values ([Appendix D.6](#)); under the setup of [Figure C.1](#), b 's lies inside c 's, so as either force grows both programs offer, then only c , then neither. Beyond both exit thresholds the assignment with conditional offers alone is an equilibrium, and the only one at every point we compute.

Unconditional offers grant insurance to students uncertain about their placement; this can benefit a program when it attracts students who would otherwise have enrolled above it, though unconditional offers do not always benefit adopting programs relative to an economy with only conditional offers. In every panel of [Figure C.1](#), c gains relative to the assignment with conditional offers alone and a and b lose. Program c 's range holds students who would otherwise have placed at b ; b 's unconditional offers attract students from a , but this gain is offset by the loss of students to c . Though b would prefer that unconditional offers were banned, it uses unconditional offers anyway because staying out would be worse: in panel (a), b 's mean intake ability is 1.138 with conditional offers alone, 1.112 when it offers alongside c , and 1.088 if b stays out and c best responds. Private information and moral hazard both reduce the value of these offers to programs, but in different ways. Private information leads students with high v_2 to decline, reducing the expected value of an unconditional offer to the program. Moral hazard acts as a "tax" on unconditional offers, lowering the value of every unconditional offer by s times its take-up rate. Once the two forces together leave the offer at a program's ceiling unprofitable, that program stops offering altogether.

Both forces generate a gap between takers and decliners. Under private information, declining is itself the signal: the informed decliners are the students who know they would place above the program, so the gap is the distance between the strong students who leave and the average student who stays. This creates a "winner's curse" from the program's perspective: an informed student who takes an unconditional offer knows she is weak, while the uninformed take regardless. Under moral hazard, takers and decliners are the same students until the examination, and the gap is what the examination reveals. [Proposition D.9](#) gives the gap in closed form, as the sum of the tax s and a private-information term that, when some holders decline for other reasons ($q > 0$), is nonzero if and only if $p > 0$; the same gap arises from a continuum of (p, s) pairs, and so the gap itself is not sufficient to distinguish the two. The panels of [Figure C.1](#) show this: $(p, s) = (0.104, 0)$ in (b) and $(0, 0.0567)$ in (c) produce the same gap of -0.139SD between takers and decliners, the conditional gap of [Table 4](#) (column 3). Together, in panel (d), they produce -0.278SD .

In both cases, programs respond by shrinking their unconditional-offer ranges, but the two forces have different welfare consequences for programs and students, and thus policy implica-

³⁵An analytical proof is shown in the online appendix.

Figure C.1: Equilibrium unconditional-offer ranges and take-up in the (v_1, v_2) plane, by force.

Notes: Axes are observable ability v_1 and the unobserved component v_2 ; ability $A = v_1 + v_2$ is constant along diagonals. Colored bands are the conditional-admission assignment (d, c, b, a from the bottom left), separated by the solid diagonals, the equilibrium admission thresholds $\underline{\alpha}_k$. The vertical gray strips are the unconditional-offer ranges of c (left) and b (right), from floor $\underline{x}_k$ to ceiling $\bar{x}_k$; the dashed verticals in (b)–(d) mark the ranges of (a), for comparison. Within each strip, ① is the gain cell, takers who would otherwise have placed above k , each displacing a worse conditional admit; ② the neutral cell, takers who would have enrolled at k anyway; ③ the loss cell, takers who would have placed below k , each displacing a better conditional admit. Informed students, absent in (a) and (c), decline in ① and accept in ② and ③. The share q of holders who decline regardless, for reasons unrelated to ability, are spread uniformly over every cell and are not drawn, for ease of viewing. Under moral hazard, (c) and (d), takers score s less than they would have, so each strip is drawn shifted down by s , at the takers' realized v_2 . Panels use $u = (3, 2.5, 2, 1)$, $U = \ln$, $q = 0.2$, $p = 0.104$, and $s = 0.0567$. In (b) and (c), this results in a taker–decliner achievement gap of -0.139SD (Table 4, column 3). Program b makes no unconditional offers in (c) and (d). The four equilibria are detailed in Appendix D.3.

tions. Private information decreases the benefits of unconditional offers, redistributing program welfare, but leaving its total constant. Ex-post student welfare (as measured after final matching) is constant under either force, since capacities fix the placements, and so is total ex-ante student welfare (as measured during the ranking phase): every student is placed, so students' expected utilities sum to $\frac{1}{4} \sum_k U(u_k)$ in every equilibrium. Unconditional offers therefore only redistribute among students. An informed student takes an offer only when it beats her placement, so under private information informed students do better, and uninformed students worse, than with

conditional offers alone; under moral hazard alone every student is uninformed, and their total is unchanged. Moral hazard, by contrast, reduces the total welfare available to programs, in addition to redistributing it. Beyond placements, the effect of moral hazard on student welfare is ambiguous and depends on the sign of returns to age-18 exams, which the model does not price; returns encompass many components, such as human capital, non-pecuniary costs of studying for tests, and direct labor market effects. Under private information, a ban on unconditional offers redistributes among students, from informed to uninformed, and among programs, toward those that lose from the offers, a and b ; under moral hazard, a ban is welfare-improving for programs in total, with ambiguous impacts on students.

D Appendix for the Theoretical Model

D.1 Timing

- | | | |
|---------|--------|--|
| $t = 0$ | apply | Students submit applications to all four programs. |
| $t = 1$ | offer | Programs make offers: conditional (admitting if the realized A meets the program's admission threshold) or unconditional (made before the examination, and so conditioned on v_1 alone). |
| $t = 2$ | rank | Students privately learn v_2 with probability p , then rank up to four offers. Students can either rank all conditional offers, or rank a single unconditional offer and forfeit all remaining offers. A share q of unconditional-offer holders, drawn independently of ability and information, rank their conditional offers regardless. |
| $t = 3$ | accept | Students who ranked conditional offers take a test, revealing $A = v_1 + v_2$. Students are accepted to the best-ranked program for which they meet the conditions. Students who took an unconditional offer score $A - s$, and keep their place. |

Equilibrium. Students best respond by [Lemma D.2](#). Each program takes the admission thresholds as given: it values an offer at target x by $V_k(x)$, the change in its payoff from a marginal offer there at the prevailing admission thresholds ([Lemma D.3](#)), and makes unconditional offers exactly where that value is positive.³⁶ Because unconditional-offer ranges cover a nonnegligible share of capacity, takers shift the admission thresholds, so an equilibrium is a joint fixed point of unconditional-offer ranges and market clearing.³⁷ [Proposition C.1](#) characterizes the unconditional-offer sets, [Appendix D.3](#) computes the fixed point, and [Proposition D.5](#) shows that it exists; at every parameter point we compute, it is unique.

D.2 Lemmas

We first prove three lemmas about a single unconditional offer. [Appendix D.3](#) extends them to unconditional-offer ranges of positive mass, [Appendix D.4](#) proves existence, and [Appendix D.6](#) proves [Propositions C.1](#) and [D.9](#).

Define the *target* of an unconditional offer as the observable ability $x = v_1$ of the student who receives it. A program's *ceiling* $\bar{x}_k$ is the highest target at which uninformed students accept its offer ([Lemma D.2](#)), and its *floor* $\underline{x}_k$ is the lowest target at which the offer breaks even ([Lemma D.4](#)); both depend on p and s , which we suppress. Program k 's *exit thresholds* are $\hat{p}_k$, the value of p at which its unconditional-offer range vanishes when $s = 0$, and $\hat{s}_k(p)$, the value of s at which it vanishes given p . Primed objects (admission thresholds α'_k , ceilings $\bar{x}'_k$, floors $\underline{x}'_k(p)$, exit thresholds $\hat{p}'_k$ and

³⁶A program that also accounted for the effect of its own offers on the admission thresholds would behave differently. In our computations for c , its extra takers raise its own admission threshold, which makes keeping conditional offers riskier for uninformed students and so raises its ceiling. Such a program makes unconditional offers below its break-even target and, with $p > 0$, loss-making offers above its ceiling, whose informed takers raise its admission threshold further.

³⁷Conditional admission is not strategic: admission thresholds clear the market. Strategic conditional admission would not help either. Committing to reject an available student only passes her down the ranking and fills the seat from below, since the clearing admission thresholds above do not depend on whom a program rejects beneath them.

$\hat{s}'_k(p)$ are evaluated at the conditional-only benchmark of Lemma D.1, where Lemmas D.2 and D.3 analyze a single offer; we write $\underline{x}'_k$ for $\underline{x}'_k(0)$. Unprimed objects are equilibrium values.

Our lemmas hold beyond the baseline economy (equal capacities, hence quartile admission thresholds, and $u = (3, 2.5, 2, 1)$ with $U = \ln$), which we use only for numerical evaluations. The proofs rely on three conditions, flagged where used and all satisfied at the baseline economy.

(C1) At the prevailing admission thresholds, every band is narrower than the support of v_2 , so that students who do not observe v_2 have uncertainty about their program assignment.

(C2) At the prevailing admission thresholds and every target x between k 's floor and ceiling, $\underline{\alpha}_k - x$ and $\bar{\alpha}_k - x$ lie in $(0, 1)$, so that an offeree at target x may place below, at, or above k depending on her v_2 .

(C3) Unconditional offers are viable for some k : $\bar{x}'_k > \underline{x}'_k$, so that some target is both profitable for k and accepted by uninformed students.

(C1) and (C2) hold at the benchmark and in every equilibrium (Proposition D.5). Preferences enter only through the ceilings $\bar{x}'_k$; the floors $\underline{x}'_k(p)$ are pure geometry. Two kinds of claims depend on the baseline: the exit order in Proposition C.1, which reverses when u_b is close enough to u_a (Appendix D.6), and the results at positive scale that we establish by computation (Appendix D.3). The two frictions enter in two places only. Moral hazard s is a per-taker cost in Lemma D.3 and nowhere else: a taker's place is secured, so her lower score moves no admission threshold. The random-decline share q multiplies every taker mass, in Lemma D.3 and in Equation (19), by $1 - q$, and nothing else: a random decliner is an ordinary conditional applicant. Every sign below, and every floor, ceiling, and exit threshold of the marginal analysis, is therefore invariant to q . At positive scale q changes the unconditional-offer ranges, though not always downward: it narrows b 's ($\hat{p}_b$ falls from 0.575 to 0.557 at $q = 0.2$) but can widen c 's, which at $p = s = 0$ is $[0.378, 0.516)$ with $q = 0.2$ and $[0.397, 0.524)$ with $q = 0$. The tables report $q = 0$ unless stated.

Lemma D.1 (Benchmark assignment). With conditional offers only, if each student enrolls at the highest-ranked program that admits her, the assignment that clears the market is unique and assortative: program k enrolls the students with $A \in [\underline{\alpha}'_k, \bar{\alpha}'_k)$, where the admission thresholds $(\underline{\alpha}'_a, \underline{\alpha}'_b, \underline{\alpha}'_c, \underline{\alpha}'_d) = (2 - \sqrt{\frac{1}{2}}, 1, \sqrt{\frac{1}{2}}, 0)$ are the quartiles of the distribution of A , and $\bar{\alpha}'_k$ is the admission threshold of the program ranked directly above k : $\bar{\alpha}'_c = \underline{\alpha}'_b$, $\bar{\alpha}'_b = \underline{\alpha}'_a$, and $\bar{\alpha}'_a = 2$, the top of the support. The assignment is the same for every p .

Proof. The ability A has distribution $F(\alpha) = \frac{\alpha^2}{2}$ for $\alpha \leq 1$ and $1 - \frac{(2-\alpha)^2}{2}$ for $\alpha \geq 1$, strictly increasing on $[0, 2]$. Students share the ranking $u_a > u_b > u_c > u_d$, so in any market-clearing outcome admission thresholds are ordered and each intake is the interval between consecutive admission thresholds. Capacities of $\frac{1}{4}$ then force $1 - F(\underline{\alpha}'_a) = \frac{1}{4}$, $1 - F(\underline{\alpha}'_b) = \frac{1}{2}$, $1 - F(\underline{\alpha}'_c) = \frac{3}{4}$, and $\underline{\alpha}'_d = 0$; strict monotonicity of F makes the solution unique. The argument uses only the common ranking, atomlessness of F , and capacities summing to one; with general capacities, the admission thresholds are the corresponding quantiles. Nothing is chosen before the examination, so nothing depends on p . $\square$

Write $\mu(A)$ for the program assigned to ability A under [Lemma D.1](#). Program k 's band is $[\underline{\alpha}'_k, \bar{\alpha}'_k]$: its own admission threshold is the lower edge, and the admission threshold of the program directly above is the upper edge, so an overline on α is the underline of the next program up. A taker who would have placed below k costs k its worst student, of ability $\underline{\alpha}'_k$. A taker who would have placed above k costs k its *best* student: she leaves a vacancy above, which is refilled from the top of k 's band, so k keeps her at the price of a student of ability $\bar{\alpha}'_k$. So the *net contribution* of a student of ability A who enrolls at k by taking an unconditional offer is

$$\varphi_k(A) \equiv \max\{A - \bar{\alpha}'_k, 0\} - \max\{\underline{\alpha}'_k - A, 0\},$$

her distance outside k 's band, signed: positive if she would have placed above k , zero if at k , negative if below. At other admission thresholds, φ_k is defined in the same way with those admission thresholds.

Lemma D.2 (Acceptance). Consider a single unconditional offer from $k \in \{b, c\}$ to a student with $v_1 = x$. One offer has measure zero, so admission thresholds stay at the benchmark values of [Lemma D.1](#); primes mark objects evaluated at those admission thresholds.

- (i) An uninformed holder accepts if and only if $x \leq \bar{x}'_k$. The *ceiling* $\bar{x}'_k$ is the unique solution in $(0, \underline{\alpha}'_k)$ of $W_k(x) = U(u_k)$, where $W_k(x) \equiv E[U(u_{\mu(A)}) \mid v_1 = x]$ is the *lottery utility* of keeping the conditional offers: continuous, piecewise linear, and strictly increasing in x .³⁸
- (ii) For every increasing U , an informed holder accepts if and only if she would have placed at or below k , that is, $A < \bar{\alpha}'_k$. Up to the measure-zero boundary $A = \bar{\alpha}'_k$, this is $\varphi_k(A) \leq 0$.
- (iii) At any prevailing admission thresholds that satisfy (C1), including those induced by a positive mass of offer holders, an uninformed holder accepts if and only if $W_k(x) \leq U(u_k)$, so the targets she accepts form an interval that ends at the ceiling, and an informed holder if and only if $A < \bar{\alpha}'_k$, with W_k , μ , φ_k , and the ceiling evaluated at those admission thresholds.

Proof. Conditional on $v_1 = x$, $A \sim U[x, x + 1]$, so the lottery utility is

$$W_k(x) = \int_0^1 U(u_{\mu(x+t)}) dt,$$

continuous and piecewise linear, with derivative $U(u_{\mu(x+1)}) - U(u_{\mu(x)})$ wherever x and $x + 1$ are interior to bands. Raising the target shifts probability from the worst reachable band to the best, so the derivative is positive as long as $\mu(x + 1)$ is ranked strictly above $\mu(x)$, which (C1) guarantees. At $x = 0$ the support $[0, 1]$ of A meets only d 's and c 's bands, so $W_c(0) = \underline{\alpha}'_c U(u_d) + (1 - \underline{\alpha}'_c) U(u_c) < U(u_c)$;³⁹ at $x =$

³⁸The ceiling carries a prime because the lottery depends on the admission thresholds through μ ; at the equilibrium admission thresholds of [Appendix D.3](#) it is a different number, $\bar{x}_k$.

³⁹In general, $W_k(0) < U(u_k)$ is the condition that some uninformed student accepts, which is part of viability (C3).

$\underline{\alpha}'_c$ the support $[\underline{\alpha}'_c, \underline{\alpha}'_c + 1]$ lies weakly above $\underline{\alpha}'_c$ and strictly above $\underline{\alpha}'_b$ with positive probability, so $W_c(\underline{\alpha}'_c) > U(u_c)$. Thus, W_c crosses $U(u_c)$ exactly once on $(0, \underline{\alpha}'_c)$, at $\bar{x}'_c$; the indifferent student at the ceiling accepts by convention. The same argument with u_b in place of u_c on $(0, \underline{\alpha}'_b)$ gives $\bar{x}'_b$: $W_b(0) = W_c(0) < U(u_c) < U(u_b)$, and at $x = \underline{\alpha}'_b$ the support lies weakly above $\underline{\alpha}'_b$ and strictly above $\underline{\alpha}'_a$ with positive probability. For (ii), an informed holder knows A and compares the certain u_k to the certain $u_{\mu(A)}$: she accepts if and only if $\mu(A)$ is k or worse, that is, $A < \bar{\alpha}'_k$, for any increasing U , and at $\mu(A) = k$ she accepts by convention. Since $\varphi_k(A) \leq 0$ exactly when $A \leq \bar{\alpha}'_k$, this is the last claim of (ii). For (iii), each student is atomistic and takes the prevailing admission thresholds as given; she compares $W_k(x)$ to $U(u_k)$ as in (i), W_k is strictly increasing by (C1), and the argument for (ii) uses only the order of the admission thresholds. $\square$

At the baseline, $\bar{x}'_c = 0.495$ and $\bar{x}'_b = 0.698$.

For an offer at target x , the offeree's v_2 falls into one of three cells: the *loss cell* $v_2 < \underline{\alpha}'_k - x$, of length $\ell_k(x) \equiv \underline{\alpha}'_k - x$, where she would have placed below k ; the *neutral cell* $\underline{\alpha}'_k - x \leq v_2 < \bar{\alpha}'_k - x$, where she would have placed at k ; and the *gain cell* $v_2 \geq \bar{\alpha}'_k - x$, of length $g_k(x) \equiv x + 1 - \bar{\alpha}'_k$, where she would have placed above k . At other admission thresholds, the cells are defined in the same way with those thresholds. Under (C2) all three cells are nonempty.

Lemma D.3 (Value of a marginal unconditional offer). Let program k make unconditional offers to every student with $v_1 \in [x, x + \varepsilon]$, with students following Lemma D.2. The prevailing admission thresholds may be the benchmark admission thresholds of Lemma D.1 or those induced by any unconditional offers, including k 's own (Appendix D.3); φ_k and the cells are evaluated at them. Holding fixed the acceptance decisions of the holders of all other unconditional offers, the change in k 's payoff is $\varepsilon V_k(x) + O(\varepsilon^2)$ at every target $x \neq \bar{x}_k$, where

$$V_k(x) = (1 - q) \left[(1 - p) \mathbf{1}\{x < \bar{x}_k\} \int_0^1 \varphi_k(x + v_2) dv_2 + p \int_0^1 \min\{\varphi_k(x + v_2), 0\} dv_2 - s \tau_k(x) \right],$$

and $\tau_k(x) = (1 - p) \mathbf{1}\{x < \bar{x}_k\} + p \int_0^1 \mathbf{1}\{\varphi_k(x + v_2) \leq 0\} dv_2$ is take-up at x among holders not subject to the random decline.

Proof. Capacities bind and clearing is assortative, so when a taker of ability A enters k 's intake exactly one student leaves it, and she is always at an admission threshold. If the taker would have placed above k , the vacancy she leaves is filled from the top down, each program taking the best student of the program below, until the program directly above k takes k 's best student, of ability $\bar{\alpha}_k$. If she would have placed below k , k sheds its worst student, of ability $\underline{\alpha}_k$, and the surplus passes down in the same way to the program she vacated. If she would have placed at k , she replaces herself. In each case k 's net is $\varphi_k(A)$, and no other program's intake enters

k 's payoff. The taker then scores $A - s$ rather than A ; since her place is secured, the shortfall changes no assignment and is a cost of s to k alone. Summing over the strip $[x, x + \varepsilon)$: by Lemma D.2, for $x \neq \bar{x}_k$ the uninformed accept if and only if $x < \bar{x}_k$ and are uniform in v_2 , while the informed accept if and only if $\varphi_k(A) \leq 0$; the share q who decline regardless are uniform in v_2 and independent of information, so they scale every term by $1 - q$ and enter the conditional pool as ordinary applicants. The takers have mass $O(\varepsilon)$, so admission thresholds move by $O(\varepsilon)$ and each displaced student has ability equal to the admission threshold up to $O(\varepsilon)$; offerees in the strip whose acceptance flips because admission thresholds moved lie within $O(\varepsilon)$ of the boundary $\varphi_k = 0$, or of the ceiling, which $x \neq \bar{x}_k$ keeps outside the strip for small ε ; either set has measure $O(\varepsilon^2)$. Holders outside the strip keep their decisions by hypothesis. $\square$

In Lemma D.3, an uninformed taker contributes her full net contribution, an informed taker, who accepts exactly when $\varphi_k \leq 0$, contributes only its negative part, and every taker costs s . The lemma gives the value of a target in closed form. With the cell lengths ℓ_k and g_k both in $(0, 1)$ (C2), $\int_0^1 \varphi_k = \frac{1}{2} (g_k^2 - \ell_k^2)$, $\int_0^1 \min\{\varphi_k, 0\} = -\frac{1}{2} \ell_k^2$, and the informed accept on the loss and neutral cells, of combined length $1 - g_k$, so $\tau_k = 1 - p g_k$. Below the ceiling $\bar{x}_k$, where uninformed students accept,

$$\frac{V_k(x)}{1 - q} = \frac{1}{2} [(1 - p) g_k(x)^2 - \ell_k(x)^2] - s(1 - p g_k(x)) = G_k(x) - p \frac{g_k(x)^2}{2} - s(1 - p g_k(x)), \quad (18)$$

where $G_k(x) \equiv \frac{1}{2} [g_k(x)^2 - \ell_k(x)^2]$ is the value with neither force: private information removes a fraction p of the gain cell and nothing else, and moral hazard charges s per taker on the take-up rate $1 - p g_k$. Above the ceiling only informed students accept, so $V_k = -(1 - q) p [\frac{1}{2} \ell_k^2 + s(1 - g_k)] \leq 0$. Raising the target by dx lengthens the gain cell and shortens the loss cell by dx each, so below the ceiling V_k is strictly increasing in x , with derivative $(1 - q) [(1 - p) g_k + \ell_k + s p] > 0$; it is strictly decreasing in s , and strictly decreasing in p wherever $g_k > 2s$, since an informed gain-cell student who declines costs k her average net contribution $g_k/2$ and saves it the s she would have forgone. We write $V_k(\bar{x}_k)$ for Equation (18) evaluated at the ceiling, the value of the last target that uninformed students accept.

Remark (a and d). For a , $\bar{\alpha}_a = 2$ is the top of the support, so the gain cell is empty: every taker has $\varphi_a \leq 0$, and since $\underline{\alpha}_a > 1$ by (C1), the loss cell is nonempty at every target. An uninformed holder therefore has lottery utility below $U(u_a)$ and accepts, so $V_a < 0$ at every target and every p . For d , $x + 1 > \underline{\alpha}_c$ for every x , since $\underline{\alpha}_c < 1$ by (C1), so an uninformed holder of d 's offer is assigned above d with positive probability if she keeps her conditional offers, and never accepts; an informed holder accepts only if she would have placed at d anyway, a wash. Neither program has a strictly profitable target at $s = 0$, and $s > 0$ only lowers every value, so by convention neither makes unconditional offers. These arguments hold at any admission thresholds that satisfy (C1).

D.3 Positive scale

Lemmas D.2 and **D.3** value a single unconditional offer at fixed admission thresholds, which is enough to sign every claim about a marginal offer in **Propositions C.1** and **D.9** (**Appendix D.6**), because an unconditional-offer set of mass ε moves every admission threshold by $O(\varepsilon)$. In equilibrium, an unconditional-offer range of positive mass moves the admission thresholds it is evaluated against. This section computes that equilibrium; **Lemma D.4** reduces each adopter's strategy to one number, and **Equation (19)** then gives the admission thresholds.

Lemma D.4 (Interval unconditional-offer sets). Fix any admission thresholds satisfying (C1) and (C2), and any p and s . For $k \in \{b, c\}$, the value $V_k(x)$ of **Lemma D.3** is nondecreasing in x below the ceiling $\bar{x}_k$, strictly so when $p < 1$, and nonpositive above it. Taking these admission thresholds as given, k makes no unconditional offers if $V_k(\bar{x}_k) \leq 0$. Otherwise its unconditional-offer set is the interval $[\underline{x}_k, \bar{x}_k)$ on which $V_k > 0$, up to its endpoints, and its floor breaks even, $V_k(\underline{x}_k) = 0$, unless $V_k > 0$ at every target below the ceiling, in which case $\underline{x}_k = 0$.

Proof. Below the ceiling, raising the target raises $\varphi_k(x + v_2)$ weakly for every v_2 , and strictly whenever $x + v_2$ lies outside k 's band, which by (C1) has positive probability; it also weakly lowers informed take-up. Hence V_k is nondecreasing, and strictly increasing when $p < 1$; under (C2), **Equation (18)** gives $V'_k = (1 - q)[(1 - p)g_k + \ell_k + sp] > 0$. Above it only informed students accept; by **Lemma D.2(ii)** each has $\varphi_k \leq 0$, and each costs s , so $V_k \leq 0$. Taking the admission thresholds as given, k values each target x at $V_k(x)$, so its payoff from an offer set S is $\int_S V_k(x) dx$, which is maximized by offering exactly where $V_k > 0$ (zero-value targets are excluded by convention). By monotonicity this set is an interval that ends at the ceiling, and since V_k is continuous below the ceiling, its floor solves $V_k(\underline{x}_k) = 0$ unless $V_k > 0$ throughout. $\square$

Market clearing pins down the admission thresholds. Let $K \subseteq \{b, c\}$ be the set of adopters, each $k \in K$ offering on the strip $[\underline{x}_k, \bar{x}_k)$ of length L_k and midpoint m_k ; every offeree in the strip not subject to the random decline takes if uninformed, and takes if informed exactly when $A < \bar{\alpha}_k$ (**Lemma D.2**); the share q who decline regardless rank their conditional offers. Write $\bar{F}(\theta) \equiv 1 - F(\theta)$ for the mass of students with $A \geq \theta$ and κ_j for capacities. Students with $A \geq \underline{\alpha}_j$ either take an unconditional offer or enroll conditionally at j or above, so for each $j \in \{a, b, c\}$, where $i < j$ means that i is ranked below j ,

$$\bar{F}(\underline{\alpha}_j) = \sum_{i \geq j} \kappa_i + (1 - q)(1 - p) \sum_{i \in K, i < j} L_i (1 - \underline{\alpha}_j + m_i) - (1 - q) \sum_{i \in K, i \geq j} L_i (\underline{\alpha}_j - m_i). \quad (19)$$

Equation (19) is a seat count, and s does not appear in it: a taker's lower score moves no admission threshold, because her place is already hers. The seats at j and above, $\sum_{i \geq j} \kappa_i$, are filled either conditionally, by students with $A \geq \underline{\alpha}_j$ who took no unconditional offer, or unconditionally, by the takers of adopters at or above j . The mass $\bar{F}(\underline{\alpha}_j)$ therefore equals the seats, plus the takers with $A \geq \underline{\alpha}_j$ at adopters below j (counted in $\bar{F}$, but filling no seat at j or above), minus the takers with

$A < \underline{\alpha}_j$ at adopters at or above j (filling a seat, but not counted in $\bar{F}$). In words, the admission threshold of j falls by the takers that adopters below j capture from above it (uninformed only, since informed takers of i have $A < \bar{\alpha}_i \leq \underline{\alpha}_j$), and rises by the takers that adopters at or above j admit from below it (informed and uninformed alike, since $A < \underline{\alpha}_j \leq \bar{\alpha}_i$).

Both terms take these closed forms because, for a taker at target t , the probability that her ability is at least $\underline{\alpha}_j$ is $t+1-\underline{\alpha}_j$ whenever $\underline{\alpha}_j$ lies strictly inside her range of abilities, $[t, t+1]$: for the adopter's own admission thresholds this is (C2), and for the others it requires $\underline{x}_c > \underline{\alpha}_a - 1$ and $\bar{x}_b < \underline{\alpha}_c$. Both conditions hold at every solution below; the two strips are also disjoint there, so no student holds two unconditional offers.

An equilibrium with adopter set K is a solution of Equation (19) for $j = a, b, c$ together with, for each $k \in K$, the acceptance condition $W_k(\bar{x}_k) = U(u_k)$ and the floor condition $V_k(\underline{x}_k) = 0$, both evaluated at the equilibrium admission thresholds: $3 + 2|K|$ equations in as many unknowns. In addition, no target may be profitable for a non-adopter $j \in \{b, c\} \setminus K$: $V_j(\bar{x}_j) \leq 0$ at the equilibrium admission thresholds (Lemma D.4).

The acceptance condition is explicit whenever the ceiling lies in $(\underline{\alpha}_a - 1, \underline{\alpha}_c)$. For targets in this interval, the lottery utility, which is the same for b and c , is $W_c(x) = U(u_d)(\underline{\alpha}_c - x) + U(u_c)(\underline{\alpha}_b - \underline{\alpha}_c) + U(u_b)(\underline{\alpha}_a - \underline{\alpha}_b) + U(u_a)(x + 1 - \underline{\alpha}_a)$, so $W_c(\bar{x}_c) = U(u_c)$ and $W_c(\bar{x}_b) = U(u_b)$ give

$$\bar{x}_c = \omega_c \underline{\alpha}_c + \omega_b(\underline{\alpha}_b - 1) + \omega_a(\underline{\alpha}_a - 1), \quad \bar{x}_b = \bar{x}_c + \omega_b, \quad (20)$$

where $\omega_k \equiv [U(u_k) - U(u_{k'})]/[U(u_a) - U(u_d)]$ is the share of the utility gap between a and d that lies between k and the program k' ranked directly below it, so the weights are positive and sum to one; at the baseline, $(\omega_a, \omega_b, \omega_c) = (0.166, 0.203, 0.631)$, and we write $\omega_{ab} \equiv \omega_a + \omega_b$.

Given the strip endpoints, Equation (19) is piecewise quadratic in $\underline{\alpha}_j$; the acceptance condition makes the ceiling affine in the admission thresholds (Equation (20)); and by Equation (18) the floor condition $V_k(\underline{x}_k) = 0$ is, at $s = 0$, $\sqrt{1-p} g_k = \ell_k$, the gain cell equal to $1/\sqrt{1-p}$ times the loss cell, which is linear in the floor: $\underline{x}_k = [\underline{\alpha}_k - \sqrt{1-p}(1 - \bar{\alpha}_k)]/(1 + \sqrt{1-p})$. With $s > 0$ the floor condition is the quadratic $(1-p)g_k^2 - \ell_k^2 = 2s(1-pg_k)$, whose left side is increasing in the floor and right side nonincreasing, so it has one admissible root, increasing in s . The equilibrium is therefore a fixed point in the $2|K|$ endpoints. Solving, at $q = 0$, along p with $s = 0$ and along s with $p = 0$:⁴⁰

⁴⁰As $L_k \rightarrow 0$ the admission thresholds return to the benchmark, so c 's unconditional-offer range vanishes exactly where the benchmark floor reaches the benchmark ceiling, $\hat{p}_c = 0.816$ (Appendix D.6). Adopter k fills $M_k = (1-p)L_k + p \int_{\underline{x}_k}^{\bar{x}_k} (\bar{\alpha}_k - t) dt$ of its seats with unconditional admits and the rest conditionally at $\underline{\alpha}_k$; the last columns report M_k as a share of capacity.

p	s	$\underline{\alpha}_a$	$\underline{\alpha}_b$	$\underline{\alpha}_c$	$\underline{x}_b$	$\bar{x}_b$	$\underline{x}_c$	$\bar{x}_c$	M_b	M_c
0	0	1.136	0.997	0.796	0.566	0.727	0.397	0.524	64%	51%
0.2	0	1.212	0.987	0.770	0.621	0.722	0.400	0.519	36%	43%
0.35	0	1.250	0.986	0.754	0.657	0.717	0.411	0.514	20%	34%
0.5	0	1.276	0.986	0.739	0.692	0.712	0.427	0.509	6%	25%
0.6	0	1.286	0.988	0.730	—	—	0.442	0.505	—	18%
0.7	0	1.290	0.994	0.720	—	—	0.463	0.501	—	10%
0.8	0	1.293	0.999	0.709	—	—	0.490	0.496	—	1%
0	0.02	1.185	0.981	0.770	0.608	0.716	0.401	0.513	43%	45%
0	0.04	1.231	0.969	0.750	0.654	0.708	0.411	0.505	22%	38%
0	0.05	1.252	0.965	0.742	0.678	0.706	0.418	0.503	11%	34%
0	0.07	1.277	0.971	0.728	—	—	0.442	0.499	—	23%
0	0.09	1.287	0.990	0.713	—	—	0.476	0.496	—	8%

The table shows 12 of the 356 parameter points at which we compute the equilibrium.⁴¹ For each candidate set of adopters we find every solution of Equation (19), with the acceptance and floor conditions, by scanning each floor on a grid of step 0.01, refined near the ceiling,⁴² and keep the solutions at which each non-adopter's offer at its ceiling has nonpositive value; exactly one remains at every point. In every equilibrium found, b makes unconditional offers only if c does. Along each line in p and in s , the length of each range falls as either force grows, and $\hat{s}_b(p)$ falls in p and exceeds $\hat{s}'_b(p)$ at every p on a grid of step 0.01 below $\hat{p}_b$: from 0.060 at $p = 0$ to 0.003 at $p = 0.55$ with $q = 0$, and from 0.056 to 0.001 with $q = 0.2$.

The signs of the two sums in Equation (19) order the admission thresholds against the benchmark. No adopter lies below c , so only the negative sum enters c 's condition and $\underline{\alpha}_c > \underline{\alpha}'_c$; no adopter lies at or above a , so only the positive sum enters a 's and $\underline{\alpha}_a < \underline{\alpha}'_a$. For b , only the positive sum enters when c is the sole adopter, so $\underline{\alpha}_b < \underline{\alpha}'_b$; when b also adopts, its own takers below $\underline{\alpha}_b$ enter with the opposite sign, and the computed fixed point settles the comparison, $\underline{\alpha}_b < \underline{\alpha}'_b$ in every row of the table. Beyond $\hat{p}_c$ the offer at each program's benchmark ceiling has nonpositive value, so making no offers is an equilibrium (Lemma D.4), and at every such point we compute it is the only one.

The box collects the four equilibria of Figure C.1: the unconditional-offer ranges, and everything market clearing implies from them. All four use $q = 0.2$, so that every panel has decliners; $p = 0.104$ and $s = 0.139/\sqrt{6} \approx 0.0567$ are chosen so that each force alone reproduces the conditional taker–decliner achievement gap of -0.139SD (Table 4, column 3), with one standard deviation of A equal to $1/\sqrt{6}$. The box's gaps and shortfalls average the comparisons of Proposition D.9 over the targets in c 's range.

⁴¹With $q = 0$, the points are every $p \in \{0, 0.01, \dots, 0.99\}$ at $s = 0$, every $s \in \{0.0025, 0.005, \dots, 0.1225\}$ at $p = 0$, and every $p \in \{0.1, 0.2, \dots, 0.8\}$ with every $s \in \{0.01, 0.02, \dots, 0.1\}$; with $q = 0.2$, every $p \in \{0, 0.05, \dots, 0.95\}$ at $s = 0$, every $s \in \{0.005, 0.01, \dots, 0.12\}$ at $p = 0$, every $p \in \{0.1, 0.2, \dots, 0.8\}$ with every $s \in \{0.01, 0.02, \dots, 0.1\}$, and the three parameter pairs of Figure C.1.

⁴²With both programs adopting, the scan covers c 's floor on $[0, 0.56]$ and b 's on $[0.45, 0.80]$ and skips overlapping ranges, which Lemma D.7 rules out. At every point a solver started from five other points finds no further solution, and at ten points a search from 60 random starting admission thresholds, using the general forms of the clearing conditions, finds exactly one equilibrium.

The equilibria of Figure C.1.

1. The equilibrium is defined by unconditional-offer ranges. Programs a and d make no unconditional offers; every uninformed offer holder accepts, an informed one accepts only if she knows she would place at or below the adopter, and a share q decline regardless.

	(a) neither $p = 0, s = 0$	(b) private information $p = 0.104, s = 0$	(c) moral hazard $p = 0, s = 0.0567$	(d) both $p = 0.104, s = 0.0567$
c offers unconditionally to $v_1 \in$	[0.378, 0.516)	[0.384, 0.514)	[0.424, 0.500)	[0.439, 0.499)
b offers unconditionally to $v_1 \in$	[0.588, 0.719)	[0.611, 0.717)	—	—

2. Through market clearing, we derive conditional admission thresholds $\underline{\alpha}_k$; these yield take-up among offer holders, the taker–decliner achievement gap, and takers’ shortfall relative to the observable projection in c ’s range (Proposition D.9; b ’s in (b) are 0.763, -0.139 SD, 0.033SD); the share of each adopter’s intake arriving unconditionally; and mean realized intake ability relative to the conditional-only benchmark (1.529, 1.138, 0.862, 0.471).

	(a) neither	(b) private information	(c) moral hazard	(d) both
Admission threshold $\underline{\alpha}_a$	1.192	1.218	1.277	1.281
$\underline{\alpha}_b$	0.983	0.983	0.970	0.978
$\underline{\alpha}_c$	0.773	0.763	0.730	0.724
Take-up in c ’s range	0.800	0.761	0.800	0.759
Taker–decliner achievement gap (SD)	0	-0.139	-0.139	-0.278
Takers’ shortfall vs. projection (SD)	0	0.033	0.139	0.173
Unconditional share of intake, b / c	42% / 44%	32% / 40%	— / 24%	— / 18%
Mean intake ability vs. benchmark, a	-0.045	-0.032	-0.004	-0.003
b	-0.026	-0.025	-0.024	-0.017
c	$+0.050$	$+0.040$	$+0.006$	$+0.004$
d	$+0.021$	$+0.018$	$+0.008$	$+0.006$
Total attainment vs. benchmark	0	0	-0.003	-0.003

Every adopter’s intake also exceeds what it would be if it alone stopped offering. Relative to the benchmark, c and d gain while a and b lose in every panel; b offers in (a) and (b) and still loses, because c ’s offers capture the top of its band; it withdraws in (c) and (d) because $s = 0.0567$ exceeds its exit threshold, which at $q = 0.2$ is $\hat{s}_b(0) = 0.056$ and $\hat{s}_b(0.104) = 0.047$. Total attainment is fixed at $s = 0$ and falls by s times the mass of takers otherwise.

With $q = 0$, program b ’s unconditional-offer range does not vanish at $\hat{p}_b' = 0.443$, where its benchmark floor reaches its benchmark ceiling (Appendix D.6) and where it still supplies 11% of b ’s intake, but at $\hat{p}_b = 0.575$, the root of $V_b(\bar{x}_b) = 0$ at the lone-adopter equilibrium admission thresholds: b exits into a market where c still makes unconditional offers, which lower both $\underline{\alpha}_a$ and, by more, $\underline{\alpha}_b$, so that at b ’s ceiling the gain cell is larger and the loss cell smaller than at the benchmark. For c the two exits coincide, $\hat{p}_c = \hat{p}_c'$, because no unconditional-offer range remains at c ’s exit, so the last offer faces the benchmark admission thresholds. The exits in s behave the same way. At $p = 0$, c ’s range vanishes at $\hat{s}_c = \hat{s}_c' = 0.100$, the value of its benchmark ceiling target per taker, $G_c(\bar{x}_c)$; b ’s vanishes at $\hat{s}_b = 0.060$ at scale against $\hat{s}_b' = 0.036$ at the margin, again because c ’s remaining offers enlarge b ’s gain cell and shrink its loss cell. The exit threshold $\hat{s}_k(p)$ falls in p : at the margin it is $\hat{s}_k'(p) = [(1-p)g_k^2 - \ell_k^2] / [2(1-pg_k)]$ at the ceiling, and at scale $(\hat{s}_b, \hat{s}_c) = (0.051, 0.092)$ at $p = 0.104$ and $(0.042, 0.084)$ at $p = 0.2$.

D.4 Existence

Appendix D.3 computes the equilibrium at chosen values of p , s , and q , and finds exactly one at each. To show that an equilibrium exists at every value, we reduce it to two numbers, the widths of b 's and c 's bands. The loss and gain cells at each floor and ceiling, and so the masses of takers, depend on the admission thresholds only through these widths, so an equilibrium is a fixed point of a map from widths to widths.

Proposition D.5 (Existence). In the baseline economy, for every $p \in [0, 1]$, $s \geq 0$, and $q \in [0, 1]$, an equilibrium exists. In every equilibrium, (C1) and (C2) hold, the two unconditional-offer ranges are disjoint, and the ability A of a student at either ceiling target, or at any target in c 's range, can fall in any of the four bands, so Equations (19) and (20) describe it.

The cells at a program's floor depend on the admission thresholds only through the width of its band, because the loss and gain cells at any target sum to one minus that width and the floor condition involves only those cells. Write $w_b \equiv \underline{\alpha}_a - \underline{\alpha}_b$ and $w_c \equiv \underline{\alpha}_b - \underline{\alpha}_c$ for the widths of b 's and c 's bands, and $w \equiv (w_b, w_c)$.

Lemma D.6 (Floor cells). Fix $p \in [0, 1]$, $s \geq 0$, a program $k \in \{b, c\}$, and a width $w_k \in (0, 1)$ of its band, and let $\rho \equiv \sqrt{1-p}$. The floor condition $(1-p)g^2 - \ell^2 = 2s(1-pg)$, with $\ell + g = 1 - w_k$, has at most one root with $\ell, g > 0$, and if it has none, k makes no unconditional offers. If it has one, write $\ell = \underline{\ell}(w_k)$. Then:

- (i) $\ell \leq \rho g$, so $\ell \leq \frac{1-w_k}{2} \leq g$; and $s < g$.
- (ii) $\underline{\ell}$ is continuously differentiable, with slope in $(-1, 0)$, and $\underline{\ell}(w_k) \rightarrow 0$ as w_k approaches a width at which the root vanishes.

Proof. The difference of the two sides, $(1-p)g^2 - (1-w_k-g)^2 - 2s(1-pg)$, has derivative $2[(1-p)g + \ell + sp] > 0$ in g , so it has at most one root. If it has none, Equation (18) is negative at every target below the ceiling, where $\ell > 0$, and the program makes no offers (Lemma D.4). (i) Since $2s(1-pg) \geq 0$, $\ell^2 \leq (1-p)g^2$. If $s \geq g$, then $(1-p)g^2 > 2g(1-pg)$, so $g(1+p) > 2$, which is impossible since $g < 1$. (ii) Since the difference rises in g , the implicit function theorem makes $\underline{\ell}$ continuously differentiable, and differentiating the floor condition gives $-\underline{\ell}' = [(1-p)g + sp]/[(1-p)g + sp + \ell]$. This lies in $(0, 1)$ because $\ell > 0$ and $(1-p)g > 0$: at $p = 1$ the floor condition reads $-\ell^2 = 2s(1-g)$ and has no root. The difference is continuous and negative at $g = 0$, so the root can leave $(0, 1 - w_k)$ only through $g = 1 - w_k$, that is, with $\underline{\ell}(w_k) \rightarrow 0$. $\square$

For an adopter k , write $(\underline{\ell}_k, \underline{g}_k)$ and $(\bar{\ell}_k, \bar{g}_k)$ for the loss and gain cells at its floor and at its ceiling. Every target has $\ell_k + g_k = 1 - w_k$, so k 's range has length $L_k = \underline{\ell}_k - \bar{\ell}_k = \bar{g}_k - \underline{g}_k$. When Equation (20) applies,

$$\bar{\ell}_c = \omega_{ab}(1 - w_c) - \omega_a w_b, \quad \bar{\ell}_b = \omega_a(1 - w_b) + \omega_c w_c, \quad (21)$$

so $\bar{\ell}_b - \bar{\ell}_c = w_c - w_b$. By Lemma D.6, $\underline{\ell}_k = \underline{\ell}(w_k)$, where the function $\underline{\ell}$ depends on p and s and is the same for both programs, and k makes unconditional offers if and only if its floor lies below its ceiling, that is, $\underline{\ell}(w_k) > \bar{\ell}_k$ (Lemma D.4); then $L_k = \underline{\ell}(w_k) - \bar{\ell}_k$.

An offeree in c 's range with cells (ℓ, g) has $A < \underline{\alpha}_c$, $A \geq \underline{\alpha}_b$, and $A \geq \underline{\alpha}_a$ with probabilities ℓ, g , and $g - w_b$; one in b 's range has $A < \underline{\alpha}_c$, $A < \underline{\alpha}_b$, and $A \geq \underline{\alpha}_a$ with probabilities $\ell - w_c, \ell$, and g . Integrating these probabilities over the ranges, Equation (19) reads $\bar{F}(\underline{\alpha}_a) = \frac{1}{4} + X_a$, $\bar{F}(\underline{\alpha}_b) = \frac{1}{2} + X_b$, and $\bar{F}(\underline{\alpha}_c) = \frac{3}{4} - T_c$, where

$$\begin{aligned} T_c &= \frac{1-q}{2} \left[\underline{\ell}_c^2 - \bar{\ell}_c^2 + (\underline{\ell}_b - w_c)^2 - (\bar{\ell}_b - w_c)^2 \right], \\ X_b &= \frac{1-q}{2} \left[(1-p) \left(\bar{g}_c^2 - \underline{g}_c^2 \right) - \underline{\ell}_b^2 + \bar{\ell}_b^2 \right], \\ X_a &= \frac{(1-q)(1-p)}{2} \left[(\bar{g}_c - w_b)^2 - (\underline{g}_c - w_b)^2 + \bar{g}_b^2 - \underline{g}_b^2 \right], \end{aligned} \quad (22)$$

and a program that makes no unconditional offers contributes no terms. Here X_a is the mass of takers with $A \geq \underline{\alpha}_a$, T_c the mass of takers with $A < \underline{\alpha}_c$, and X_b the mass of c 's takers with $A \geq \underline{\alpha}_b$ less that of b 's takers with $A < \underline{\alpha}_b$. These masses depend on the admission thresholds only through w , so an equilibrium is a fixed point of the width map $\Phi = (\Phi_b, \Phi_c)$,

$$\Phi_b(w) = \bar{F}^{-1}\left(\frac{1}{4} + X_a\right) - \bar{F}^{-1}\left(\frac{1}{2} + X_b\right), \quad \Phi_c(w) = \bar{F}^{-1}\left(\frac{1}{2} + X_b\right) - \bar{F}^{-1}\left(\frac{3}{4} - T_c\right),$$

and the equilibrium's admission thresholds are the three values of $\bar{F}^{-1}$.

Equations (20) and (22) hold only if the two ranges are disjoint and the ability of a student at either ceiling target, or at any target in either range, can fall in any of the four bands, that is, only if these targets lie in $(\underline{\alpha}_a - 1, \underline{\alpha}_c)$. Bounding the two widths shows that this holds in every equilibrium.

Lemma D.7 (Where equilibria lie). In every equilibrium, w lies in

$$\begin{aligned} \Omega_0 &\equiv \{w : 0.025 \leq w_b \leq 0.362, \\ &\quad 0.156 - 0.2 w_b \leq w_c \leq \min\{0.34 - 0.07 w_b, 2 - \sqrt{2} - w_b\}\}, \end{aligned}$$

both ceilings lie in $(\underline{\alpha}_a - 1, \underline{\alpha}_c)$, $x_c > \underline{\alpha}_a - 1$, and the two unconditional-offer ranges are disjoint.

Proof. Let $n_b \equiv \bar{F}(\underline{\alpha}_b) - \bar{F}(\underline{\alpha}_a)$ and $n_c \equiv \bar{F}(\underline{\alpha}_c) - \bar{F}(\underline{\alpha}_b)$ be the masses of the two bands.⁴³

Band masses. Program c enrolls $\frac{1}{4}$: the students in its band, less those who take b 's offer, plus its own takers from outside its band. An offeree at a target t in c 's range has A uniform on $[t, t+1]$, which contains c 's band because $t < \underline{\alpha}_c$ and $t+1 - \underline{\alpha}_b \geq \underline{g}_c > 0$, so she lies outside the band with probability $1 - w_c$; an offeree of b lies in it with probability at most w_c . A student who holds both offers takes b 's, so c 's takers come from targets of total length at most L_c . Hence, and in the same way for b ,

$$\frac{1}{4} - (1 - w_c)L_c \leq n_c \leq \frac{1}{4} + w_c L_b, \quad \frac{1}{4} - (1 - w_b)L_b \leq n_b \leq \frac{1}{4} + w_b L_c. \quad (23)$$

⁴³The argument uses market clearing and best responses, not the forms of Equations (19) and (20).

Takers only raise $\underline{\alpha}_c$ and only lower $\underline{\alpha}_a$, so every admission threshold lies in $[\underline{\alpha}'_c, \alpha'_a] = [1/\sqrt{2}, 2 - 1/\sqrt{2}]$, and $w_b + w_c \leq 2 - \sqrt{2}$. On that interval the density f of A is at most one, concave, and symmetric about one, so a band of width y has mass at least $\underline{n}(y) \equiv \int_{1/\sqrt{2}}^{1/\sqrt{2}+y} f$, and $\underline{n}(w_k) \leq n_k \leq w_k$.

Range lengths. By Lemma D.6(i), $L_k \leq \frac{1-w_k}{2} - \bar{\ell}_k$. With $U(u_d) = 0$, the lottery utility at target x is $U(u_a)[\omega_c \Pr(A \geq \underline{\alpha}_c) + \omega_b \Pr(A \geq \underline{\alpha}_b) + \omega_a \Pr(A \geq \underline{\alpha}_a)]$. At c 's ceiling these probabilities are $1 - \bar{\ell}_c$, $\max\{1 - \bar{\ell}_c - w_c, 0\}$, and $\max\{1 - \bar{\ell}_c - w_c - w_b, 0\}$, so $W_c(\bar{x}_c) = U(u_c)$ reads $\omega_c \bar{\ell}_c = \omega_b \max\{1 - \bar{\ell}_c - w_c, 0\} + \omega_a \max\{1 - \bar{\ell}_c - w_c - w_b, 0\}$. The right side falls in $\bar{\ell}_c$ and is at least its value without the maxima, so $\bar{\ell}_c$ is at least its value in Equation (21), with equality when $\bar{x}_c > \underline{\alpha}_a - 1$. The same argument at b 's ceiling gives $\bar{\ell}_b = \min\{\omega_a(1 - w_b) + \omega_c w_c, \frac{\omega_a}{\omega_{ab}}(1 - w_b)\}$, with the first term exactly when $\bar{x}_b \leq \underline{\alpha}_c$, that is, when $w_c \leq \frac{\omega_a}{\omega_{ab}}(1 - w_b)$. Hence, with $\eta_c \equiv \frac{1}{2} - \omega_{ab}$ and $\eta_b \equiv \frac{1}{2} - \omega_a$, which are 0.131 and 0.334,

$$\begin{aligned} L_c &\leq \hat{L}_c, & \hat{L}_c &\equiv \eta_c(1 - w_c) + \omega_a w_b, \\ L_b &\leq \max\{\hat{L}_b, 0.051(1 - w_b)\}, & \hat{L}_b &\equiv \eta_b(1 - w_b) - \omega_c w_c. \end{aligned} \quad (24)$$

Bounds on w .

- (a) By Equations (23) and (24), $\underline{n}(w_b) \leq \frac{1}{4} + w_b(\eta_c + \omega_a w_b)$. The difference of the two sides rises in w_b , since $\underline{n}' \geq 1/\sqrt{2}$, and is positive at 0.37, so $w_b < 0.37$.
- (b) $w_c \geq n_c \geq \frac{1}{4} - (1 - w_c)\hat{L}_c$. The difference of the two sides rises in w_c , and at $w_c = 0.156 - 0.2 w_b$ it is a convex quadratic in w_b that is negative at 0 and at 0.37, so $w_c \geq 0.156 - 0.2 w_b$.
- (c) With (b), $\underline{n}(w_b) \leq \frac{1}{4} + w_b[\eta_c(0.844 + 0.2 w_b) + \omega_a w_b]$, which fails at $w_b = 0.362$ and, as in (a), above it.
- (d) $w_b \geq n_b \geq \frac{1}{4} - (1 - w_b)L_b$. If the second term of Equation (24) binds, $w_b > 0.19$. Otherwise, by (b), $w_b \geq \frac{1}{4} - (1 - w_b)[\eta_b(1 - w_b) - \omega_c(0.156 - 0.2 w_b)]$, which fails on $[0, 0.025]$: the difference of the two sides is a convex quadratic that is negative at 0 and at 0.025.
- (e) $\underline{n}(w_c) \leq n_c \leq \frac{1}{4} + w_c L_b$. The first term of Equation (24) gives $w_c \hat{L}_b \leq \eta_b^2(1 - w_b)^2/(4\omega_c)$. The second applies only if $w_c \geq \frac{\omega_a}{\omega_{ab}}(1 - w_b)$, which with $w_b + w_c \leq 2 - \sqrt{2}$ requires $w_b \leq 0.248$, and then gives less. So $\underline{n}(w_c) \leq \frac{1}{4} + \eta_b^2(1 - w_b)^2/(4\omega_c)$, and $w_c \leq 0.34 - 0.07 w_b$ because $\underline{n}(0.34 - 0.07 w_b)$ minus the right side is concave in w_b and positive at 0.025 and at 0.362.

So $w \in \Omega_0$.

Ceilings and floors. At target $\underline{\alpha}_a - 1$, c 's loss cell is $1 - w_b - w_c$ and the lottery utility is $U(u_a)[\omega_c(w_b + w_c) + \omega_b w_b]$. On Ω_0 , $\omega_c(1 - w_c) > (1 - \omega_a)w_b$, so this is below $U(u_c)$ and $\bar{x}_c > \underline{\alpha}_a - 1$. Also $w_c < \min\{\frac{\omega_a}{\omega_{ab}}(1 - w_b), 1 - 2w_b\}$, so $\bar{x}_b < \underline{\alpha}_c$ and, since $\underline{g}_c \geq \frac{1-w_c}{2} > w_b$, $\bar{x}_c > \underline{\alpha}_a - 1$. The probabilities behind Equation (22) are then exact, and Equation (21) holds with equality.

Disjoint ranges. The ranges overlap if and only if $\underline{x}_b < \bar{x}_c$, that is, $\underline{\ell}(w_b) > w_c + \bar{\ell}_c$, which by Lemma D.6(i) requires $\omega_c w_c + \eta_b w_b < \eta_c$. A student who holds both offers takes b 's, so c 's takers come from $[\underline{x}_c, \underline{x}_b]$, of length $\underline{\ell}(w_c) - \underline{\ell}(w_b) + w_c$, and b 's takers include $(1-q) w_c L_b$ in c 's band. As for Equation (23), $n_c \geq \frac{1}{4} - (1-q)[(1-w_c)(\underline{\ell}(w_c) - \underline{\ell}(w_b) + w_c) - w_c(\underline{\ell}(w_b) - \bar{\ell}_b)]$, which rises in $\underline{\ell}(w_b)$. Setting $\underline{\ell}(w_b) = w_c + \bar{\ell}_c$ and $\underline{\ell}(w_c) \leq \frac{1-w_c}{2}$ gives $w_c \geq n_c \geq \frac{1}{4} - \max\{(1-w_c)\bar{L}_c - \omega_b w_c, 0\}$. The difference of the two sides rises in w_c , and at $w_c = (\eta_c - \eta_b w_b)/\omega_c$, where the overlap condition binds, it is a convex quadratic in w_b that is negative at 0.025 and at η_c/η_b , beyond which the ranges cannot overlap. So on Ω_0 the ranges are disjoint. $\square$

The Poincaré–Miranda theorem gives a fixed point of a map on a four-sided region if, on each side, the map does not push the width that side constrains outward. Ω_0 has a fifth side, $w_b + w_c = 2 - \sqrt{2}$, so we work on a larger region. Let $\Omega \equiv \{w : 0.025 \leq w_b \leq 0.362, 0.156 - 0.2 w_b \leq w_c \leq 0.34 - 0.07 w_b\} \supseteq \Omega_0$, and let $\tilde{\Phi}$ be the map on Ω given by the formulas for Φ , with ceiling cells from Equation (21) and takers from Equation (22), whether or not the conditions of Lemma D.7 hold. By Lemma D.7, every equilibrium is a fixed point of $\tilde{\Phi}$ in Ω_0 .

Lemma D.8 (The width map). For every $p \in [0, 1]$, $s \geq 0$, $q \in [0, 1]$, and $w \in \Omega$:

(i) $X_a \geq 0$, $T_c \geq 0$, and $|X_b| \leq 0.1$.

(ii) $\tilde{\Phi}_c(w) \in [0.156 - 0.2 w_b, 0.34 - 0.07 w_b]$; $\tilde{\Phi}_b(w) \geq 0.025$ if $w_b = 0.025$, and $\tilde{\Phi}_b(w) \leq 0.362$ if $w_b = 0.362$.

Proof. Part (i). On Ω , $X_a \geq 0$, because $\underline{g}_c \geq \frac{1-w_c}{2}$ and $\bar{g}_c + \frac{1-w_c}{2} > 2w_b$. Only the b -terms of T_c , $(1-q)L_b[\frac{1}{2}(\underline{\ell}_b + \bar{\ell}_b) - w_c]$, can be negative, and only if $\bar{\ell}_b < w_c$, which on Ω requires $w_b > 0.288$ and $w_c > 0.286$, both above ω_b . Since $\underline{\ell}$ has slope in $(-1, 0)$ (Lemma D.6(ii)), there $L_c \geq \underline{\ell}(w_b) - \max\{w_c - w_b, 0\} - \bar{\ell}_b + w_c - \omega_b > L_b$, so $T_c \geq (1-q)[L_c \bar{\ell}_c + L_b(\bar{\ell}_b - w_c)] \geq (1-q)L_b(2\bar{\ell}_c - \omega_b) \geq 0$, as $\bar{\ell}_c > \omega_b/2$ on Ω . By Lemma D.6(i), X_b lies between $-\frac{1}{2}\hat{L}_b(\frac{1-w_b}{2} + \bar{\ell}_b)$ and $\frac{1}{2}\hat{L}_c(\bar{g}_c + \frac{1-w_c}{2})$; each factor is monotone in w_b and w_c , and the products of their maxima over Ω are below 0.1, so $|X_b| \leq 0.1$. The admission thresholds that $\tilde{\Phi}$ assigns therefore lie in $[1/\sqrt{2}, 2 - 1/\sqrt{2}]$.

Part (ii). The new bands of c and b have widths $\tilde{\Phi}_c(w)$ and $\tilde{\Phi}_b(w)$ and masses $\frac{1}{4} - T_c - X_b$ and $\frac{1}{4} + X_b - X_a$, which satisfy Equation (23) with $L_k \leq \hat{L}_k$. Steps (b) and (e) of the proof of Lemma D.7 then bound $\tilde{\Phi}_c(w)$, and steps (d) and (c) bound $\tilde{\Phi}_b(w)$ at the two ends. $\square$

Proof of Proposition D.5. Let $\Psi(w) \equiv w - \tilde{\Phi}(w)$ on Ω , a continuous map.⁴⁴ By Lemma D.8(ii), $\Psi_b \leq 0$ on the left edge of Ω , where $w_b = 0.025$, and $\Psi_b \geq 0$ on the right edge, where $w_b = 0.362$; and $\Psi_c \leq 0$ on the lower edge, where $w_c = 0.156 - 0.2 w_b$, and $\Psi_c \geq 0$ on the upper edge, where $w_c = 0.34 - 0.07 w_b$.

⁴⁴By Lemma D.6(ii), each floor cell $\underline{\ell}(w_k)$ is continuous where it exists and tends to zero where it vanishes. Since $\bar{\ell}_k > 0$ on Ω (Equation (21)), a program therefore starts or stops making unconditional offers only where $L_k = 0$, and there its terms in Equation (22) vanish.

Rescaling w_c linearly for each w_b , so that the lower edge goes to 0 and the upper edge to 1, maps Ω homeomorphically onto the rectangle $[0.025, 0.362] \times [0, 1]$, edges to sides, so by the Poincaré–Miranda theorem Ψ has a zero in Ω .⁴⁵

Let w^* be a zero of Ψ in Ω . There $X_a, T_c \geq 0$ (Lemma D.8(i)), so $w_b^* + w_c^* \leq 2 - \sqrt{2}$ and $w^* \in \Omega_0$. The steps of the proof of Lemma D.7 on ceilings and floors and on disjoint ranges then apply at w^* : the first uses only $w \in \Omega_0$, and the bound on c 's band mass in the second also holds for the mass $\frac{1}{4} - T_c - X_b$ that $\tilde{\Phi}$ assigns, since $L_c \leq \hat{L}_c$ and an overlap would make $L_b > \omega_b$. So at w^* the ranges are disjoint, the formulas describe actual takers, and w^* is an equilibrium. The other claims hold in every equilibrium by Lemma D.7: (C1) because both widths are below one, and (C2) because every target in a range has cells in $(0, 1)$. $\square$

D.5 The taker–decliner achievement gap

In Proposition C.1, programs choose their unconditional-offer ranges knowing p and s . Even so, the students who take an offer differ from what their observables predict, and from those who decline it. Fix an adopter k and a target x in its range, and write $g_k(x)$ for the length of the gain cell, the share of offerees at x who would have placed above k .

Proposition D.9 (The taker–decliner achievement gap). In any equilibrium of Proposition C.1, at every target x in adopter k 's unconditional-offer range:

- (i) Take-up is $(1 - q)(1 - pg_k(x))$: private information lowers it, moral hazard does not.
- (ii) Takers fall short of the ability their observables project by

$$\underbrace{\frac{pg_k(1 - g_k)}{2(1 - pg_k)}}_{\text{private information}} + \underbrace{s}_{\text{moral hazard}},$$

where the first term is positive if and only if $p > 0$. Either force alone produces a shortfall, and, at given admission thresholds, the same shortfall arises from a continuum of (p, s) pairs.

- (iii) If $p > 0$ or $q > 0$, so that some holders decline, takers score below decliners by

$$\Delta_k(x) = \underbrace{\frac{pg_k(1 - g_k)}{2} \left[\frac{1}{1 - pg_k} + \frac{1 - q}{q + (1 - q)pg_k} \right]}_{\text{private information}} + \underbrace{s}_{\text{moral hazard}}.$$

If $q > 0$, then at given admission thresholds the same gap arises from a continuum of (p, s) pairs.

⁴⁵If $f = (f_1, f_2)$ is continuous on a rectangle $[a_1, b_1] \times [a_2, b_2]$, with $f_i \leq 0$ where $x_i = a_i$ and $f_i \geq 0$ where $x_i = b_i$, then f has a zero in the rectangle. See Kulpa (1997).

- (iv) Each adopter's floor breaks even in realized ability, so its range already accounts for the shortfall. A program above the highest adopter enrolls a weaker intake than in the benchmark whenever a positive mass of uninformed takers has ability in its benchmark band; under the setup of [Figure C.1](#) this holds for every program above the highest adopter, and, by computation, b 's intake is also weaker when it adopts. Total attainment falls by s times the mass of takers.

Under private information the decliners include the informed students who would have placed above k ; under moral hazard alone they are the q who decline for other reasons, and the gap is s . At the calibration of [Figure C.1](#), $p = 0.104$ alone and $s = 0.0567$ alone each produce a gap of -0.139SD , and together -0.278SD . Private information redistributes: at $p = 0.104$, c gains 0.040 in mean intake ability (0.050 with neither force), a loses 0.032, and b loses 0.025 although it adopts, because c 's offers capture the top of b 's band. Moral hazard also lowers the total: at $s = 0.0567$, c 's gain from adopting shrinks to 0.006, b withdraws and still loses 0.024, and total attainment falls by s times the mass of takers (the box in [Appendix D.3](#)).

D.6 Proofs of the propositions

Proof of [Proposition C.1](#). Step 1: who adopts. Programs a and d make no offers by the remark after [Equation \(18\)](#). For $k \in \{b, c\}$, [Equation \(18\)](#) gives $V_k \geq 0$ if and only if

$$(1 - p)g_k(x)^2 - \ell_k(x)^2 \geq 2s(1 - pg_k(x)),$$

which at $s = 0$ is $\sqrt{1 - p}g_k(x) \geq \ell_k(x)$, the gain cell at least $1/\sqrt{1 - p}$ times the loss cell. We take $s = 0$ for the closed forms and return to s in Step 2. Since V_k is strictly increasing in x below the ceiling and nonpositive above it, the strictly profitable targets form the interval $(\underline{x}'_k(p), \bar{x}'_k)$ with the floor

$$\underline{x}'_k(p) = \frac{\underline{\alpha}'_k - \sqrt{1 - p}(1 - \bar{\alpha}'_k)}{1 + \sqrt{1 - p}},$$

the break-even target, excluded by convention. At $p = 0$ the floor is $\underline{x}'_k = \frac{1}{2}(\underline{\alpha}'_k + \bar{\alpha}'_k) - \frac{1}{2}$, the target whose two cells are equal; at the baseline, $\underline{x}'_c = 0.354 < \bar{x}'_c = 0.495$ and $\underline{x}'_b = 0.646 < \bar{x}'_b = 0.698$, so both middle programs adopt (C3).⁴⁶ The two ranges are disjoint, c 's below b 's: $\bar{x}'_c < \underline{x}'_b$ at the benchmark, and $\bar{x}_c < \underline{x}_b$ at every computed equilibrium ([Appendix D.3](#)). These are the unconditional-offer ranges of the marginal analysis; the equilibrium ranges differ, because they shift the admission thresholds (b 's is wider and c 's narrower at $p = s = 0$), and are reported in [Appendix D.3](#).

Step 2: shrinking and exit. The floor is strictly increasing in s and in p . It solves

⁴⁶(C3) is a joint condition on preferences and geometry: a low enough u_k pushes $\bar{x}'_k$ below the floor and removes k from the adopter set, and for some utility vectors so does weak enough risk aversion, though not for $u = (3, 2.5, 2, 1)$, where the ceilings stay above the floors even for a risk-neutral student. It fails for a under every preference, since a 's gain cell is empty and no target is profitable for a (the remark after [Equation \(18\)](#)).

$h(x) \equiv (1-p)g_k(x)^2 - \ell_k(x)^2 - 2s(1-pg_k(x)) = 0$, where h is increasing in x and decreasing in s , and $\partial h / \partial p = -g_k(g_k - 2s)$, which is negative at the root because there $g_k - 2s = [g_k(1-g_k) + \ell_k^2] / (1-pg_k) > 0$. The ceiling depends on neither force, so the range shrinks from below as either force grows. At $s = 0$ it is nonempty exactly while the ceiling target still pays, $\sqrt{1-p}g_k(\bar{x}'_k) > \ell_k(\bar{x}'_k)$, that is,

$$p < \hat{p}'_k = 1 - \left[\frac{\ell_k(\bar{x}'_k)}{g_k(\bar{x}'_k)} \right]^2.$$

At the baseline, $\hat{p}'_b = 0.443$ and $\hat{p}'_c = 0.816$, so the ladder is $\{b, c\} \rightarrow \{c\} \rightarrow \emptyset$. Program c 's exit is exact at any scale, $\hat{p}_c = \hat{p}'_c$, because b exits first at the baseline, so no unconditional-offer range remains at c 's exit and the marginal offer faces the benchmark admission thresholds; b 's is not, since b exits into a market where c still makes unconditional offers, and [Appendix D.3](#) computes $\hat{p}_b = 0.575$. In s , the ceiling target pays exactly while $s < \hat{s}'_k(p) = [(1-p)g_k^2 - \ell_k^2] / [2(1-pg_k)]$ evaluated at $\bar{x}'_k$, an exit threshold that is positive exactly for $p < \hat{p}'_k$ and strictly decreasing in p , since

$$\frac{d\hat{s}'_k}{dp} = -\frac{g_k[g_k(1-g_k) + \ell_k^2]}{2(1-pg_k)^2} < 0;$$

at $p = 0$ it is $G_k(\bar{x}'_k)$, 0.100 for c and 0.036 for b . As with p , c 's exit threshold is exact at scale and b 's is not: [Appendix D.3](#) computes $\hat{s}_b = 0.060$ at $p = 0$.

Step 3: no offers beyond the exit, at any scale. At $p = 1$ every holder is informed, so by [Lemma D.3](#), at any admission thresholds and every target, $V_k = (1-q) \left[\int_0^1 \min\{\varphi_k(x + v_2), 0\} dv_2 - s\tau_k(x) \right] \leq 0$. No target is strictly profitable, so no program adopts, and the outcome is [Lemma D.1](#).⁴⁷ For $p \geq \hat{p}_c$, or $s \geq \hat{s}_c(p)$ for each k , the offer at each program's benchmark ceiling has nonpositive value (for b because $\hat{p}'_b < \hat{p}_c$ and, at every p we compute, $\hat{s}'_b(p) < \hat{s}_b(p)$), so by [Lemma D.4](#) the benchmark is an equilibrium, and at every such point we compute it is the only one ([Appendix D.3](#)).

Positive scale. Steps 1 and 2 establish the shape of the ranges at the margin, for the exit thresholds $\hat{p}'_k$ and $\hat{s}'_k$. At positive scale, the two ranges in (i) are disjoint by [Lemma D.7](#), and in (ii) the facts that the ranges are longest at $p = s = 0$, that b offers if and only if $s < \hat{s}_b(p)$, that c offers if and only if $s < \hat{s}_c(p)$ (which needs the computed fact that b never offers alone), that no program offers beyond the exit thresholds (Step 3), and that $\hat{s}_b$ falls in p , rest on the computation in [Appendix D.3](#), which covers $q = 0$ and the setup of [Figure C.1](#), $q = 0.2$. $\square$

Remark (the order of exit). The order of exit is the same under both forces, and one number decides it. With equal capacities the bands have a common width $\bar{w} = \underline{\alpha}'_a - \underline{\alpha}'_b = \underline{\alpha}'_b - \underline{\alpha}'_c$, so at any target $\ell_k + g_k = 1 - \bar{w}$ for $k \in \{b, c\}$, and both exits are decreasing functions of the loss cell at the ceiling alone: $\hat{p}'_k = 1 - [\ell_k / (1 - \bar{w} - \ell_k)]^2$, and $\hat{s}'_k(p)$ has numerator $(1-p)(1 - \bar{w} - \ell_k)^2 - \ell_k^2$, falling in

⁴⁷If v_2 were public, an unconditional offer could condition on A and would be a conditional offer, so [Lemma D.1](#) would again apply.

ℓ_k , over denominator $2[1 - p(1 - \bar{w} - \ell_k)]$, rising in it. Hence b exits first, in p and in s at every p , if and only if $\ell_b(\bar{x}'_b) > \ell_c(\bar{x}'_c)$: its ceiling lies deeper below its own admission threshold, because the insurance it sells, against falling to c , is worth less to an uninformed student than the insurance c sells, against falling to d . At the baseline $0.302 > 0.212$; the inequality, and with it both orders at once, reverses at $u_b = 2.63$. The nesting is a consequence of equal capacities, not of the two forces. The s -intercept $G_k(\bar{x}'_k) = \frac{1}{2}(g_k^2 - \ell_k^2)$ scales with the size of the cells, while the p -intercept $1 - (\ell_k/g_k)^2$ depends only on their ratio; with a common $\ell_k + g_k$ the two move together. With unequal capacities $\ell_k + g_k$ differs across programs, a program can have both cells longer than the other's, and the exit thresholds can cross.⁴⁸

Proof of Proposition D.9. Fix a target x in k 's range, so $x < \bar{x}_k$, with cells evaluated at the equilibrium admission thresholds (Lemma D.2(iii)). Of the holders at x , a share q decline regardless; of the rest, the uninformed accept, and the informed accept exactly on the loss and neutral cells, $v_2 < 1 - g_k$, by Lemma D.2(ii). Take-up is therefore $(1 - q)[(1 - p) + p(1 - g_k)] = (1 - q)(1 - pg_k)$, which is (i); s does not enter because a taker's score is realized after she accepts.

For (ii), v_2 is uniform on $[0, 1]$ at every target, so the observable projection of a holder's v_2 is $\frac{1}{2}$. Takers have mass $(1 - q)(1 - pg_k)$ and mean

$$v_2 = \frac{(1 - p)\frac{1}{2} + p\frac{(1 - g_k)^2}{2}}{1 - pg_k} = \frac{1}{2} - \frac{pg_k(1 - g_k)}{2(1 - pg_k)},$$

and score s less than that. The private-information part is positive exactly when $p > 0$, since $g_k \in (0, 1)$ by (C2); at given admission thresholds, so that g_k is fixed, the shortfall is continuous and strictly increasing in both p and s , so each of its level sets in the (p, s) plane is a curve.

For (iii), decliners have mass $q + (1 - q)pg_k$: the q who decline regardless, with mean v_2 of $\frac{1}{2}$, and the informed on the gain cell, with mean $1 - g_k/2$; their mean

$$v_2 = \frac{\frac{q}{2} + (1 - q)pg_k(1 - \frac{g_k}{2})}{q + (1 - q)pg_k} = \frac{1}{2} + \frac{(1 - q)pg_k(1 - g_k)}{2(q + (1 - q)pg_k)}.$$

Subtracting the takers' realized mean gives $\Delta_k(x)$. At $p = 0$ the decliners are the q alone and the gap is s ; at $q = 0$ the private-information term is $(1 - g_k)/[2(1 - pg_k)]$, which does not vanish as $p \rightarrow 0$, because the only decliners are then the informed on the gain cell however few they are. For $q > 0$ and at given admission thresholds, the gap Δ_k is continuous and strictly increasing in both p and s , so the same gap arises along a curve of (p, s) pairs.

For (iv), the floor condition $V_k(\underline{x}_k) = 0$ of Lemma D.4 is stated in realized values, so the equilibrium range already nets out both the informed decliners and the s . Let j be a program above the highest adopter. Only the captured-from-above sum en-

⁴⁸At capacities $(0.4, 0.2, 0.2, 0.2)$ and $u_b = 2.6$, c exits first in s ($\hat{s}_c(0) = 0.089 < \hat{s}_b(0) = 0.100$) but b exits first in p (the exit thresholds cross near $p = 0.6$), marginally and at scale alike; such crossings occur only within a narrow band of u_b around the point where the ordering flips.

ters j 's condition in Equation (19), so $\underline{\alpha}_j < \underline{\alpha}'_j$ (Appendix D.3), and the same holds for every program above j , so $\bar{\alpha}_j \leq \bar{\alpha}'_j$. Its intake is therefore a set of mass $\frac{1}{4}$ inside $\{A < \bar{\alpha}'_j\}$ that excludes, by hypothesis, a positive mass of students in its benchmark band $[\underline{\alpha}'_j, \bar{\alpha}'_j)$: uninformed takers of adopters below j . Under the setup of Figure C.1 such takers exist for every such j , because at every equilibrium we compute, every target in an adopter's range exceeds $\underline{\alpha}'_a - 1 = 1 - \sqrt{1/2}$, so an uninformed taker's ability, uniform on $[x, x + 1]$, reaches every benchmark band above the adopter's. The benchmark intake $[\underline{\alpha}'_j, \bar{\alpha}'_j)$ is the unique mass- $\frac{1}{4}$ subset of $\{A < \bar{\alpha}'_j\}$ with the highest mean, so j 's mean realized ability is below it. For b as an adopter the comparison is numerical: its mean intake ability is below the benchmark in panels (a) and (b) of Figure C.1 and at every p in the table of Appendix D.3. Finally, every student enrolls somewhere and capacities sum to one, so the sum of realized intake abilities is the total ability of the population less s times the mass of takers. $\square$

E Optimality of the Application Portfolio under Correlated Admissions

The Stage-1 estimator of [Section 4.3](#) requires the observed portfolio to satisfy the one-shot-swap (OSS) inequalities of [Larroucau and Rios \(2020\)](#). As noted in [Section 4.3.2](#), under nested admission ([Assumption 2](#)) these inequalities are necessary for optimality but not sufficient; the exact conditions are the dynamic-programming inequalities of [Ali and Shorrer \(2025\)](#). This appendix writes both sets of inequalities in one notation, characterizes the gap between them, and uses it to justify our use of the OSS inequalities in estimation.

E.1 Setup

Fix a student and drop the subscript i . Following [Implication 1](#), program s has Stage-1 utility $u_s \equiv U_{is}$ and admission probability $q_s \equiv q_{is}$, the joint probability that s makes her an offer and admits her ([Section 4.3.1](#)). As in [Implication 1](#), admission events are nested: a student admitted to a program is also admitted to every program with a higher q_s .

Nested events with these probabilities are generated by a single score. Let S be uniform on $[0, 1]$, and let the student be admitted to s if and only if $S < q_s$, so a more selective program has a lower q_s . Because one score governs every admission, rejection is informative. After rejection by program s' , the score is uniform on $[q_{s'}, 1]$, so for any less selective program s ($q_s > q_{s'}$) the admission probability given rejection by s' is

$$\frac{q_s - q_{s'}}{1 - q_{s'}}.$$

After rejection by every program in a set $\mathcal{R}$, the score is uniform on $[\max_{s \in \mathcal{R}} q_s, 1]$, which depends only on the least selective program that rejected the student.

For a portfolio $A = \{s_1, \dots, s_m\}$ with $q_{s_1} \leq \dots \leq q_{s_m}$, and a student who knows only that $S \geq t$, the expected value of the best admission is

$$V(A; t) = \mathbb{E} \left[\max_{s \in A, S < q_s} u_s \mid S \geq t \right] = \frac{1}{1-t} \sum_{l=1}^m [\max\{q_{s_l}, t\} - \max\{q_{s_{l-1}}, t\}] \max_{j \geq l} u_{s_j}, \quad (25)$$

with $q_{s_0} \equiv 0$. At $t = 0$ this is the portfolio value of [Implication 1](#). Let $A^* = \{s_1^*, \dots, s_K^*\}$ be the observed portfolio, indexed so that $q_{s_1^*} < \dots < q_{s_K^*}$; by [Implication 2](#), $u_{s_1^*} > \dots > u_{s_K^*}$, so the inner maximum in [Equation \(25\)](#) is $u_{s_l^*}$. Let $\tilde{V}(n; t) = \max_{|A|=n} V(A; t)$ be the value of the best n -program portfolio for a student who knows that $S \geq t$, and let $A_{m+1:K}^* = \{s_{m+1}^*, \dots, s_K^*\}$ be the observed programs less selective than s_m^* .

E.2 The two sets of inequalities

E.2.1 Larroucau–Rios one-shot swaps (LR)

When admissions are independent across programs, the Marginal Improvement Algorithm of [Chade and Smith \(2006\)](#) solves the portfolio problem, and [Larroucau and Rios \(2020\)](#) show that optimality of A^* can be characterized by a polynomial-sized set of one-shot-swap inequalities: for each

$s \in A^*$ and each $\ell \notin A^*$,

$$V(A^*; 0) \geq V((A^* \setminus \{s\}) \cup \{\ell\}; 0). \quad (\text{LR})$$

These $K(|\mathcal{S}| - K)$ inequalities compare portfolios that differ in a single program, each valued at the prior. They are the inequalities of [Equation \(16\)](#), which our estimator imposes with V evaluated under nested admission, as here.

E.2.2 Ali-Shorrer dynamic programming (AS)

Under nested admission, the optimal portfolio instead satisfies the dynamic-programming recursion of [Ali and Shorrer \(2025\)](#): at each level $m \in \{1, \dots, K\}$, the choice s_m^* must be optimal given rejection by all more selective programs already chosen. Let $t_m \equiv q_{s_{m-1}^*}$, with $q_{s_0^*} \equiv 0$. Then, for each level m and each alternative j with $q_j > t_m$,

$$\begin{aligned} \frac{q_{s_m^*} - t_m}{1 - t_m} u_{s_m^*} + \frac{1 - q_{s_m^*}}{1 - t_m} \tilde{V}(K - m; q_{s_m^*}) \\ \geq \frac{q_j - t_m}{1 - t_m} u_j + \frac{1 - q_j}{1 - t_m} \tilde{V}(K - m; q_j). \end{aligned} \quad (\text{AS})$$

The left-hand side is the value of choosing s_m^* at level m : it admits the student with probability $(q_{s_m^*} - t_m)/(1 - t_m)$, and upon rejection she continues *optimally* knowing that $S \geq q_{s_m^*}$. The right-hand side is the same value for choosing j in place of s_m^* .

E.3 The gap

The LR inequalities hold the observed tail $A_{m+1:K}^*$ fixed, while the AS inequalities reoptimize the continuation after rejection. To compare them, condition on rejection by $s_1^*, \dots, s_{m-1}^*$, which both portfolios share, and write the swap of s_m^* for j with the tail held fixed:

$$\begin{aligned} \frac{q_{s_m^*} - t_m}{1 - t_m} u_{s_m^*} + \frac{1 - q_{s_m^*}}{1 - t_m} V(A_{m+1:K}^*; q_{s_m^*}) \\ \geq \frac{q_j - t_m}{1 - t_m} u_j + \frac{1 - q_j}{1 - t_m} V(A_{m+1:K}^*; q_j). \end{aligned} \quad (\text{LR}')$$

When $t_m < q_j \leq q_{s_{m+1}^*}$ and $u_{s_{m+1}^*} \leq u_j \leq u_{s_{m-1}^*}$, (LR') is the one-shot swap of s_m^* for j in [Equation \(LR\)](#), divided by $1 - t_m$; the upper bound on u_j holds whenever A^* satisfies [Equation \(LR\)](#), since otherwise j would dominate s_{m-1}^* . Comparing [Equation \(LR'\)](#) with [Equation \(AS\)](#), the right-hand side of (AS) exceeds that of (LR') by

$$\Delta_m(j) \equiv \frac{1 - q_j}{1 - t_m} [\tilde{V}(K - m; q_j) - V(A_{m+1:K}^*; q_j)] \geq 0, \quad (26)$$

since $\tilde{V}(K - m; q_j)$ is the value of the *optimal* $(K - m)$ -program portfolio given rejection by j , which weakly exceeds the value of the fixed tail. The left-hand sides coincide when A^* satisfies (AS) at every level: by backward induction the observed tail is then the optimal continuation, $\tilde{V}(K - m; q_{s_m^*}) = V(A_{m+1:K}^*; q_{s_m^*})$. Hence (AS) implies (LR'), and the slack in (LR') exceeds the slack in (AS)

by exactly $\Delta_m(j)$: the OSS inequalities check only whether a single swap improves on the observed portfolio holding the rest fixed, whereas the exact condition lets the continuation be reoptimized after the swap.

The gap arises only because rejection is informative: when admissions are independent, rejection carries no information, and the OSS inequalities characterize the optimum exactly (Larroucau and Rios, 2020). Under nested admission, $\Delta_m(j)$ is scaled by $(1 - q_j)/(1 - t_m)$, the probability that program j rejects the student given rejection by s_{m-1}^* . Because the bracket in Equation (26) is at most $\max_s u_s$, $\Delta_m(j) \leq \frac{1-q_j}{1-t_m} \max_s u_s$, which is small for safety programs with high admission probability, whatever the reoptimization gain.

Rejection by a program with admission probability t moves the chance of admission to any other program by at most t : for $q > t$ the change is $t(1-q)/(1-t) \leq t$, and for $q \leq t$ the chance falls from q to zero. Because students condition on rejection by their most selective programs, where $\hat{q}_{is}$ is lowest, the posteriors in Equation (AS) stay close to the prior, and the OSS inequalities closely approximate the exact conditions. Equivalently, one may interpret the estimator as identifying preferences from the requirement that the observed portfolio be a *local* optimum of the portfolio problem (for which the OSS inequalities are necessary) rather than a global one, with Equation (26) bounding the difference between the two.

F Structural Model Estimation Details

This appendix records how the demand-side estimator of Section 4.3 is implemented: the construction of the revealed-preference cone, the Gibbs sampler, the treatment of estimated admission beliefs, and the warm start. We write $\mathcal{L}_i(p)$ for the matrix of student i 's revealed-preference inequalities (the constraints $\mathcal{L}_i$ of Section 4.3.2) evaluated at the offer probabilities p , so the augmented utilities satisfy the cone $\mathcal{L}_i(p) U_i > 0$.

F.1 Constraint construction

For each student the observed portfolio implies three blocks of rows (Section 4.3.2). The ranking rows order utilities inversely to the admission probabilities $\hat{q}_{is}$ (Implication 2). The positivity rows require $U_{is} > 0$ for $s \in A_i$. For a student with five programs, the one-shot-swap rows of Equation (16) compare A_i with every portfolio that replaces one chosen program by one non-chosen program, so the comparison set is the complement of the portfolio rather than the set of all portfolios; each swap row involves the chosen programs and exactly one non-chosen alternative, which has a single negative coefficient. For a student who applies to fewer than five programs, the addition rows instead compare A_i with every portfolio that adds one non-chosen program ℓ ; under nested admission each reduces to $U_{i\ell} \leq U_{is}$, where s is the chosen program just less selective than ℓ , or to $U_{i\ell} \leq 0$ if ℓ is less selective than every chosen program. The swap and addition rows use the total acceptance probabilities $\hat{q}_{is} = \hat{p}_{is}[\hat{g}_{is} + (1 - \hat{g}_{is})\hat{a}_{is}^o]$ of Section 4.3.1, built from the Firth-penalized offer and conversion probits, so both the offer and the conversion probability enter the portfolio stage, as Implication 1 requires. For the ranking stage we also form, for each observed pair of ranked choices, the probability that the second choice converts given that the first does not, by truncating the common μ_i above the first choice's threshold in Equation (8).

F.2 Gibbs sampler

We draw from the joint posterior of $(\delta, \bar{\delta}, \sigma_\delta^2, \bar{\beta}^c, \Gamma^c, \sigma_\eta^2, \sigma_v^2, \{\eta_i^c\}, \{U_i\})$ by a blocked sampler. Each sweep alternates:

- (a) **Utilities** $U_i \mid \delta, \beta_i^c, \sigma_v^2$, truncated to $\mathcal{L}_i(\hat{p}) U_i > 0$: conditional on the chosen utilities the alternatives are independent and drawn as exact one-dimensional truncated normals, while the (at most five) chosen utilities are drawn jointly by hit-and-run within the cone. The program fixed effects δ and the coefficient $\beta_i^c = \bar{\beta}^c + \Gamma^c \text{Low-SES}_i + \eta_i^c$ enter only through this draw's mean.
- (b) **Program fixed effects** $\delta \mid U, \beta^c, \sigma_v^2$: a conjugate Normal-means update. The residual $U_{is} + 0.01 d_{is} - \beta_i^c \text{close}_{is} - \beta^m m_{is}$ is a noisy observation of δ_s with precision σ_v^{-2} , so δ_s is drawn from a Normal with posterior precision $n_s \sigma_v^{-2} + \sigma_\delta^{-2}$, where n_s counts the latent utilities in which s appears. The hyperparameters of the prior $\delta_s \sim N(\bar{\delta}, \sigma_\delta^2)$ are themselves drawn (Normal and inverse-Gamma), shrinking thinly-applied-to programs toward the population mean rather than letting them take extreme values fit to a handful of students.
- (c) **Preference parameters**: η_i^c (Normal), $(\bar{\beta}^c, \Gamma^c)$ in one joint Normal draw, σ_v^2 (inverse-Gamma), and σ_η^2 under the half- t prior described below. The nesting of Equation (13) is imposed by

masking: η^c and σ_η^2 are sampled only in the specification carrying an i.i.d. component, and Γ^c only where the close coefficient loads on Low-SES status.

The cone is homogeneous in U , so the scale is pinned by fixing the linear distance coefficient at $\beta^d = -0.01$ and estimating σ_v^2 freely, which denominates utilities in units of 100 kilometers of travel. For the heterogeneity scales we use a weakly informative half- t prior and update the $(\eta_i^c, \sigma_\eta^2)$ block by an Ancillarity-Sufficiency Interweaving Strategy (ASIS). The same interweave is applied to the $(\delta, \sigma_\delta^2)$ block to improve mixing for both popular programs and unpopular programs.

F.3 Estimated beliefs

On real data $\mathcal{L}_i(\hat{q})$ is built from the estimated joint acceptance probabilities $\hat{q}_{is}$ of [Section 4.3.1](#), which enter the row *coefficients*, so sampling error that reorders two nearby programs mis-specifies rows. As in [Section 4.3.2](#) we treat q as latent and resample it jointly with U , keeping every utility draw over an exact cone. We augment the sampler with an independence-Metropolis step: propose q' , recomputed from the same joint formula, by drawing the offer-probit parameters (γ_s, c_s) from their asymptotic distribution, holding the conversion-probit parameters $(\gamma_s^{\text{admit}}, \alpha_s, c_s^R)$ fixed at their point estimate, then accept iff the observed portfolio is still optimal under the proposal — with this proposal the Metropolis ratio reduces to that feasibility indicator — rebuilding the rows for accepted students and otherwise keeping the incumbent. The test is reorder-aware, and initializing at $\hat{q}$ nests the fixed-belief sampler. Proposals are drawn separately for each program-cohort cell; cells with perfect separation draw from a $\text{Beta}(r+1, n-r+1)$ distribution, where r of the cell's n applicants received offers, and cells without a fitted probit keep a belief of 0.3.

F.4 Warm start

The free-scale sampler is initialized from a pooled preliminary run in which all students share one close-to-home coefficient. The results of this run provide a population mean near the true parameter, per-student deviations of each student's pooled utilities on their characteristics, and a feasible utility vector to begin the estimation procedure. The fixed effects are started at the program-level means of the pooled utilities, shrunk toward the overall mean so that unpopular programs start near $\bar{\delta}$ and popular ones start near their own mean. The pooled run is estimated on a free scale and handed off by a single global rescaling onto $\beta^d = -0.01$, exact by scale-invariance of the cone.

F.5 Ranking stage

The ranking stage uses the same device. Each candidate ranking's expected value is linear in the program latents, so the requirement that the realized pair maximize expected utility gives linear winner inequalities $\mathcal{W}_i U_i \geq 0$, which we sample with the same hit-and-run kernel. Conditional on the latents, we draw the ranking-shock variance σ_ϵ^2 (inverse-Gamma) and the coefficient λ_i on the unconditional-offer premium (a multivariate-normal regression on z_i of the utility gap that carries the premium, as in [Equation \(9\)](#)). This stage also carries the private signal: ψ_i is augmented per student, and σ_ψ is drawn by slice sampling on $(0, \sigma_r)$ under a half- t prior, given $\{\psi_i\}$ and the realized $\{r_i\}$, with the target set to zero unless every observed ranking remains feasible. Because ψ_i enters the ranking inequalities through $\Phi(\cdot)$ rather than linearly, its draw is a Metropolis step within the

block. Stage-1 uncertainty is propagated by drawing each β_i^c and the fixed effects δ from their Stage-1 posterior. This procedure was validated on simulated data.

F.6 Validity of the sequential estimator

Within each block, the estimator is a data-augmentation Gibbs sampler of the kind [McCulloch and Rossi \(1994\)](#) introduced for discrete choice. Where the full conditional is not conjugate (the belief resampling at Stage 1 and the signal draw at Stage 3), we use Metropolis steps. The resulting Metropolis-within-Gibbs kernel leaves the target posterior invariant and is ergodic under the conditions of [Tierney \(1994\)](#). We report posterior means and standard deviations after burn-in. We assess convergence using split- $\hat{R}$, effective sample sizes and Geweke diagnostics, reported in [Table A.7](#).

We estimate the three blocks in order: the supply side (Stages 2 and 4), then Stage 1, then Stage 3. Each block conditions on the estimates of the blocks before it and carries forward part of their uncertainty. The Stage-1 sampler resamples the offer-probit component of $\hat{q}_{is}$ from its asymptotic distribution and holds the conversion-probit component at its point estimate ([Appendix F.3](#)); because a student's belief draw is accepted only if her portfolio stays feasible, her applications restrict the beliefs used in Stage 1, but they do not change the supply-side estimates used anywhere else. The Stage-3 sampler draws β_i^c and δ from their Stage-1 posterior ([Appendix F.5](#)), and the rankings do not update the Stage-1 parameters.

This cut rests on two features of the model. First, students hold true beliefs about admission ([Section 4.1.2](#)), so the supply-side estimates are also estimates of students' beliefs. Second, Stage-1 beliefs depend on the supply side only through the scalar q_{is} . Its marginal components $\hat{p}_{is}$ and $\hat{a}_{is}$ are correctly specified whatever the correlation across programs, because the common shock μ_i is integrated out of each marginal ([Section 4.3.1](#)). Combining them into $\hat{q}_{is}$ through the conditional conversion probability $\hat{a}_{is}^o$ additionally uses the bivariate-normal structure of μ_i .

The blocks follow estimation order, not the game's timing. The conversion probit is fit on first-ranked choices, so it uses a Stage-3 decision, and its fitted values enter Stage-1 beliefs. This creates no circularity, because $\hat{q}_{is}$ is evaluated for every applicant rather than at the student's own realized ranking. The cost is that the conversion probit uses only first-ranked choices, not the full at-risk sample of [Section 4.3.1](#).

F.7 Run lengths and priors

The Stage-1 block is a Gibbs sampler over $(\delta, \bar{\delta}, \sigma_\delta^2, \bar{\beta}^c, \Gamma^c, \sigma_\eta^2, \sigma_v^2, \{\eta_i^c\})$ and the utilities $\{U_i\}$, run for 50,000 iterations with 5,000 discarded as burn-in. Priors are Normal-inverse-Gamma for $(\delta, \sigma_\delta^2)$, Normal for $(\bar{\beta}^c, \Gamma^c)$ with a half- t prior on σ_η , and inverse-Gamma for σ_v^2 . The Stage-3 block is run for 250,000 iterations with 25,000 discarded.

F.8 Counterfactuals

We detail the application system designs used in [Section 4.6](#).

For each program-cohort we compute the *expected capacity* $\hat{N}_{st}$ that program s expects to fill in cohort t , given the offers it made at Stage 2 in the estimated baseline and its own beliefs about yield

and cutoff clearing. Summing over the applicants who hold an offer from s , this is the probability that an applicant ranks s first and clears its condition, plus the probability that she ranks s second, misses her first-choice condition, and clears s 's. The second term is computed as a joint rather than as a product of marginals: missing one condition and clearing another are both driven by the same realization of r_i , so the two events are negatively dependent and multiplying the marginals overstates the rebound. We hold $\hat{N}_{st}$ fixed across designs.

For each design, we compute a fixed point of the best-response map in student and program strategies. We iterate until three conditions hold: the weighted average gap between each program's expected intake and $\hat{N}_{st}$ is below a tolerance, aggregate intake is within tolerance of total capacity, and every program's intake is within three sampling standard deviations of $\hat{N}_{st}$. We cap the loop at 80 iterations and keep the best iterate among those that meet these conditions. Students choose portfolios at Stage 1 and rankings at Stage 3 to maximize expected utility, as in [Section 4.1.4](#) and [Section 4.1.2](#). Their Stage-1 admission beliefs use the offer cutoffs and unconditional-offer shares that programs play in the design, computed on a preliminary pool in which students apply on tastes alone rather than on the pool that actually applies. Programs choose scalar cutoffs, taking the applicant pool as given, to maximize the expected sum of enrolling students' quality subject to expected enrollment not exceeding $\hat{N}_{st}$, where a student's quality to program s is measured by s 's own estimated weights on her characteristics.

Two features of the program side matter. First, the relevant index differs by stage, because the program's information does. At Stage 2 the program has not yet observed r_i (which is why it makes conditional offers), so it maximizes expected quality given its Stage-2 information, and the offer cutoff $\bar{\pi}_s$ is a threshold on π_{is} in [Equation \(11\)](#). At Stage 4, r_i is realized, and the conversion cutoff $\bar{\pi}_s^R$ is a threshold on the admissions index in [Equation \(8\)](#). Second, following [Kapor \(2025\)](#), a program that maximizes expected enrolled quality subject to capacity uses a threshold rule on its index, so its choice reduces to a scalar cutoff.

Except where noted, the data-generating process in each design is identical to the setup described in [Section 4](#). Program fixed effects $\{\delta_s\}$ are recalibrated in the simulated baseline to match observed application counts and then held fixed across designs; all other preference parameters are held at their estimated values, and the choice set consists of the same set of programs as used in parameter estimation. Some offer holders decline all their offers, or rank no second choice, for reasons outside the model; in each design we add such random declines so that the decline and no-second-choice rates, by number of offers held, match the data.

In simulation, the offer score of [Equation \(11\)](#) is implemented as $\pi_{is} = \kappa_s \tilde{z}_{is} + \mu_i$, where $\tilde{z}_{is}$ is the index $z_i^{\text{offer}} \gamma_s$ centered within each program-cohort and κ_s is a per-program scale fit to how sharply offers separate offer holders from other applicants in the data. The offer threshold varies by program-cohort, $\bar{\pi}_{st}$, so an offer from s is the event $\mu_i \geq L_{is} \equiv \bar{\pi}_{st} - \kappa_s \tilde{z}_{is}$; students update on these thresholds at Stage 3.

1. **Ban on unconditional offers.** In Stage 2 we set $\bar{\pi}_s^U = \infty$ for all program-cohorts, so that every offer is conditional and every first choice is tested against $\bar{\pi}_s^R$. Programs re-optimize $\bar{\pi}_s$ and $\bar{\pi}_s^R$.
2. **Unconditional behavior consistent with 2018.** In Stage 2, $\bar{\pi}_s^U$ is set program-cohort by

program-cohort so that the share of s 's offers that are unconditional equals its 2018 share; what is held fixed across cohorts is the share rather than the threshold, since the applicant pool differs by cohort. Programs re-optimize $\bar{\pi}_s$ and $\bar{\pi}_s^R$ subject to that constraint.

3. **Post-qualification admissions.** There is a single application stage, taken after age-18 results are released, so that r_i is common knowledge. Three things follow. The signal ψ_i is moot, since students and programs observe r_i alike. The conditional–unconditional distinction disappears, since there is no future performance left to condition on: every place a program offers is unconditional, and the single cutoff $\bar{\pi}_s^R$ on the admissions index of Equation (8) is the admission decision. Because application and ranking now coincide, a student knows her full idiosyncratic taste $v_{is} + \varepsilon_{is}$ when she chooses, rather than only v_{is} . Programs rank the full pool of applicants by that index and admit to capacity; students rank programs; the outcome is the stable matching, which we compute as the student-proposing deferred acceptance outcome. The only uncertainty a student faces at the point of choosing is her own μ_i , which programs observe and she does not.

Post-qualification admissions are also the policy analogue of removing students' informational advantage: once results are public, the asymmetry that drives selection into unconditional offers disappears institutionally rather than by assumption.